# IWASAWA THEORY OVER $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ AND WEIL-ÉTALE COHOMOLOGY

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Abstract. For a rational prime $p$, we establish key ring-theoretical properties of the completed group-ring $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ and also strengthen the theory of pro-discrete $\mathbb{Z}[\![\mathbb{Z}_p]\!]$-modules introduced in [7]. We then apply the resulting theory to obtain a variety of new, and explicit, results concerning divisor class groups and the Weil-étale cohomology of $\mathbb{G}_m$ over general $\mathbb{Z}_p$-extensions of a global function field $k$ of characteristic $p$. In particular, we formulate and, under mild technical hypotheses, prove a 'main conjecture of integral Iwasawa theory' over finite abelian extensions $\mathcal{F}$ of the constant $\mathbb{Z}_p$-extension of $k$ that is significantly finer than the (simultaneous) validity of the equivariant Tamagawa number conjecture for $\mathbb{G}_m$ over every finite subextension of $\mathcal{F}/k$.

## 1. Introduction

For a commutative ring $A$, rational prime $p$ and finite group $H$, the completed group-ring $A[\![\mathbb{Z}_p \times H]\!]$ is the inverse limit of the group-rings $A[(\mathbb{Z}/(p^n)) \times H]$ with respect to the canonical projection maps $A[(\mathbb{Z}/(p^{n+1})) \times H] \to A[(\mathbb{Z}/(p^n)) \times H]$.

Arithmetic modules over the ring $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ arise naturally from the limits of families of modules in $\mathbb{Z}_p$-extensions of global fields. However, since $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ is neither Noetherian nor has a natural compact topology, such modules are usually considered, following Iwasawa, by passing to pro-$p$ completions and then working over the associated (Noetherian and compact) algebra $\mathbb{Z}_p[\![\mathbb{Z}_p]\!]$. Nevertheless, this passage to pro-$p$ completion can lose significant information and previous authors have either stressed the benefits, and intrinsic difficulties, in concrete situations of working over either $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ or $\mathbb{Z}_\ell[\![\mathbb{Z}_p]\!]$ for a prime $\ell \neq p$, or have tried to understand particular aspects of the problem (see, for instance, Washington [34] and Coleman [10] and, for a more recent example, Bandini and Longhi [3]). With these general issues in mind, in previous work with Daoud [7] we introduced a category of 'pro-discrete' $\mathbb{Z}[\![\mathbb{Z}_p]\!]$-modules for which one can prove analogues of both Nakayama's Lemma and Roiter's Lemma and thereby establish criteria for finite-presentability. By using these results we were then able to prove that the ring $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ is not coherent (in the classical sense of Chase [9] and Bourbaki [5, Chap. 1]) but is 2-coherent (in the sense of Costa [12]) and also has weak Krull dimension 2 (in the sense of Tang [28]). In this article, we further clarify the basic ring-theoretical properties of $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ before extending key aspects of the general theory of pro-discrete modules and then applying the results to study Iwasawa-theoretic modules in the setting of global function fields.

In a little more detail, after reviewing the basic theory of pro-discrete modules in §2, we shall start our investigations in §3 by studying the explicit interplay between ideals of $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ and of $\mathbb{Z}_\ell[\![\mathbb{Z}_p]\!]$ for each prime $\ell$. In this way, for example, we prove $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ has infinite Krull dimension and that ideals of the classical Iwasawa algebra $\mathbb{Z}_p[\![\mathbb{Z}_p]\!]$ are uniquely determined by their pull-backs to $\mathbb{Z}[\![\mathbb{Z}_p]\!]$. We then discuss the maximal spectrum of $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ and, in particular, classify all maximal ideals that are pro-discrete and also show, using a classical result of Zsigmondy, that there are infinitely many maximal ideals whose residue field is a rational vector space of uncountable dimension.

In §4 we then establish criteria for finitely-generated pro-discrete modules over rings of the form $\mathbb{Z}[\![\mathbb{Z}_p \times H]\!]$ for a finite abelian group $H$ to be projective and to be torsion. We also obtain a method for computing the initial Fitting ideals of finitely-presented pro-discrete torsion modules over such rings, before establishing basic properties of the derived limits of perfect complexes in the

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theory of pro-discrete modules. In the remainder of the article, we shall then apply the resulting theory to study the $\mathbb{Z}[\![\mathbb{Z}_p \times H]\!]$-modules that naturally arise over $(\mathbb{Z}_p \times H)$-extensions of a global function field $k$ of characteristic $p$ from the limits of divisor class groups (in §5) and from the limits of Lichtenbaum's Weil-étale cohomology of $\mathbb{G}_m$ (in §6).

To be more precise about these arithmetic applications, in Theorem 5.1 we prove that, if only finitely many places ramify in a given $\mathbb{Z}_p$-extension of $k$, then the inverse limit of divisor class groups over this extension is both pro-discrete and torsion over $\mathbb{Z}[\![\mathbb{Z}_p]\!]$. By using these facts, we can then also establish criteria in terms of explicit growth conditions that are equivalent to the limit module being pseudo-coherent.

Assuming these growth conditions to be satisfied by a given $(\mathbb{Z}_p \times H)$-extension $\mathcal{F}/k$, we shall then prove in Theorem 6.5 that the $\mathbb{Z}[\![\mathbb{Z}_p \times H]\!]$-module arising from the cohomology in degree 1 of the derived inverse limit of Weil-étale cohomology complexes of $\mathbb{G}_m$ over the tower $\mathcal{F}/k$ is both finitely-presented and torsion and that its initial Fitting ideal can be described in terms of the values of Dirichlet $L$-series. We stress here that this link between Dirichlet $L$-series and the structure of Weil-étale cohomology is established over rings $\mathbb{Z}[\![\mathbb{Z}_p \times H]\!]$ that have infinite Krull dimension and are neither coherent, compact nor regular and that it incorporates information about full divisor class groups rather than just their $p$-primary parts. The latter feature reflects the underlying fact that our techniques permit the first unified treatment of the different $\ell$-primary components of profinite Iwasawa modules, the advantages of any potential such approach having essentially already been made clear elsewhere (see, for example, the discussions in the introductions to [34] and [35] and, more recently, in [3, §1.4]). A further point of interest is that the use of derived limit complexes is forced in this context since it can be shown that the inverse limits of Weil-étale cohomology groups are themselves finitely-presented over $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ if and only if the corresponding inverse systems of unit groups and norm maps are Mittag-Leffler and that this almost never happens (see Corollary 6.6).

The growth conditions on divisor class groups that our approach requires seem very mild (from the point of view of general Cohen-Lenstra-Martinet type heuristics). In particular, we can verify them unconditionally over the constant $\mathbb{Z}_p$-extension $k^{\mathrm{con}}$ of $k$ and so, in this setting, the theory can be further developed. In particular, in §7 the results we obtain for $(\mathbb{Z}_p \times H)$-extensions $\mathcal{F}/k$ with $k^{\mathrm{con}} \subseteq \mathcal{F}$ motivate the formulation of a 'main conjecture of integral Iwasawa theory' for the Weil-étale cohomology of $\mathbb{G}_m$ over $\mathcal{F}/k$. This conjecture is considerably finer than the (already known) validity of the equivariant Tamagawa number conjecture for $\mathbb{G}_m$ over all finite subextensions of $\mathcal{F}/k$ and suggests links between Zeta functions and Weil-étale cohomology that appear stronger than those observed previously. We provide concrete evidence for the conjecture and, in particular, prove it fully for several natural families, including the case that $H$ is any $p$-group and also (under mild conditions) for fields $\mathcal{F} = F^{\mathrm{con}}$ in which $F/k$ is any finite subextension of the Drinfeld modular towers of $k$ studied by Bley and Popescu [4] (and hence of the Carlitz-Hayes cyclotomic extensions of $k$ studied by Anglès et al [1]). For more details see, in particular, Theorem 7.1, Remark 7.2, Conjecture 7.8 and Remark 7.9.

It is, of course, widely believed, following the philosophy of Iwasawa, that constant $\mathbb{Z}_p$-extensions of global function fields of characteristic $p$ are in key respects strongly analogous to cyclotomic $\mathbb{Z}_p$-extensions of number fields (see, for example, the discussion starting on [25, p. 188]). For this reason, the above results naturally suggest a variety of new and interesting questions over cyclotomic $\mathbb{Z}_p$-extensions of number fields that concern the structure over $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ of inverse limits of both (full) ideal class groups and the Selmer groups of $\mathbb{G}_m$, and their possible consequences concerning asymptotic growth formulae. We consider such issues elsewhere.

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## 2. Pro-discrete modules

For the reader's convenience, in this section we shall quickly review relevant aspects of the theory of pro-discrete modules developed in [7]. Throughout, we write $\mathbb{N}_0$ for the set $\mathbb{N} \cup \{0\}$ of non-negative integers and set

$$R := \mathbb{Z}[\![\mathbb{Z}_p]\!] \quad \text{and} \quad R_n := \mathbb{Z}[\mathbb{Z}/(p^n)] \text{ for } n \in \mathbb{N}_0. \tag{1}$$

We also fix a finite abelian group $H$ and consider modules over the group ring $R[H]$. In particular, for each $R[H]$-module $M$ and $n \in \mathbb{N}_0$ we define an $R_n[H]$-module

$$M_{(n)} := R_n \otimes_R M,$$

where the tensor product is with respect to the natural (surjective) homomorphism $R \to R_n$.

If $M$ is a module over any ring $A$, then we write $\mu_A(M)$ for the cardinality of a minimal generating set, with $\mu_A(M)$ regarded as $\infty$ if $M$ is not finitely-generated. For an abelian group $M$ we write $M_{\mathrm{tor}}$ for its torsion subgroup and $M_{\mathrm{tf}}$ for the quotient of $M$ by $M_{\mathrm{tor}}$.

We write $\mathcal{P}$ for the set of rational primes. For $\ell \in \mathcal{P}$ we write $\mathbb{F}_\ell$ for the field of cardinality $\ell$ and, for an abelian group $M$, we set $M/\ell := \mathbb{F}_\ell \otimes_{\mathbb{Z}} M$ and write $M[\ell^\infty]$ for the maximal $\ell$-primary subgroup of $M_{\mathrm{tor}}$.

We first recall the definition of a pro-discrete $R[H]$-module.

**Definition 2.1.** An inverse system $(M_n, \pi_n)_{n\in\mathbb{N}_0}$ of $R[H]$-modules is a *pro-discrete system* if each $M_n$ is naturally an $R_n[H]$-module and, for each $n \in \mathbb{N}$, the transition morphism $\pi_n$ induces an isomorphism $\overline{\pi}_n : (M_n)_{(n-1)} \cong M_{n-1}$ of $R_{n-1}[H]$-modules. An $R[H]$-module is said to be *pro-discrete* if it is equal to the limit of a pro-discrete system of $R[H]$-modules.

**Remark 2.2.** The ring $R$ is itself a pro-discrete $R$-module since it is the limit of the pro-discrete inverse system $(R_n, \varrho^n_{n-1})_{n\in\mathbb{N}_0}$ in which $\varrho^n_{n-1}$ for $n \in \mathbb{N}$ denotes the natural map $R_n \to R_{n-1}$. In general, any $R[H]$-module $M$ gives rise to a pro-discrete system $(M_{(n)}, \pi_n)_n$, with $\pi_n$ for $n \in \mathbb{N}$ the canonical map $M_{(n)} \to M_{(n-1)}$, and hence to a pro-discrete module $\varprojlim_n M_{(n)}$ for which there exists a canonical map $\pi_M : M \to \varprojlim_n M_{(n)}$. However, many finitely-presented $R[H]$-modules are not pro-discrete and so the category of pro-discrete $R[H]$-modules is not abelian (for more details see [7, Rem. 3.12]).

The next result provides analogues of Nakayama's Lemma and Roiter's Lemma for the category of pro-discrete $R[H]$-modules.

Before stating it, we recall that an $R_n[H]$-module $M$ is said to be 'locally-free (of rank $d$)' if for every rational prime $\ell$ the $(\mathbb{Z}_\ell \otimes_{\mathbb{Z}} R_n)[H]$-module $\mathbb{Z}_\ell \otimes_{\mathbb{Z}} M$ is free (of rank $d$).

**Proposition 2.3** ([7, Th. 3.8])**.** *Let $M$ be a pro-discrete $R[H]$-module that arises as the limit of a pro-discrete system $(M_n, \pi_n)_{n\in\mathbb{N}_0}$. Then the following claims are valid.*

(i) *$\mu_{R[H]}(M)$ is finite if and only if each $\mu_{R_n}(M_n)$ is finite and bounded independently of $n$.*
(ii) *If $\mu_{R[H]}(M)$ is finite, then $\mu_{\mathbb{Z}[H]}(M_0)$ is finite and there exists a pro-discrete $R[H]$-submodule $M'$ of $M$ with both of the following properties:*

(a) $\mu_{R[H]}(M') \leq \mu_{\mathbb{Z}[H]}(M_0)$;
(b) $M/M'$ *is pro-discrete and, for every* $n$, *the order of* $(M/M')_{(n)}$ *is finite and prime-to-p. In particular, if each* $R_n[H]$*-module* $M_n$ *is locally-free, then for any given natural numbers* $m$ *and* $t > 1$, *the* $R[H]$*-module* $M'$ *can be chosen to be free and such that* $\big(M/M'\big)_{(m)}$ *is finite and of order prime-to-pt, respectively such that* $M'_m = M_m$ *if* $M_m$ *is a free* $R_n[H]$*-module.*

We next recall criteria for the finite-presentability of finitely-generated pro-discrete modules. To state the result we fix a topological generator $\gamma$ of $\mathbb{Z}_p$ and, for each $n \in \mathbb{N}_0$, set

$$\varpi_n := \gamma^{p^n} - 1 \in R \quad \text{and} \quad \gamma_n := \varrho_n(\gamma) \in R_n,$$

with $\varrho_n$ the group-ring homomorphism $R \to R_n$ induced by the canonical projection $\mathbb{Z}_p \to \mathbb{Z}/p^n$. In the sequel, for $m \in \mathbb{N}$ with $m > n$, we similarly write $\varrho_n^m$ for the natural ring homomorphism $R_m \to R_n$.

**Proposition 2.4** ([7, Th. 3.11])**.** *Let* $M$ *be a finitely-generated* $R[H]$*-module.*

(i) *Then* $M$ *is pro-discrete if and only if the natural map* $\pi_M : M \to \varprojlim_n M_{(n)}$ *is bijective. In particular, if* $M$ *is the limit of a pro-discrete system* $(M_n, \pi_n)_n$ *then for every* $n$ *the natural map* $M_{(n)} \to M_n$ *is bijective.*
(ii) *If* $M$ *is pro-discrete, then it is finitely-presented if and only if* $\mu_{R_n}(\mathsf{Tor}_1^R(R_n, M))$ *is bounded independently of* $n$. *The latter condition is satisfied if, for example, each* $M_n$ *is finite and the quantities* $\mu_R\big(M^{\varpi_n=0}\big)$ *are finite and bounded independently of* $n$.
(iii) *If* $M$ *is a submodule of* $R[H]^a$ *for some natural number* $a$, *then* $M$ *is pro-discrete. In addition, any such module* $M$ *is finitely-presented if each group* $\big(R[H]^a/M\big)^{\varpi_n=0}[p^\infty]$ *has bounded exponent.*

## 3. Ideal theory

In this section we investigate the ideal theory of $R$. More precisely, we clarify aspects of the interplay between the ideal theories of $R$ and of $R^\ell := \mathbb{Z}_\ell[\![\mathbb{Z}_p]\!]$ for each prime number $\ell$, before proving that $R$ has infinite Krull dimension and then determining the structure of $\mathrm{MaxSpec}(R)$.

For each $n \in \mathbb{N}_0$, we write $I(n)$ for the ideal $\ker(\varrho_n)$ of $R$ and recall from [7, Prop. 2.1(ii)] the existence of a canonical short exact sequence of $R$-modules

$$0 \to R \cdot \varpi_n \xrightarrow{\subset} I(n) \xrightarrow{\xi_n} \big(\mathbb{Z}_p/\mathbb{Z}\big) \otimes_{\mathbb{Z}} R_n \to 0, \tag{2}$$

in which $R$ acts on the third module via $1 \otimes_R \varrho_n$ and $\xi_n$ is a natural connecting homomorphism that arises in the theory of pro-discrete modules.

These sequences play a key role in the theory developed in [7] and, to help set context, we record the following result which shows that their classification is not straightforward.

**Lemma 3.1.** *The group* $\mathrm{Ext}^1_R\big(\big(\mathbb{Z}_p/\mathbb{Z}\big) \otimes_{\mathbb{Z}} R_n, R \cdot \varpi_n\big)$ *is isomorphic to* $\mathrm{Ext}^1_R\big(\big(\mathbb{Z}_p/\mathbb{Z}\big) \otimes_{\mathbb{Z}} R_n, R\big)$ *and is both uniquely* $p$*-divisible and contains a rational vector space of uncountable dimension.*

*Proof.* The isomorphism is induced by the fact that the $R$-module $R \cdot \varpi_n$ is free of rank one (since $R$ is a domain and $\varpi_n \neq 0$). To prove the remaining assertions, we set $R_n^{\langle p \rangle} := \big(\mathbb{Z}_p/\mathbb{Z}\big) \otimes_{\mathbb{Z}} R_n$ and $\mathcal{E}_n^1 := \mathrm{Ext}^1_R\big(R_n^{\langle p \rangle}, R \cdot \varpi_n\big)$. Then, as the group $\mathbb{Z}_p/\mathbb{Z}$ is uniquely $p$-divisible, so is $\mathcal{E}_n^1$. In addition, since $I(n)$ has no non-zero $p$-divisible elements, the group $\mathrm{Hom}_R\big(R_n^{\langle p \rangle}, I(n)\big)$ vanishes and so the canonical connecting homomorphism $\mathrm{Hom}_R\big(R_n^{\langle p \rangle}, R_n^{\langle p \rangle}\big) \to \mathcal{E}_n^1$ induced by (2) is injective. In particular, since the torsion subgroup of $\mathbb{Z}_p/\mathbb{Z}$ is $\mathbb{Z}_{(p)}/\mathbb{Z}$, with $\mathbb{Z}_{(p)}$ the localisation of $\mathbb{Z}$ at $p$, and $\mathbb{Z}_p/\mathbb{Z}_{(p)}$ is a rational vector space of uncountable dimension, it is enough to note that the group

$$\mathrm{Hom}_R\big((\mathbb{Z}_p/\mathbb{Z}_{(p)}) \otimes_{\mathbb{Z}} R_n, R_n^{\langle p \rangle}\big) \cong \mathrm{Hom}_{\mathbb{Z}}\big(\mathbb{Z}_p/\mathbb{Z}_{(p)}, R_n^{\langle p \rangle}\big) = \mathrm{Hom}_{\mathbb{Q}}(\mathbb{Z}_p/\mathbb{Z}_{(p)}, \mathbb{Z}_p/\mathbb{Z}_{(p)} \otimes_{\mathbb{Z}} R_n)$$

is isomorphic to a subgroup of $\mathrm{Hom}_R\big(R_n^{\langle p\rangle}, R_n^{\langle p\rangle}\big)$. $\square$

3.1. **Ideals of $R$ and of each $R^\ell$.** For $\ell \in \mathcal{P}$ and $n \in \mathbb{N}$, we set $R_n^\ell := \mathbb{Z}_\ell[\![\mathbb{Z}/p^n]\!]$ and abuse notation by continuing to write $\varrho_n$ and $\varrho_n^m$ for $m > n$ for the homomorphisms $R^\ell \to R_n^\ell$ and $R_m^\ell \to R_n^\ell$ that are respectively induced by the projections $\varrho_n$ and $\varrho_n^m$. We also note that each ring $R^\ell = \varprojlim_{\varrho_{n-1}^n} R_n^\ell$ has a natural (compact) inverse limit topology and, for a subset $X$ of $R^\ell$, we write $X^c$ for its closure with respect to this topology.

**Proposition 3.2.** *The following claims are valid for every ideal $J$ of $R$.*

(i) $\bigcap_{n\in\mathbb{N}_0}(J + I(n)) = \bigcap_{\ell\in\mathcal{P}}\big(R \cap (R^\ell \cdot J)^c\big)$.

(ii) *If the natural map $\pi_J : J \to \varprojlim_n J_{(n)}$ is bijective (as is automatically the case, by Proposition 2.4(i), if $J$ is finitely-generated), then one has*

$$\Big(\bigcap\nolimits_{n\in\mathbb{N}_0}(J + I(n))\Big)^2 \subseteq J \subseteq \bigcap\nolimits_{n\in\mathbb{N}_0}(J + I(n)).$$

*In particular, if $J$ is a prime ideal for which $\pi_J$ is bijective, then $J = \bigcap_{\ell\in\mathcal{P}}\big(R \cap (R^\ell \cdot J)^c\big)$.*

(iii) *There exists an infinite chain $J(1) \supsetneq J(2) \supsetneq \cdots$ of principal (and hence pro-discrete) ideals of $R$ with both of the following properties.*

(a) *For $i \in \mathbb{N}$ and $n \in \mathbb{N}_0$, one has $J(i) + I(n) = I(0)$, and hence $\varrho_n(J(i)) = R_n \cdot \varrho_n(\varpi_0)$.*

(b) *For $i \in \mathbb{N}$ and $\ell \in \mathcal{P}$, one has $R^\ell \cdot J(i) = R^\ell \cdot \varpi_0$.*

*Proof.* We first claim that, for each $\ell \in \mathcal{P}$ and ideal $X$ of $R^\ell$, one has

$$X^c = \varprojlim_{\varrho_{n-1}^n} \varrho_n(X). \tag{3}$$

To show this, we note that each $\varrho_n(X)$ is a $\mathbb{Z}_\ell$-submodule of $R_n^\ell$, and hence compact. The limit $\varprojlim_{\varrho_{n-1}^n} \varrho_n(X)$ is therefore a compact submodule of $R^\ell = \varprojlim_{\varrho_{n-1}^n} R_n^\ell$, and hence closed. In particular, since it is clear $X \subseteq \varprojlim_{\varrho_{n-1}^n} \varrho_n(X)$, one also has $X^c \subseteq \varprojlim_{\varrho_{n-1}^n} \varrho_n(X)$. It is therefore enough to show that any closed submodule $Y$ of $R^\ell$ that contains $X$ must contain $\varprojlim_{\varrho_{n-1}^n} \varrho_n(X)$. To do this we use the obvious exact commutative diagrams

$$\begin{array}{ccccccccc}
0 & \longrightarrow & Y \cap (R^\ell \cdot \varpi_n) & \xrightarrow{\subseteq} & Y & \xrightarrow{\varrho_n} & \varrho_n(Y) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle\subseteq} & & \| & & \downarrow{\scriptstyle\varrho_{n-1}^n} & & \\
0 & \longrightarrow & Y \cap (R^\ell \cdot \varpi_{n-1}) & \xrightarrow{\subseteq} & Y & \xrightarrow{\varrho_{n-1}} & \varrho_{n-1}(Y) & \longrightarrow & 0.
\end{array}$$

Since $Y$ and each $R^\ell \cdot \varpi_n$ are closed submodules of $R^\ell$, the first terms in both rows are compact (and the inclusion map between them is continuous with respect to these topologies). Recalling that limits are exact for inverse systems of compact modules (indexed by $\mathbb{N}$), it follows that the derived limit $\varprojlim_n^1(Y \cap (R^\ell \cdot \varpi_n))$ vanishes. Upon passing to the inverse limit over $n$ of the above diagrams, and noting that $\bigcap_n(R^\ell \cdot \varpi_n) = (0)$, one therefore deduces that the inclusion map $Y \to \varprojlim_{\varrho_{n-1}^n} \varrho_n(Y)$ is bijective. In particular, since $\varrho_n(X) \subseteq \varrho_n(Y)$ for each $n$, it follows that $\varprojlim_{\varrho_{n-1}^n} \varrho_n(X) \subseteq Y$, as required to prove (3).

To prove (i) we use the fact that, for every $\ell \in \mathcal{P}$ and $n \in \mathbb{N}_0$, one has $\varrho_n(R^\ell \cdot J) = \mathbb{Z}_\ell \otimes_{\mathbb{Z}} \varrho_n(J)$. This implies, in particular, that there exists an exact sequence of $R_n$-modules

$$0 \to \varrho_n(J) \xrightarrow{\subseteq} R_n \cap \varrho_n(R^p \cdot J) \to R_n/\varrho_n(J) \xrightarrow{\phi} R_n^p/\varrho_n(R^p \cdot J),$$

in which $\phi$ sends each element $x$ to $1 \otimes x$ in $\mathbb{Z}_p \otimes_{\mathbb{Z}} (R_n/\varrho_n(J)) \cong R_n^p/\varrho_n(R^p \cdot J)$. In particular, since the kernel of $\phi$ is

$$\bigoplus\nolimits_{\ell\in\mathcal{P}\setminus\{p\}}\big(R_n/\varrho_n(J)\big)[\ell^\infty] = \bigoplus\nolimits_{\ell\in\mathcal{P}\setminus\{p\}}\big(R_n^\ell/\varrho_n(R^\ell \cdot J)\big)[\ell^\infty],$$

one obtains an exact commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \varrho_n(J) & \xrightarrow{\subseteq} & R_n \cap \varrho_n(R^p \cdot J) & \xrightarrow{(\beta_n^\ell)_\ell} & \bigoplus_{\ell \in \mathcal{P}\setminus\{p\}} \big(R_n^\ell / \varrho_n(R^\ell \cdot J)\big) \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \varrho_{n-1}(J) & \xrightarrow{\subseteq} & R_{n-1} \cap \varrho_{n-1}(R^p \cdot J) & \xrightarrow{(\beta_{n-1}^\ell)_\ell} & \bigoplus_{\ell \in \mathcal{P}\setminus\{p\}} \big(R_{n-1}^\ell / \varrho_{n-1}(R^\ell \cdot J)\big), \end{array}$$

where all vertical maps are induced by $\varrho_{n-1}^n$ and each $\beta_n^\ell$ by the composite $R_n \subset R_n^\ell \to R_n^\ell/\varrho_n(R^\ell \cdot J)$. By passing to the limit of these diagrams, and taking account of (3) with $X = R^p \cdot J$, we therefore obtain an exact sequence of $R$-modules

$$0 \to \varprojlim_{\varrho_{n-1}^n} \varrho_n(J) \xrightarrow{\subseteq} R \cap (R^p \cdot J)^c \xrightarrow{(\varprojlim_n \beta_n^\ell)_\ell} \prod\nolimits_{\ell \in \mathcal{P}\setminus\{p\}} \varprojlim_{\varrho_{n-1}^n} \big(R_n^\ell / \varrho_n(R^\ell \cdot J)\big). \tag{4}$$

In addition, for each $\ell \in \mathcal{P} \setminus \{p\}$, by passing to the limit over $n$ of the exact commutative diagrams

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \varrho_n(R^\ell \cdot J) & \xrightarrow{\subseteq} & R_n^\ell & \longrightarrow & R_n^\ell / \varrho_n(R^\ell \cdot J) & \longrightarrow & 0 \\ & & \varrho_{n-1}^n \downarrow & & \varrho_{n-1}^n \downarrow & & \varrho_{n-1}^n \downarrow & & \\ 0 & \longrightarrow & \varrho_{n-1}(R^\ell \cdot J) & \xrightarrow{\subseteq} & R_{n-1}^\ell & \longrightarrow & R_{n-1}^\ell / \varrho_{n-1}(R^\ell \cdot J) & \longrightarrow & 0 \end{array}$$

(in which all occurring modules are compact), and applying (3) with $X = R^\ell \cdot J$, one obtains a natural isomorphism of $R^\ell$-modules $R^\ell/(R^\ell \cdot J)^c \cong \varprojlim_{\varrho_{n-1}^n} \big(R_n^\ell / \varrho_n(R^\ell \cdot J)\big)$. Upon combining these isomorphisms with (4), one obtains an exact sequence of $R$-modules

$$0 \to \varprojlim_{\varrho_{n-1}^n} \varrho_n(J) \xrightarrow{\subseteq} R \cap (R^p \cdot J)^c \xrightarrow{\beta} \prod\nolimits_{\ell \in \mathcal{P}\setminus\{p\}} \big(R^\ell / (R^\ell \cdot J)^c\big)$$

in which $\beta = (\beta^\ell)_\ell$ and each $\beta^\ell$ is induced by the natural composite map $R \subset R^\ell \to R^\ell/(R^\ell \cdot J)^c$. In particular, one has $\ker(\beta) = \bigcap_{\ell \in \mathcal{P}} \big(R \cap (R^\ell \cdot J)^c\big)$. To deduce the equality in (i) from the above exact sequence, it is thus enough to show that the ideals $\varprojlim_{\varrho_{n-1}^n} \varrho_n(J)$ and $\bigcap_{n \in \mathbb{N}_0} (J + I(n))$ of $R$ coincide. In addition, this is true since the explicit definition of each ideal $I(n)$ as the kernel of the projection $\varrho_n : R \to R_n$ ensures that both of these ideals are equal to the unique largest ideal $\tilde{J}$ of $R$ with the property that $\varrho_n(\tilde{J}) = \varrho_n(J)$ for every $n$.

To prove (ii) we use the ideal $\tilde{J} = \varprojlim_{\varrho_{n-1}^n} \varrho_n(J) = \bigcap_{n \in \mathbb{N}_0} (J + I(n))$ and also, for each $n \in \mathbb{N}_0$, set $Q_n(J) := \mathsf{Tor}_1^R(R/J, R_n)$. Then, for each $n \in \mathbb{N}$, the argument of [7, Th. 3.11(i)] gives an exact commutative diagram of $R_n$-modules

$$\begin{array}{ccccccccc} 0 & \longrightarrow & Q_n(J) & \longrightarrow & J_{(n)} & \longrightarrow & \varrho_n(J) & \longrightarrow & 0 \\ & & \alpha_n \downarrow & & \alpha_n' \downarrow & & \varrho_{n-1}^n \downarrow & & \\ 0 & \longrightarrow & Q_{n-1}(J) & \longrightarrow & J_{(n-1)} & \longrightarrow & \varrho_{n-1}(J) & \longrightarrow & 0 \end{array} \tag{5}$$

in which $\alpha_n'$ is the canonical (surjective) map. In particular, if $\pi_J$ is bijective, then by passing to the inverse limit over $n$ of these sequences we obtain an exact sequence of $R$-modules

$$0 \to J \xrightarrow{\subseteq} \tilde{J} \to \varprojlim\nolimits^1_{\alpha_n} Q_n(J) \to 0.$$

We note next that the action on $Q_n(J)$ of an element $x$ of $R$ coincides both with the functorial map induced by multiplication by $x$ on the module $R/J$ and also with the functorial map induced by multiplication by $\varrho_n(x)$ on $R_n$, and hence that $Q_n(J)$ is annihilated (as an $R_n$-module) by elements of $\varrho_n(J)$. In particular, the action of $R$ on $\varprojlim^1_{\alpha_n} Q_n(J)$ factors through $R/\tilde{J}$. From the last displayed

exact sequence we may therefore deduce that $\tilde{J}^2 \subseteq J$, as required to prove the first assertion of (ii). Given this, the second assertion of (ii) then follows directly from the fact that, if $J$ is a prime ideal, then the inclusions $\tilde{J}^2 \subseteq J \subseteq \tilde{J}$ combine to imply $J = \tilde{J}$.

To prove (iii) we fix a subset $\{n_i\}_{i\in\mathbb{N}}$ of $\mathbb{N}$ such that each $n_i$ is prime-to-$p$ and divides $n_{i+1}$, and then consider the principal ideals $J(i) := R \cdot (\gamma^{n_i} - 1)$. Since $\gamma^{n_i} - 1$ is a divisor of $\gamma^{n_{i+1}} - 1$ in $R$, one has $J(i+1) \subseteq J(i)$. It is also easily checked that $\varrho_n(J(i)) = R_n \cdot (\gamma_n - 1)$ for each $n$ and $R^\ell \cdot J(i) = R^\ell \cdot (\gamma - 1)$ for each $\ell$ (so that (iii)(a) and (iii)(b) are valid). It therefore suffices to prove $J(i) \neq J(j)$ for $i \neq j$ and, for this, it is enough to show that the index of $J(i)$ in $R \cdot (\gamma - 1)$ is equal to $n_i$. To do this, we use the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R\cdot(\gamma-1) & \xrightarrow{\subset} & I(0) & \xrightarrow{\xi_0} & \mathbb{Z}_p/\mathbb{Z} & \longrightarrow & 0 \\ & & \subseteq\uparrow & & \| & & \uparrow{\scriptstyle \times n_i} & & \\ 0 & \longrightarrow & J(i) & \xrightarrow{\subset} & I(0) & \xrightarrow{\xi_{0,i}} & \mathbb{Z}_p/\mathbb{Z} & \longrightarrow & 0. \end{array} \tag{6}$$

Here the upper row is (2) with $n = 0$ and the lower row is the exact sequence constructed by the argument of [7, Prop. 2.1(ii)] after replacing $\gamma - 1$ by $\gamma^{n_i} - 1 = (\sum_{a=0}^{a=n_i-1} \gamma^a)(\gamma - 1)$. The commutativity of the first square is then clear and that of the second square is checked by explicitly comparing the respective connecting homomorphisms $\xi_0$ and $\xi_{0,i}$. Hence, since the kernel of multiplication by $n_i$ on $\mathbb{Z}_p/\mathbb{Z}$ is cyclic of order $n_i$, the Snake Lemma applies to the above diagram to imply that the index of $J(i)$ in $R \cdot (\gamma - 1)$ is $n_i$, as claimed. □

**Remark 3.3.** (i) For each $\ell \in \mathcal{P}$, a finitely generated ideal $X$ of $R^\ell$ is compact, and hence closed, so that $X = X^c$. In particular, since $R^p$ is Noetherian, all of its ideals are closed. If $\ell \neq p$, then Bandini and Longhi have shown that an ideal of $R^\ell$ is closed if and only if it is principal (see [3, Prop. 2.3]).
(ii) A finitely generated ideal of $R$ is pro-discrete (by Proposition 2.4(iii)), but not all pro-discrete ideals are finitely generated. For example, $I(0)$ is pro-discrete (by [7, Lem. 3.6]) but the surjectivity of the map $\xi_0$ in (2) implies $I(0)$ cannot be generated by any countable set of elements.
(iii) Proposition 3.2 (ii) and (iii) combine to imply the ideals $J(i)$ of $R$ are not prime, and this also follows from the fact the lower row in (6) implies $I(0)^2 \subseteq J(i)$ and $I(0) \not\subseteq J(i)$. Whilst these ideals show that even principal ideals of $R$ are not determined by their extensions to $R^\ell$ for all $\ell \in \mathcal{P}$, they are still distinguishable in these rings since, if $i < j$, then for any prime divisor $\ell$ of $n_j/n_i$ the images in $R^\ell$ of the completions $\varprojlim_n (J(i)/\ell^n)$ and $\varprojlim_n (J(j)/\ell^n)$ can be seen to differ. In particular, in relation to later arithmetic questions, it would be interesting to know if finitely-generated (or perhaps, more generally, pro-discrete) ideals $J$ of $R$ are determined by knowledge of both $\varrho_n(J)$ for all $n \in \mathbb{N}$ and of $\varprojlim_n (J/\ell^n)$ for all $\ell \in \mathcal{P}$.

We next show that an ideal of the classical Iwasawa algebra $R^p = \mathbb{Z}_p[\![\mathbb{Z}_p]\!]$ is uniquely determined by its restriction to $R$.

**Proposition 3.4.** *For every ideal $I$ of $R^p$, one has $R^p \cdot (I \cap R) = I$.*

*Proof.* We can fix $n \in \mathbb{N}$ and a subset $\{f_i\}_{1\leq i\leq n}$ of $I$ such that $I = \sum_{i=1}^{i=n} R^p \cdot f_i$. Then, assuming that $R^p \cdot ((R^p \cdot f_i) \cap R) = R^p \cdot f_i$ for every $i$, one has

$$I = \sum_{i=1}^{i=n} R^p \cdot f_i = \sum_{i=1}^{i=n} R^p \cdot ((R^p \cdot f_i) \cap R) \subseteq R^p \cdot \sum_{i=1}^{i=n} ((R^p \cdot f_i) \cap R) \subseteq R^p \cdot (I \cap R) \subseteq I$$

and hence $R^p \cdot (I \cap R) = I$. It is therefore enough to prove the stated equality in the case that $I = R^p \cdot f$ for some $f \in R^p$.

For each $i \in \mathbb{N}_0$, we write $\Phi_i(X)$ for the $p^i$-th cyclotomic polynomial in $\mathbb{Z}[X]$. We recall $\Phi_i(X)$ is irreducible as an element of the polynomial ring $\mathbb{Z}_p[X]$ and further that, for each $n \in \mathbb{N}_0$, one

has $X^{p^n}-1=\prod_{i=0}^{i=n}\Phi_i(X)$ in $\mathbb{Z}[X]$. Now if, for any $m\in\mathbb{N}_0$, the quotient of $R^p$ by the ideal $R^p\cdot f+R^p\cdot\varpi_m$ is infinite, then $f$ must be divisible in $R^p$ by $\Phi_i(\gamma)$ for some $i$ with $0\leq i\leq m$. Hence, after dividing $f$ by a finite (possibly empty) product $g$ of elements of the form $\Phi_i(\gamma)$, we obtain an ideal $I':=R^p\cdot(fg^{-1})$ with the property that $\varrho_n(I')$ has finite index in $R_n^p$ for every $n\in\mathbb{N}_0$. In addition, as $g$ belongs to $R$ and is a non-zero divisor in $R^p$, one has $R^p\cdot((R^p\cdot f)\cap R)=R^p\cdot f$ if $R^p\cdot((R^p\cdot(fg^{-1}))\cap R)=R^p\cdot fg^{-1}$. It follows that, after replacing $f$ by $fg^{-1}$ if necessary, we can in the sequel assume that $R_n^p/\varrho_n(I)$ is finite for every $n\in\mathbb{N}_0$.

In this case, for each $n$ one has $R_n+\varrho_n(I)=R_n^p$ and hence also

$$R_n^p/\varrho_n(I)=R_n/(R_n\cap\varrho_n(I))=\mathbb{Z}_p\otimes_{\mathbb{Z}}[R_n/(R_n\cap\varrho_n(I))]=R_n^p/[R_n^p(R_n\cap\varrho_n(I))],$$

so that $R_n^p\cdot(R_n\cap\varrho_n(I))=\varrho_n(I)$. This in turn gives an exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R_n\cap\varrho_n(I) & \xrightarrow{x\to(x,x)} & R_n\oplus\varrho_n(I) & \xrightarrow{(x,y)\mapsto x-y} & R_n^p & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle\alpha_n} & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & R_{n-1}\cap\varrho_{n-1}(I) & \xrightarrow{x\to(x,x)} & R_{n-1}\oplus\varrho_{n-1}(I) & \xrightarrow{(x,y)\mapsto x-y} & R_{n-1}^p & \longrightarrow & 0 \end{array} \tag{7}$$

in which every vertical map is induced by $\varrho_{n-1}^n$. In addition, since $\varrho_n(I)$ is a free $R_n^p$-module of rank one (under the current hypotheses), the natural map $(\varrho_n(I))_{(n-1)}\to\varrho_{n-1}(I)$ is bijective. Using this last fact, an easy diagram chase implies that the map $\alpha_n$ is surjective.

Now, by using (3) (and Remark 3.3(i)) with $X=I$, the inverse limit over $n$ of the above diagrams gives an exact sequence

$$0\to\varprojlim_{\varrho_{n-1}^n}(R_n\cap\varrho_n(I))\xrightarrow{x\to(x,x)}R\oplus I\xrightarrow{(x,y)\mapsto x-y}R^p,$$

and thereby an equality $R\cap I=\varprojlim_{\varrho_{n-1}^n}(R_n\cap\varrho_n(I))$. For each $m$, it then follows that

$$\varrho_m\big(R^p\cdot(R\cap I)\big)=R_m^p\cdot\varrho_m\big(\varprojlim_{\varrho_{n-1}^n}(R_n\cap\varrho_n(I))\big)=R_m^p\cdot(R_m\cap\varrho_m(I))=\varrho_m(I),$$

where the second equality follows from the surjectivity of each map $\alpha_n$ in (7) and the third was observed just before (7).

Hence, since (3) (and Remark 3.3(i)) implies that both

$$R^p\cdot(R\cap I)=\varprojlim_m\varrho_m\big(R^p\cdot(R\cap I)\big)\quad\text{and}\quad I=\varprojlim_m\varrho_m(I),$$

it therefore follows that $R^p\cdot(I\cap R)=I$, as claimed. □

3.2. **Observations on** Spec$(R)$**.** In this subsection, we make several observatons about prime ideals of $R$. As a first step, we determine all relevant Krull dimensions.

In the sequel, for $m\in\mathbb{N}_0$ we shall write $\mathcal{O}_m$ for the ring of algebraic integers of $\mathbb{Q}(e^{2\pi i/p^m})$. For $n\in\mathbb{N}$, we also set $[n]:=\{m\in\mathbb{N}:1\leq m\leq n\}$.

**Proposition 3.5.**

(i) $\dim(R^p)=2$ *and* $\dim(R/p)=1$.
(ii) $\dim(R^\ell)=\infty$ *and* $\dim(R/\ell)=0$ *for every* $\ell\in\mathcal{P}\setminus\{p\}$.
(iii) $\dim(R)=\infty$.

*Proof.* For $\ell\in\mathcal{P}$ one has $R^\ell=\mathbb{Z}_\ell[\![\mathbb{Z}_p]\!]$ and $R/\ell=\mathbb{F}_\ell[\![\mathbb{Z}_p]\!]$. Claim (i) is therefore a result of classical Iwasawa theory, but to prove (ii) and (iii) we shall use results of Maroscia and of Gilmer and Heinzer. To do this, for each $n\geq m$ we write $\tau_m^n$ for the ring homomorphism $R_n\to\mathcal{O}_m$ induced by sending $\gamma_n$ to $e^{2\pi i/p^m}$. Then, for each $n\in\mathbb{N}$, there exists an induced isomorphism of $\mathbb{Z}[1/p]$-algebras

$$R_n[1/p]\cong\prod_{m\in\{0\}\cup[n]}\mathcal{O}_m[1/p],\quad x\mapsto(\tau_m^n(x))_m. \tag{8}$$

In particular, if we fix $\ell \in \mathcal{P} \setminus \{p\}$, then these decompositions combine to induce isomorphisms of $\mathbb{Z}_\ell$-algebras

$$R^\ell \cong \prod_{m \in \mathbb{N}_0} (\mathbb{Z}_\ell \otimes_{\mathbb{Z}} \mathcal{O}_m) \quad \text{and} \quad R^\ell/\ell \cong R/\ell \cong \prod_{m \in \mathbb{N}_0} \big(\mathcal{O}_m/\ell\big). \tag{9}$$

Now, as $\dim(\mathbb{Z}_\ell \otimes_{\mathbb{Z}} \mathcal{O}_m) = 1$ for all $m$, the first isomorphism combines with a general result [15, Th. 3.3] of Gilmer and Heinzer to imply $\dim(R^\ell) = \infty$. On the other hand, since the Chinese Remainder Theorem implies each ring $\mathcal{O}_m/\ell$ is a finite direct product of fields (as $\ell \neq p$), the second isomorphism combines with a result [23, Prop. 2.6] of Maroscia to imply $\dim(R/\ell) = 0$. This proves (ii).

To prove (iii), we write $R(\ell)$ for the direct product ring $\prod_{m \in \mathbb{N}_0} \big(\mathcal{O}_m/\ell^m\big)$. Then, since the image of $\ell$ in each ring $\mathcal{O}_m/\ell^m$ is nilpotent, but the image of $\ell$ under the diagonal map $R \to R(\ell)$ is not nilpotent, the result of [15, Th. 3.4] implies that $\dim(R(\ell)) = \infty$. To prove (iii), it is therefore enough to construct a surjective ring homomorphism from $R$ to $R(\ell)$. To do this, for each $n \in \mathbb{N}$ we write $R(\ell; n)$ for the ring $\prod_{m \in [n]} \big(\mathcal{O}_m/\ell^m\big)$ and consider the composite ring homomorphism

$$\pi_n : R_n \subset R_n[1/p] \cong \prod_{m \in \{0\} \cup [n]} \mathcal{O}_m[1/p] \to \prod_{m \in [n]} \big(\mathcal{O}_m[1/p]\big)/\ell^m = R(\ell; n), \tag{10}$$

where the isomorphism is (8), the second arrow is the natural projection and the equality is valid since $\ell \neq p$. For each $n \in \mathbb{N}$, one then has an exact commutative diagram of $R$-modules

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \ker(\pi_n) & \xrightarrow{\subseteq} & R_n & \xrightarrow{\pi_n} & R(\ell; n) & \longrightarrow & 0 \\ & & \big\downarrow{\scriptstyle \tilde{\varrho}^n_{n-1}} & & \big\downarrow{\scriptstyle \varrho^n_{n-1}} & & \big\downarrow{\scriptstyle \rho^n_{n-1}} & & \\ 0 & \longrightarrow & \ker(\pi_{n-1}) & \xrightarrow{\subseteq} & R_{n-1} & \xrightarrow{\pi_{n-1}} & R(\ell, n-1) & \longrightarrow & 0 \end{array} \tag{11}$$

in which $\varrho^n_{n-1}$ is the restriction of $\varrho^n_{n-1}$ and $\rho^n_{n-1}$ the natural projection map. Note here that $\pi_n$ (and similarly $\pi_{n-1}$) is surjective since $R_n[1/p]/R_n$ is a $p$-torsion group and $p$ acts invertibly on $R(\ell; n)$ (as $\ell \neq p$). It is then enough to show the maps $\tilde{\varrho}^n_{n-1}$ are surjective since, if true, upon passing to the inverse limit over $n$ of these diagrams this would imply the induced composite map

$$R = \varprojlim_{\varrho^n_{n-1}} R_n \xrightarrow{(\pi_n)_n} \varprojlim_{\rho^n_{n-1}} R(\ell; n) = R(\ell)$$

is a surjective ring homomorphism, as required. Now, as $\varrho^n_{n-1}$ is surjective, the Snake Lemma implies that $\tilde{\varrho}^n_{n-1}$ is surjective if $\pi_n(\ker(\varrho^n_{n-1})) = \ker(\rho^n_{n-1})$. To check this we write $e_1$ for the idempotent $(1/p)\sum_{i \in [p]} \gamma_n^{p^{n-1}i}$ of $R_n[1/p]$. Then $\tau^n_n((1-e_1)) = 1 \in \mathcal{O}_n$ and $p(1-e_1)$ belongs to $R_n \cdot (\gamma_n^{p^{n-1}} - 1) = \ker(\varrho^n_{n-1})$. Hence, since $\ker(\rho^n_{n-1}) = \mathcal{O}_n/\ell^n$ and $\ell \neq p$, it follows that $\pi_n(\ker(\varrho^n_{n-1})) = \ker(\rho^n_{n-1})$, as required to complete the proof. □

**Remark 3.6.** The proof of Proposition 3.5(ii) shows that, contrary to the result of Proposition 3.4, for $\ell \in \mathcal{P} \setminus \{p\}$ there are infinitely many prime ideals of $R^\ell$ whose intersection with $R$ is (0).

Proposition 3.5 implies that it is difficult to obtain a fully explicit description of $\mathrm{Spec}(R)$. Nevertheless, it is still possible to make several useful observations about prime ideals.

**Lemma 3.7.** *The following claims are valid for any prime ideal $J$ of $R$.*

(i) *Assume that $\varpi_m \in J$ for some $m \in \mathbb{N}_0$. Then $J$ is the full pre-image under $\varrho_m$ of a prime ideal of $R_m$ and equal to both $\varprojlim_{\varrho^n_{n-1}} \varrho_n(J)$ and $\bigcap_{\ell \in \mathcal{P}} \big(R \cap (R^\ell \cdot J)^c\big)$. In addition, $J$ is finitely-generated if and only if it is a maximal ideal that contains a rational prime.*

(ii) *Assume that $J$ is disjoint from $\{\varpi_n\}_{n \in \mathbb{N}_0}$. Then $J$ is finitely-generated if and only if it is equal to $R \cap (R^p \cdot f_J(\gamma))$ for an irreducible distinguished polynomial $f_J(X)$ of $\mathbb{Z}_p[\![X]\!]$ that is (uniquely determined by $J$ and) not equal to $\Phi_i(X)$ for any $i \in \mathbb{N}_0$.*

*Proof.* We assume first that $\varpi_m \in J$. Then, as the exact sequence (2) implies $I(m)^2 \subseteq R \cdot \varpi_m$, one has $I(m)^2 \subseteq J$. Hence, since $J$ is a prime ideal, one has $I(m) \subseteq J$ and so $\varrho_m(J) \cong J/I(m)$ is a prime ideal of $R_m \cong R/I(m)$ and $J$ is its full pre-image under $\varrho_m$. It also follows that $J = J + I(m')$ for all $m' \geq m$, and hence that

$$J = \bigcap\nolimits_{n \in \mathbb{N}_0} (J + I(n)) = \varprojlim\nolimits_{\varrho^n_{n-1}} \varrho_n(J) = \bigcap\nolimits_{\ell \in \mathcal{P}} \big(R \cap (R^\ell \cdot J)^c\big),$$

where the second and third equalities follow from (the proof of) Proposition 3.2(i). To complete the proof of (i), it is now enough to recall that [7, Th. 4.2(iii)] proves that any such prime ideal $J$ is finitely-generated if and only if it is a maximal ideal that contains a rational prime.

To prove (ii), we assume $J \cap \{\varpi_n\}_{n \in \mathbb{N}_0} = \emptyset$. Then, by [7, Th. 4.2(iv)], there exists a distinguished polynomial $f_J(X)$ of $\mathbb{Z}_p[X]$ of the stated kind such that $J = (R^p \cdot f_J(\gamma)) \cap R$. In addition, since Proposition 3.4 implies that

$$R^p \cdot f_J(\gamma) = R^p \cdot ((R^p \cdot f_J(\gamma)) \cap R) = R^p \cdot J,$$

the polynomial $f_J$ is uniquely determined by $J$. □

To obtain a fully explicit description of $\mathrm{Spec}(R)$, we are therefore reduced to characterising the non-finitely-generated prime ideals of $R$ that are disjoint from $\{\varpi_n\}_{n \in \mathbb{N}_0}$ and, in many respects, these remain mysterious. However, in the case of maximal ideals, some further analysis is possible. To state a result in this direction, we note that the maximum spectrum $\mathrm{MaxSpec}(R)$ of $R$ decomposes as a disjoint union, parametrised by prime ideals $\mathfrak{a}$ of $\mathbb{Z}$, of the subspaces $\mathrm{MaxSpec}_{\mathfrak{a}}(R)$ comprising maximal ideals whose residue characteristic generates $\mathfrak{a}$. The following result (some of which is well-known) describes these individual subspaces and, in particular, obtains a complete description of the set of pro-discrete maximal ideals, thereby extending the classification of finitely-generated prime ideals of $R$ that is obtained in [7, Th. 4.2].

**Proposition 3.8.** *The following claims are valid.*

(i) $\mathrm{MaxSpec}_{(p)}(R)$ *is a singleton comprising the ideal generated by* $\{p, \varpi_0\}$ *which is pro-discrete and has residue field* $\mathbb{F}_p$.

(ii) $\mathrm{MaxSpec}_{(\ell)}(R)$ *for* $\ell \in \mathcal{P} \setminus \{p\}$ *is homeomorphic to the Stone-Čech compactification of* $\mathbb{N}$ *and is compact, Hausdorff and non-Noetherian. In addition, the following conditions on an ideal* $\mathfrak{m}$ *in* $\mathrm{MaxSpec}_{(\ell)}(R)$ *are equivalent (and imply its residue field is finite):* $\mathfrak{m}$ *is pro-discrete;* $\mathfrak{m}$ *is finitely-generated;* $\mathfrak{m}$ *is principal;* $\mathfrak{m} \cap \{\varpi_n\}_{n \in \mathbb{N}_0} \neq \emptyset$*;* $(R^\ell \cdot \mathfrak{m})^c \neq R^\ell$.

(iii) *If* $\mathfrak{m}$ *belongs to* $\mathrm{MaxSpec}_{(0)}(R)$*, then* $\mathfrak{m}$ *is not pro-discrete and* $(R^\ell \cdot \mathfrak{m})^c = R^\ell$ *for all* $\ell \in \mathcal{P}$*. In addition, the residue fields of infinitely many ideals in* $\mathrm{MaxSpec}_{(0)}(R)$ *are of uncountable dimension over* $\mathbb{Q}$.

*Proof.* The unique maximal ideal of $R/pR \cong \mathbb{F}_p[\![\mathbb{Z}_p]\!]$ is generated by the image of $\varpi_0$ and has residue field $\mathbb{F}_p$. The ideal generated by $\{p, \varpi_0\}$ is therefore the unique element of $\mathrm{MaxSpec}_{(p)}(R)$ and is automatically pro-discrete (by Proposition 2.4(iii)) since it is finitely-generated. This proves (i).

To prove (ii) we fix $\ell \in \mathcal{P} \setminus \{p\}$ and note the second isomorphism in (9) implies points in $\mathrm{MaxSpec}_{(\ell)}(R)$ correspond to maximal ideals of a ring $\prod_{m \in \mathbb{N}_0} \big(\mathcal{O}_m/\ell\big)$ that is isomorphic to an infinite direct product of (finite) fields. Such maximal ideals therefore correspond bijectively with the set of ultrafilters on $\mathbb{N}$ and, except perhaps for the assertion of non-Noetherianity, the first sentence of (ii) is a well-known consequence of this fact (see, for example, [29] and the references contained therein). In addition, every ideal of $R/\ell$ is radical (by [29, Th. 3.1(iii)]) and every prime ideal of $R/\ell$ is maximal (by Proposition 3.5(ii)) and so the non-Noetherianity of the space $\mathrm{MaxSpec}_{(\ell)}(R)$ follows directly from that of the ring $R/\ell$.

To prove the remainder of (ii) we note first that [29, Th. 3.1(i)] implies a prime ideal $\mathfrak{m}$ of $R/\ell$ is finitely-generated if and only if it is principal (and thereby corresponds to a principal ultrafilter), in which case $R/\mathfrak{m}$ is finite and so $\mathfrak{m}$ contains $\varpi_n$ for all sufficiently large $n$. Conversely, if $\mathfrak{m}$ contains both $\ell$ and $\varpi_n$ for some $n$, then Lemma 3.7(i) implies $\mathfrak{m}$ is the full pre-inage of a maximal ideal of $R_n/\ell$ and so is finitely-generated since [7, Prop. 2.1(iv)] implies that the kernel of the projection $R/\ell \to R_n/\ell$ is generated by the image of $\varpi_n$. Next we show that $\mathfrak{m}$ is pro-discrete if and only if it is finitely-generated. Taking account of Proposition 2.4(iii), it is enough for this to assume $\mathfrak{m}$ is pro-discrete, and prove $\overline{\mathfrak{m}} := \mathfrak{m}/(\ell)$ is finitely-generated. To do this, we note that the tautological exact sequence $0 \to (\ell) \xrightarrow{\iota} \mathfrak{m} \to \overline{\mathfrak{m}} \to 0$ induces, upon passing to coinvariants, a family of exact commutative diagrams of $R_n$-modules

$$\begin{array}{ccccccccc} 0 & \longrightarrow & (\ell)_{(n)} & \xrightarrow{\iota_{(n)}} & \mathfrak{m}_{(n)} & \longrightarrow & \overline{\mathfrak{m}}_{(n)} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & (\ell)_{(n-1)} & \xrightarrow{\iota_{(n-1)}} & \mathfrak{m}_{(n-1)} & \longrightarrow & \overline{\mathfrak{m}}_{(n-1)} & \longrightarrow & 0. \end{array}$$

Here all vertical maps are the natural projections and injectivity of each $\iota_{(n)}$ follows from the fact $(\ell)_{(n)} \cong R_{(n)} \cong R_n$ is torsion-free whilst $\overline{\mathfrak{m}}$, and hence also $\mathrm{Tor}_1^R(R_n, \overline{\mathfrak{m}})$, is annihilated by $\ell$. Now, since the first vertical maps in the above diagrams are surjective, exactness is preserved upon passing to the inverse limit, thus giving (as both $(\ell) \cong R$ and $\mathfrak{m}$ are pro-discrete) an exact sequence $0 \to (\ell) \xrightarrow{\iota} \mathfrak{m} \to \varprojlim_n \overline{\mathfrak{m}}_{(n)} \to 0$. Upon comparing this sequence with the original, one deduces that the natural map $\overline{\mathfrak{m}} \to \varprojlim_n \overline{\mathfrak{m}}_{(n)}$ is bijective. Since $\overline{\mathfrak{m}}_{(n)}$ identifies with the image of $\overline{\mathfrak{m}}$ under the projection $R/\ell \to \prod_{m\in\{0\}\cup[n]}(\mathcal{O}_m/\ell)$ induced by (8), one therefore has $\overline{\mathfrak{m}} = \prod_{m\in\mathbb{N}_0} \mathrm{im}(\kappa_m)$, with each $\kappa_m$ the map $\overline{\mathfrak{m}} \subset R/\ell \to \mathcal{O}_m/\ell$ induced by $\tau_m^n$. The fact $\overline{\mathfrak{m}}$ is a maximal ideal of $R/\ell$ now implies that $\mathrm{im}(\kappa_m) = \mathcal{O}_m/\ell$ for all but one value of $m$, and it is then clear that $\prod_{m\in\mathbb{N}_0} \mathrm{im}(\kappa_m)$ is finitely-generated over $R/\ell$, as required. To complete the proof of (ii) it is now enough to prove that $\mathfrak{m}\cap\{\varpi_n\}_{n\in\mathbb{N}_0} = \emptyset \iff (R^\ell\cdot\mathfrak{m})^c = R^\ell$. The forward implication here is valid since if $\mathfrak{m}\cap\{\varpi_n\}_{n\in\mathbb{N}_0} = \emptyset$, then the maximality of $\mathfrak{m}$ implies $\mathfrak{m}+I(n) = R$ for all $n$ and hence that $(R^\ell\cdot\mathfrak{m})^c = R^\ell$ as a consequence of Proposition 3.2(i). On the other hand, the converse implication follows from the fact that, for any $n$, every ideal of the ring $R_n/\ell$ is closed for the $\ell$-adic topology.

To prove (iii) we fix $\mathfrak{m}$ in $\mathrm{MaxSpec}_{(0)}(R)$. Then, since no ring $R_n$ can have a non-zero uniquely divisible quotient, Lemma 3.7(i) implies $\mathfrak{m} \cap \{\varpi_n\}_{n\in\mathbb{N}_0} = \emptyset$ and then, just as above, this implies $\mathfrak{m}+I(n) = R$ for every $n$. The latter fact has two consequences. Firstly, it combines with Proposition 3.2(i) to imply that $(R^\ell \cdot \mathfrak{m})^c = R^\ell$ for every $\ell \in \mathcal{P}$, as claimed. Secondly, it implies that $\varrho_n(\mathfrak{m}) = R_n$ for every $n$, and hence that the argument in the paragraph following diagram (5) (with $J = \mathfrak{m}$) implies each module $Q_n(\mathfrak{m})$ vanishes. From the exactness of the upper row of (5), we can therefore deduce that $\mathfrak{m}_{(n)}$ is a free $R_n$-module of rank 1. In particular, if $\mathfrak{m}$ is equal to the limit of a pro-discrete system $(\mathfrak{m}_n)_n$, then the induced surjective maps $\mathfrak{m}_{(n)} \to \mathfrak{m}_n$ would imply that $\mu_R(\mathfrak{m}_n) \leq 1$ for every $n$ and hence, by Proposition 2.3(i), that $\mathfrak{m}$ is a finitely-generated $R$-module. From the first assertion of Proposition 3.2(ii), it would then follow that $\mathfrak{m}$ contains $R = \left(\bigcap_{n\in\mathbb{N}_0}(\mathfrak{m} + I(n))\right)^2$ and, from this contradiction, we deduce that $\mathfrak{m}$ cannot be pro-discrete, as claimed.

To complete the proof of (iii), we shall now show that, for each integer $a > 1$, there exists a maximal ideal $\mathfrak{m}_a$ of $R$ (with $\mathfrak{m}_a \neq \mathfrak{m}_{a'}$ if $a \neq a'$) that contains $\gamma - a$ and is such that $R/\mathfrak{m}_a$ has characteristic zero and uncountably infinite dimension over $\mathbb{Q}$. To do this, we use the ideal $X(a) := R \cdot (\gamma - a)$ and, for $n \in \mathbb{N}_0$ and $\ell \in \mathcal{P}$, set $X(a)_n := \varrho_n(X(a))$, $X(a)^\ell = R^\ell \cdot X(a)$ and $X(a)^\ell_n = \varrho_n(R^\ell \cdot X(a))$. Then, since $R_n/X(a)_n$ is the cokernel of the endomorphism of $R_n$ given by multiplication by $\gamma_n - a$, a simple determinant computation shows $R_n/X(a)_n$ to be finite and of order $a^{p^n} - 1$. In a similar way, one checks that $\mathsf{Tor}_1^R(R_n, R/X(a))$ vanishes and hence that the

natural map $X(a)_{(n)} \to X(a)_n$ is bijective. These facts then combine to give identifications

$$R/X(a) = \varprojlim_n (R_n/X(a)_n) = \varprojlim_n \big(\bigoplus_{\ell\in\mathcal{P}} (R_n^\ell/X(a)_n^\ell)\big) = \prod_{\ell\in\mathcal{P}} \varprojlim_n \big(R_n^\ell/X(a)_n^\ell\big) = \prod_{\ell\in\mathcal{P}(a)} N^\ell(a).$$

Here all inverse limits are with respect to the maps induced by the projections $\varrho_{n-1}^n$. In particular, the first equality is obtained by passing to the (inverse) limit over the tautological exact sequences $0 \to X(a)_n \to R_n \to R_n/X(a)_n \to 0$ and noting $X(a) = \varprojlim_n X(a)_{(n)} = \varprojlim_{\varrho_{n-1}^n} X(a)_n$ and $\varrho_{n-1}^n(X(a)_n) = X(a)_{n-1}$ for all $n \in \mathbb{N}$. In addition, we have set

$$\mathcal{P}(a) := \{\ell \in \mathcal{P} : \exists\, n \in \mathbb{N}_0,\, \ell \mid (a^{p^n} - 1)\}$$

and $N^\ell(a) := R^\ell/X(a)^\ell$ for $\ell \in \mathcal{P}$. Hence, since (3) implies that $X(a)^\ell = \varprojlim_{\varrho_{n-1}^n} X(a)_n^\ell$, and also $\varrho_{n-1}^n(X(a)_n^\ell) = X(a)_{n-1}^\ell$ for all $n \in \mathbb{N}$, one has

$$N^\ell(a) = \varprojlim_n \big(R_n^\ell/X(a)_n^\ell\big) = \varprojlim_n \big(\mathbb{Z}_\ell \otimes_{\mathbb{Z}} (R_n/X(a)_n)\big)$$

and therefore $N^\ell(a) \neq (0) \iff \ell \in \mathcal{P}(a)$.

Now, as a special case of Zsigmondy's Theorem [36], the set $\mathcal{P}(a)$ is infinite. It follows that $\bigoplus_{\ell\in\mathcal{P}(a)} N^\ell(a)$ is a proper ideal of $R/X(a)$ and so we can choose a maximal ideal $\overline{\mathfrak{m}}_a$ of $R/X(a)$ containing $\bigoplus_{\ell\in\mathcal{P}(a)} N^\ell(a)$, and then write $\mathfrak{m}_a$ for the maximal ideal of $R$ given by the full pre-image of $\overline{\mathfrak{m}}_a$. To show that the field $R/\mathfrak{m}_a$ has characteristic 0, we argue by contradiction and so assume it is annihilated by multiplication by $\ell_0$ for some $\ell_0 \in \mathcal{P}$. Then, since multiplication by $\ell_0$ acts invertibly on $\prod_{\ell\in\mathcal{P}(a)\setminus\{\ell_0\}} N^\ell(a)$, one has $\prod_{\ell\in\mathcal{P}(a)\setminus\{\ell_0\}} N^\ell(a) \subseteq \overline{\mathfrak{m}}_a$ and hence also

$$R/X(a) = N^{\ell_0}(a) \oplus \prod_{\ell\in\mathcal{P}(a)\setminus\{\ell_0\}} N^\ell(a) \subseteq \overline{\mathfrak{m}}_a.$$

This contradicts the fact $\overline{\mathfrak{m}}_a$ is a proper ideal of $R/X(a)$ and so $R/\mathfrak{m}_a$ must have characteristic 0, as claimed.

To prove $\dim_{\mathbb{Q}}(R/\mathfrak{m}_a)$ is uncountably infinite, it is enough to show $R/\mathfrak{m}_a$ is uncountable, and to do this we label the elements of $\mathcal{P}(a)$ in ascending order as $\{p_i\}_{i\in\mathbb{N}}$. It is then enough to note that the function $\{0,1\}^{\mathbb{N}} \to R/\mathfrak{m}_a$ sending each $f$ to the image of $(\sum_{n=1}^{n=m(k)} f(n)2^{n-1})_k \in \prod_k R^{p_k}$ in $R/\mathfrak{m}_a = \big(\prod_k (R^{p_k}/X(a)^{p_k})\big)/\overline{\mathfrak{m}}_a$ is injective, where $m(k) \in \mathbb{N}$ is defined via the inequalities $2^{m(k)} - 1 < p_k \leq 2^{m(k)+1} - 1$. (This argument is motivated by the StackExchange discussion [26].)

Finally, we note that if $a > a'$ and $\mathfrak{m}_a = \mathfrak{m}_{a'}$, then $\mathfrak{m}_a$ contains $a - a' = (\gamma - a) - (\gamma - a')$. This would imply the field $R/\mathfrak{m}_a$ is annihilated by multiplication by $a - a'$ and this contradicts the fact it has characteristic 0. □

**Remark 3.9.** (i) The proof of Proposition 3.8(iii) shows that, for each $\mathfrak{m}$ in $\mathrm{MaxSpec}_{(0)}(R)$, one has $\varrho_n(\mathfrak{m}) = R_n$ for all $n \in \mathbb{N}_0$ and, since $R/\mathfrak{m}$ is uniquely divisible, also $\varprojlim_n (\mathfrak{m}/\ell^n) = R^\ell$ for all $\ell \in \mathcal{P}$. It follows that the question in Remark 3.3(iii) has a negative answer if one omits the hypothesis that $J$ is pro-discrete.

(ii) Set $I := I(0)$ and write $J(\subset I)$ for the $R$-ideal generated by $\{p\varpi_0\} \cup \{\varpi_n - p\varpi_{n+1}\}_{n\in\mathbb{N}_0}$ and $\pi_J$ for the projection $R \to R/J$. Then, since any prime ideal of $R$ containing $J$ contains either $p$ or $\varpi_0$, Lemma 3.7(i) implies $\mathrm{Spec}(R/J) = \{\pi_J(I)\} \cup \{\pi_J(I + R\cdot\ell)\}_{\ell\in\mathcal{P}}$ and hence $\dim(R/J) = 1$. In addition, the upper row of (6) combines with the fact $\pi_J(\varpi_n) = p \cdot \pi_J(\varpi_{n+1})$ has order $p^{n+1}$ to imply that $R/J$ has a subgroup isomorphic to $\mathbb{Q}/\mathbb{Z}$, thereby complementing the uniquely divisible quotients of $R$ arising from Proposition 3.8(iii).

## 4. Module theory

In this section, we extend, or refine, several general results about pro-discrete modules from [7]. Throughout the section, we fix a finite abelian group $H$ and set

$$\mathbb{A}_n := R_n[H] \quad \text{for} \quad n \in \mathbb{N}_0 \quad \text{and} \quad \mathbb{A} := R[H] = \varprojlim_{\varrho^n_{n-1}} \mathbb{A}_n.$$

4.1. **Projective modules.** We first establish a criterion for a pro-discrete $\mathbb{A}$-module to be projective. Before stating it, we recall that, in [7], a prime $p$ is said-to-be 'exceptional' if it is irregular, validates Vandiver's Conjecture and is such that the $p$-adic $\lambda$-invariant of every (odd) isotypic factor of the ideal class group of $\mathbb{Q}(e^{2\pi i/p})$ is at least $p-1$. We further recall that heuristic arguments suggest there should be only finitely many exceptional $p$ [21, Chap. 10, App.] and that, in any case, results of Hart et al [19] imply such a prime must be greater than $2^{31}$.

**Proposition 4.1.** *Let $M = \varprojlim_n M_n$ be a finitely-generated pro-discrete $\mathbb{A}$-module. Then, unless $p$ is exceptional, $M$ is projective if and only if, for every $n$, the $\mathbb{A}_n$-module $M_n$ is projective.*

*Proof.* It is enough to show that, if $p$ is not exceptional, then $M$ is a projective $\mathbb{A}$-module if every $M_n$ is a projective $\mathbb{A}_n$-module.

We first aim to prove that, for each $\ell$, the $\mathbb{A}/\ell$-module $M/\ell$ is free. To do this we note that, since each $M_n$ is torsion-free, one has $M/\ell = \varprojlim_n (M_n/\ell)$ and we further recall that, by Swan's Theorem (cf. [13, Th. 32.11]), each $\mathbb{A}_n$-module $M_n$ is both locally-free and of rank independent of $n$ since $M_{n-1}$ is isomorphic to $(M_n)_{(n-1)}$.

To deal with the case $\ell = p$, we note the final assertion of Proposition 2.4(i) implies that the $\mathbb{F}_p[H]$-module $(M/p)_0 := \mathbb{F}_p[H] \otimes_{(\mathbb{A}/p)} (M/p)$ is equal to $(\mathbb{Z}_p[H] \otimes_{\mathbb{A}} M)/p \cong M_0/p$. In particular, since $\mathbb{Z}_p \otimes_{\mathbb{Z}} M_0$ is a free $\mathbb{Z}_p[H]$-module, the $\mathbb{F}_p[H]$-module $M_0/p = (\mathbb{Z}_p \otimes_{\mathbb{Z}} M_0)/p$ is free, and we write $t$ for its rank. By applying Nakayama's Lemma for the Noetherian ring $\mathbb{A}/p$, we can thus deduce that $M/p$ is generated as a $\mathbb{A}$-module by any set $\mathfrak{b} := \{b_i\}_{1 \le i \le t}$ that maps to a basis of $(M/p)_0$. It follows that $\mathfrak{b}' = \bigcup_{h \in H} \{h(b_i)\}_{1 \le i \le t}$ is a minimal set of generators of $M/p$ over the principal ideal domain $R/p$. In addition, since the assumption $p$ is not exceptional combines with [7, Th. 3.8(iii)(a)] to imply $M$ is a projective $R$-module, its quotient $M/p$ is a projective, and hence free, module over $R/p$ and so $\mathfrak{b}'$ must be a basis. It follows that $M/p$ is a free $\mathbb{A}/p$-module (with basis $\mathfrak{b}$), as claimed.

We next assume $\ell \neq p$. In this case, the second isomorphism in (9) induces algebra decompositions $\mathbb{A}/\ell \cong \prod_{m \in \mathbb{N}_0} (\mathcal{O}_m[H]/\ell)$ and $\mathbb{A}_n/\ell \cong \prod_{m=0}^{m=n} (\mathcal{O}_m[H]/\ell)$ for each $n \in \mathbb{N}_0$. In particular, since each $\mathbb{A}_n/\ell$-module $M_n/\ell = (\mathbb{Z}_\ell \otimes_{\mathbb{Z}} M_n)/\ell$ is free, and of rank that is independent of $n$, it is straightforward to check directly that $M/\ell$ is a free $\mathbb{A}/\ell$-module, as required.

At this point, since $M$ is finitely generated over $\mathbb{A}$, we can fix an exact sequence of $\mathbb{A}$-modules $0 \to K \to F \to M \to 0$ in which $F$ is free of finite rank. We set $X := \mathrm{Hom}_R(M, K)$ and note that, since $M$ is a projective $R$-module (as observed above), the induced sequence

$$0 \to X \to \mathrm{Hom}_R(M, F) \to \mathrm{Hom}_R(M, M) \to 0 \tag{12}$$

is exact. In addition, as $K$ is torsion-free, for $\ell \in \mathcal{P}$ the short exact sequence $0 \to K \xrightarrow{\ell} K \to K/\ell \to 0$ induces an isomorphism of $\mathbb{A}$-modules

$$X/\ell \cong \mathrm{Hom}_R(M, K/\ell) \cong \mathrm{Hom}_{R/\ell}(M/\ell, K/\ell).$$

Hence, since $M/\ell$ is a free $\mathbb{A}/\ell$-module, and thus induced as an $H$-module, the $H$-module $X/\ell$ is also induced (cf. the proof of [2, §9, Cor.]) and thus cohomologically-trivial. In particular, by considering the exact sequence of Tate cohomology of the exact sequence $0 \to X \xrightarrow{\ell} X \to X/\ell \to 0$ we can now deduce that, for each integer $q$ and subgroup $J$ of the Sylow $\ell$-subgroup $H(\ell)$ of $H$, the map $\hat{H}^q(J, X) \xrightarrow{\ell} \hat{H}^q(J, X)$ is bijective. Since each $\hat{H}^q(J, X)$ is an $\ell$-group, it therefore follows that $X$

is a cohomologically-trivial $H(\ell)$-module. Since this is true for all $\ell$, it then follows that $X$ is a cohomologically-trivial $H$-module (cf. [2, §9, Th. 9]). In particular, by taking $H$-invariants of the exact sequence (12), we can now deduce the map $\mathrm{Hom}_{\mathbb{A}}(M, F) \to \mathrm{Hom}_{\mathbb{A}}(M, M)$ is surjective and hence that the identity map of $M$ extends to a map $M \to F$ of $\mathbb{A}$-modules. It follows that $M$ is a direct summand of the (free) $\mathbb{A}$-module $F$ and so is projective, as required. □

**Remark 4.2.** Whilst the argument used to prove Proposition 4.1 relies on the hypothesis that $p$ is not exceptional (via an application of [7, Th. 3.8(iii)]), we do not know if its statement becomes false upon removal of this hypothesis.

4.2. **Torsion modules.** We next establish criteria for a pro-discrete module to be torsion.

**Proposition 4.3.** *A pro-discrete $\mathbb{A}$-module $M = \varprojlim_n M_n$ is torsion over $R$ in the following cases.*

(i) *Every $\mu_{R_n}(M_n)$ is finite and, for any sufficiently large $n$, the $\mathbb{Q}[\mathbb{Z}/p^n]$-module spanned by $M_n$ has no non-zero free quotient.*

(ii) *The ($\mathbb{Z}$-)torsion-subgroup of each $M_n$ has finite exponent, $M_0$ is a finitely generated abelian group and the $R^p$-module $\varprojlim_n(\mathbb{Z}_p \otimes_{\mathbb{Z}} M_n)$ is torsion.*

*Proof.* We first assume the hypotheses of (i). Then, for each $n$, the given isomorphism $M_{n-1} \cong (M_n)_{(n-1)}$ combines with a standard property of Fitting ideals to imply that $\varrho^n_{n-1}(\mathrm{Fitt}^0_{R_n}(M_n)) = \mathrm{Fitt}^0_{R_{n-1}}(M_{n-1})$. In particular, since $\mathrm{Fitt}^0_{R_n}(M_n)$ annihilates $M_n$, the limit $\varprojlim_n \mathrm{Fitt}^0_{R_n}(M_n)$ (with respect to the transition maps $\varrho^n_{n-1}$) is an ideal of $R$ that annihilates $M$. To prove (i) it is therefore enough to show that this limit is non-zero, or equivalently that the ideal $\mathrm{Fitt}^0_{R_n}(M_n)$ is non-zero for some $n$ (and hence for all sufficiently large $n$). In addition, since Fitting ideals commute with scalar extension, the ideal $\mathrm{Fitt}^0_{R_n}(M_n)$ vanishes if and only if

$$\mathbb{Q} \otimes_{\mathbb{Z}} \mathrm{Fitt}^0_{R_n}(M_n) = \mathrm{Fitt}^0_{\mathbb{Q}[\mathbb{Z}/p^n]}(\mathbb{Q} \otimes_{\mathbb{Z}} M_n)$$

vanishes. Claim (i) is therefore true since $\mathrm{Fitt}^0_{\mathbb{Q}[\mathbb{Z}/p^n]}(\mathbb{Q} \otimes_{\mathbb{Z}} M_n)$ vanishes if and only if $\mathbb{Q} \otimes_{\mathbb{Z}} M_n$ has a non-zero free $\mathbb{Q}[\mathbb{Z}/p^n]$-module quotient.

We now assume the conditions of (ii) and set $\mathbb{Z}' := \mathbb{Z}[1/p]$ and $M'_n := \mathbb{Z}' \otimes_{\mathbb{Z}} M_{n,\mathrm{tor}}$. Then, by passing to the limit over $n$ of the obvious exact sequence $0 \to M'_n \to M_n \to \mathbb{Z}_p \otimes_{\mathbb{Z}} M_n$ we obtain an exact sequence of $R$-modules $0 \to M' \to M \to \widehat{M}$, with $M' := \varprojlim_n M'_n$ and $\widehat{M} := \varprojlim_n(\mathbb{Z}_p \otimes_{\mathbb{Z}} M_n)$ It is therefore enough to show that $M'$ and $\widehat{M}$ are both torsion $R$-modules.

To deal with $M'$ we write $\Delta_n$ for the subgroup of $\mathbb{Z}/p^n$ of order $p$ and $e_{(n)}$ for the idempotent $p^{-1}\sum_{h\in\Delta_n} h$ of $R_n[1/p]$. Then, upon taking $\Delta_n$-coinvariance of the tautological short exact sequence $0 \to M_{n,\mathrm{tor}} \to M_n \to M_{n,\mathrm{tf}} \to 0$ one obtains an exact sequence

$$\mathrm{Tor}_1^{\mathbb{Z}[\Delta_n]}(\mathbb{Z}, M_{n,\mathrm{tf}}) \to (M_{n,\mathrm{tor}})_{(n-1)} \to M_{n-1} \to (M_{n,\mathrm{tf}})_{(n-1)} \to 0.$$

In particular, since $\mathrm{Tor}_1^{\mathbb{Z}[\Delta_n]}(\mathbb{Z}, M_{n,\mathrm{tf}}) \cong \hat{H}^{-2}(\Delta_n, M_{n,\mathrm{tf}})$ and the torsion subgroup $\hat{H}^{-1}(\Delta_n, M_{n,\mathrm{tf}})$ of $(M_{n,\mathrm{tf}})_{(n-1)}$ are both either trivial or of exponent $p$, the isomorphism $M_{n-1} \cong (M_n)_{(n-1)}$ restricts to give an isomorphism $M'_{n-1} \cong (M'_n)_{(n-1)}$. This isomorphism in turn induces a direct sum decomposition of $R_n$-modules

$$M'_n = (1 - e_{(n)})M'_n \oplus e_{(n)}M'_n \cong (1 - e_{(n)})M'_n \oplus M'_{n-1}$$

and one can use this to construct, by induction on $n$, a non-zero element of $R = \varprojlim_n R_n$ that annihilates $M'$. Explicitly, if $x_{n-1}$ is a non-zero element of $R_{n-1}$ that annihilates $M'_{n-1}$ and $y_n \in R_n$ is any pre-image of $x_{n-1}$ under $\varrho^n_{n-1}$, then the above decomposition implies $y_n e_{(n)}$ annihilates $M'_n$. We now fix an integer $c$ that is divisible by $p$ and such that $c \equiv 1 \pmod{t_n}$, where $t_n$ is the exponent

of $M'_n$. Then $x_n := y_n - cy_n(1-e_{(n)})$ belongs to $R_n$, satisfies $\varrho^n_{n-1}(x_n) = x_{n-1}$ and annihilates $M'_n$ since, for each $m \in M'_n$, one has

$$x_n(m) = y_n(1-e_{(n)})(m) - cy_n(1-e_{(n)})(m) = y_n(1-e_{(n)})((1-c)m) = y_n(1-e_{(n)})(0) = 0.$$

To consider $\widehat{M}$, we note that the inverse system $(\mathbb{Z}_p \otimes_{\mathbb{Z}} M_n)_n$ induced by $(M_n)_n$ is pro-discrete, and hence that $\mathbb{Z}_p \otimes_{\mathbb{Z}} M_0$ identifies with $\mathbb{Z}_p \otimes_{R^p} \widehat{M}$. Since the given assumptions imply $\mathbb{Z}_p \otimes_{\mathbb{Z}} M_0$ is finitely-generated over $\mathbb{Z}_p$, we can therefore apply Nakayama's Lemma to the Noetherian ring $R^p$ to deduce that $\widehat{M}$ is finitely generated. Since $\widehat{M}$ is also assumed to be a torsion $R^p$-module, we may therefore fix an element $f$ of $R^p$ that annihilates $\widehat{M}$. It is then enough to note that Proposition 3.4 implies $R \cap (R^p \cdot f) \neq (0)$ and hence that $M$ is a torsion $R$-module. □

**Remark 4.4.** If $M = \varprojlim_n M_n$ is a finitely-generated torsion pro-discrete $R$-module, then the argument of Proposition 4.3(ii) implies that $\widehat{M} := \varprojlim_n (\mathbb{Z}_p \otimes_{\mathbb{Z}} M_n)$ is a finitely-generated torsion $R^p$-module. In particular, if we define the 'characteristic ideal' $\mathrm{char}_R(M)$ of $M$ to be $R \cap \mathrm{char}_{R^p}(\widehat{M})$, then Proposition 3.4 implies this ideal uniquely determines $\mathrm{char}_{R^p}(\widehat{M})$. It is an interesting problem to understand if $\mathrm{char}_R(M)$ can be directly interpreted in terms of the structure of $M$ as an $R$-module.

4.3. **Fitting ideals.** In this subsection, we clarify, under suitable hypotheses, the link between the initial Fitting ideal of a $\mathbb{\Lambda}$-module $M$ and the initial Fitting ideals of the $\mathbb{\Lambda}_n$-modules $M_{(n)}$.

Following [31, Def. 15.65.1], we refer to a $\mathbb{\Lambda}$-module $M$ as 'pseudo-coherent' if, for every $t \in \mathbb{N}$, there exists an exact sequence of $\mathbb{\Lambda}$-modules

$$M(t) \to M(t-1) \to \cdots \to M(0) \to M \to 0$$

in which each module $M(i)$ is both finitely-generated and free. (In some articles, such as [14], the phrase 'finitely $\infty$-presented' is used in place of pseudo-coherent.)

For $n \in \mathbb{N}_0$ and a finitely-presented $\mathbb{\Lambda}_n$-module $N$ we also set

$$\mathcal{F}^0_n(N) := \mathrm{Fit}^0_{\mathbb{\Lambda}_n}(N).$$

We observe that, for a finitely-generated $\mathbb{\Lambda}$-module $M$, one has $\varrho^n_{n-1}(\mathcal{F}^0_n(M_{(n)})) = \mathcal{F}^0_{n-1}(M_{(n-1)})$ for every $n \in \mathbb{N}$ (cf. the beginning of the proof of Proposition 4.3) and hence that the ideals $\mathcal{F}^0_n(M_{(n)})$ form an inverse system with respect to the maps $\varrho^n_{n-1}$.

**Proposition 4.5.** *Fix a natural number $d$ and an exact sequence of $\mathbb{\Lambda}$-modules*

$$P \to \mathbb{\Lambda}^d \to M \to 0 \tag{13}$$

*in which $P$ is the limit of a pro-discrete system $(P_n)_n$ in which each $\mathbb{\Lambda}_n$-module $P_n$ is locally-free of rank $d$. Then the following claims are valid.*

(i) *The $\mathbb{\Lambda}$-module $M$ is pseudo-coherent.*
(ii) *The $\mathbb{\Lambda}$-module $\mathrm{Fit}^0_{\mathbb{\Lambda}}(M)$ is (well-defined and) pro-discrete.*
(iii) *If $M$ is torsion over $R$, then there exists an inverse system $(\mathcal{T}_n(M), \nu_n)_{n\in\mathbb{N}}$ of $\mathbb{\Lambda}$-modules with both of the following properties.*
  (a) *Each $\mathcal{T}_n(M)$ is a $\mathbb{\Lambda}_n$-lattice that lies in the same genus as $\mathrm{Ann}_{\mathbb{\Lambda}_n}(\mathcal{F}^0_n(M_{(n)}))$, respectively is isomorphic to $\mathrm{Ann}_{\mathbb{\Lambda}_n}(\mathcal{F}^0_n(M_{(n)}))$ if $P_n$ is a free $\mathbb{\Lambda}_n$-module.*
  (b) *There exists a canonical short exact sequence of $\mathbb{\Lambda}$-modules*

$$0 \to \mathrm{Fit}^0_{\mathbb{\Lambda}}(M) \xrightarrow{\subseteq} \varprojlim_{\varrho^n_{n-1}} \mathrm{Fit}^0_{\mathbb{\Lambda}_n}(M_{(n)}) \xrightarrow{\xi_M} \varprojlim^1_{\nu_n} \mathcal{T}_n(M) \to 0 \tag{14}$$

*in which $\xi_M$ is a natural connecting homomorphism.*

*Proof.* At the outset, we note the given assumptions on $P$ combine with the Swan-Forster Theorem (via [7, Rem 3.14]) to imply that $\mu_{\mathbb{A}_n}(P_n) \leq d+1$ for every $n$. Hence, by Proposition 2.3(ii), one knows $P$ is a finitely-generated pro-discrete $\mathbb{A}$-module. Applying Proposition 2.4(i), one then deduces that each $\mathbb{A}_n$-module $P_{(n)}$ identifies with $P_n$ and so is locally-free of rank $d$.

Now, to prove (i), the result of [7, Prop. 5.1(ii)] reduces us to proving $P$ is finitely-presented over $\mathbb{A}$ and hence, by Proposition 2.4(ii), to showing $\mu_{\mathbb{A}_n}(\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, P))$ is bounded independently of $n$. In fact, we shall show that each group $\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, P)$ vanishes. To do this we fix $n$ and note that, for each integer $t > 1$, Proposition 2.3(ii) implies the existence of a free $\mathbb{A}$-submodule $P' = P'(n,t)$ of $P$ that has rank $d$ and is such that $P/P'$ is pro-discrete and the cokernel $(P/P')_{(n)}$ of the natural map $P'_{(n)} \xrightarrow{\nu_n} P_{(n)}$ is finite of order prime-to-$pt$, respectively vanishes if $P_n$ is a free $\mathbb{A}_n$-module. Since $P'_{(n)}$ and $P_{(n)}$ are both locally-free of rank $d$, the map $\nu_n$ is therefore injective in all cases and bijective if $P_n$ is a free $\mathbb{A}_n$-module. The tautological short exact sequence $0 \to P' \to P \to P/P' \to 0$ therefore gives rise to an exact sequence

$$\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, P') \to \mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, P) \xrightarrow{\nu'_n} \mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, P/P') \to 0,$$

in which the first term vanishes (as $P'$ is free) and so $\nu'_n$ is bijective. We set $\mathcal{F}^0(N) := \mathrm{Fit}^0_{\mathbb{A}}(N)$ for a finitely-presented $\mathbb{A}$-module $N$. Then, since $\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, P/P')$ is annihilated by $\varrho_n\big(\mathcal{F}^0(P/P')\big) = \mathcal{F}^0_n((P/P')_{(n)})$ (cf. the argument in the first paragraph after (5)), one deduces that $\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, P)$ is annihilated by a natural number $|(P/P')_{(n)}|$ that is prime-to-$t$. In particular, since $t$ can be chosen arbitrarily, the group $\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, P)$ must vanish, as suffices to prove (i).

We next note that, since $M$ is finitely-presented, the $\mathbb{A}$-ideal $\mathcal{F}^0(M)$ is (well-defined and) finitely-generated and hence, by Proposition 2.4(i) and (iii), both pro-discrete (thereby verifying claim (ii)) and equal to the natural limit $\varprojlim_n \mathcal{F}^0(M)_{(n)}$. We shall prove (iii) by analysing this limit and, for this purpose, we first fix $n$ and, for each integer $t > 1$, use the free $\mathbb{A}$-submodule $P' = P'(n,t)$ of $P$ constructed above. We also write $\theta$ for the map $P \to \mathbb{A}^d$ in (13) and $M'$ for the cokernel of the composite map $P' \subseteq P \xrightarrow{\theta} \mathbb{A}^d$ and claim there exists an isomorphism of $\mathbb{A}_n$-modules

$$\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, \mathbb{A}/\mathcal{F}^0(M')) \cong \mathcal{A}_n(M) := \mathrm{Ann}_{\mathbb{A}_n}(\mathcal{F}^0_n(M_{(n)})). \tag{15}$$

To verify this we note $\mathcal{F}^0(M')$ is a principal $\mathbb{A}$-ideal (since $P'$ is free) and choose a generator $\lambda$. Then, since $M$ and $P/P'$ are torsion $R$-modules (the former by assumption and the latter as a consequence of Proposition 4.3(i) and the finiteness of $(P/P')_{(n)}$) the exact sequence

$$0 \to P/P' \to M' \to M \to 0 \tag{16}$$

implies $M'$ is a torsion $R$-module and hence that $\lambda$ is a non-zero divisor in $\mathbb{A}$. The associated short exact sequence $0 \to \mathbb{A} \xrightarrow{\times\lambda} \mathbb{A} \to \mathbb{A}/\mathcal{F}^0(M') \to 0$ then induces an isomorphism of $\mathbb{A}$-modules

$$\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, \mathbb{A}/\mathcal{F}^0(M')) \cong \ker\big(\mathbb{A}_n \xrightarrow{\varrho_n(\lambda)} \mathbb{A}_n\big) = \{r \in \mathbb{A}_n : r \cdot \varrho_n(\lambda) = 0\} = \mathrm{Ann}_{\mathbb{A}_n}(\varrho_n(\mathcal{F}^0(M')),$$

where the last equality is valid since $\varrho_n(\mathcal{F}^0(M')) = \mathbb{A}_n \cdot \varrho_n(\lambda)$. To deduce (15), it is then enough to note $\varrho_n(\mathcal{F}^0(M')) = \mathcal{F}^0_n(M'_{(n)})$ and hence that

$$\mathrm{Ann}_{\mathbb{A}_n}(\varrho_n(\mathcal{F}^0(M'))) = \mathrm{Ann}_{\mathbb{A}_n}(\mathcal{F}^0_n(M'_{(n)})) = \mathcal{A}_n(M),$$

where the last equality follows from the fact $\mathbb{Q} \cdot M'_{(n)} = \mathbb{Q} \cdot M_{(n)}$ since $(P/P')_{(n)}$ is finite.

The surjective map $M' \to M$ in (16) implies $\mathcal{F}^0(M') \subseteq \mathcal{F}^0(M)$ and hence that there exists a tautological short exact sequence $0 \to \mathcal{F}^0(M)/\mathcal{F}^0(M') \to \mathbb{A}/\mathcal{F}^0(M') \to \mathbb{A}/\mathcal{F}^0(M) \to 0$. Taken in conjunction with (15), this sequence induces an exact sequence of $\mathbb{A}_n$-modules

$$\mathsf{Tor}_1^{\mathbb{A}}(\mathbb{A}_n, \mathcal{F}^0(M)/\mathcal{F}^0(M')) \to \mathcal{A}_n(M) \to \mathcal{T}_n(M) \to \big(\mathcal{F}^0(M)/\mathcal{F}^0(M')\big)_{(n)}, \tag{17}$$

in which we set

$$\mathcal{T}_n(M) := \mathsf{Tor}_1^{\mathbb{\Lambda}}(\mathbb{\Lambda}_n, \mathbb{\Lambda}/\mathcal{F}^0(M)).$$

Now, a standard multiplication property of Fitting ideals combines with (16) to imply $\mathcal{F}^0(M)/\mathcal{F}^0(M')$ is annihilated by $\mathcal{F}^0(P/P')$ and hence that the two end terms in the above sequence are annihilated by $\varrho_n(\mathcal{F}^0(P/P')) = \mathcal{F}_n^0((P/P')_{(n)})$. In particular, since $(P/P')_{(n)}$ is finite and of order prime-to-$t$, respectively vanishes if $P_n$ is a free $\mathbb{\Lambda}_n$-module, the two end terms in (17) are annihilated by an integer prime-to-$t$, respectively are trivial if $P_n$ is a free $\mathbb{\Lambda}_n$-module. Hence, since $t$ can be chosen arbitrarily in this argument, the sequence (17) implies $\mathcal{T}_n(M)$ is a $\mathbb{\Lambda}_n$-lattice lying in the same genus as $\mathcal{A}_n(M)$, respectively is isomorphic to $\mathcal{A}_n(M)$ if $P_n$ is a free $\mathbb{\Lambda}_n$-module, thereby verifying (iii)(a).

At this point, we note that the tautological short exact sequence

$$0 \to \mathcal{F}^0(M) \to \mathbb{\Lambda} \to (\mathbb{\Lambda}/\mathcal{F}^0(M)) \to 0$$

combines with the fact $\varrho_n(\mathcal{F}^0(M)) = \mathcal{F}_n^0(M_{(n)})$ to induce, for every $n$, a canonical exact commutative diagram of $\mathbb{\Lambda}_n$-modules

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{T}_n(M) & \longrightarrow & \mathcal{F}^0(M)_{(n)} & \longrightarrow & \mathcal{F}_n^0(M_{(n)}) & \longrightarrow & 0 \\
 & & {\scriptstyle \nu_n}\big\downarrow & & \big\downarrow & & {\scriptstyle \varrho_{n-1}^n}\big\downarrow & & \\
0 & \longrightarrow & \mathcal{T}_{n-1}(M) & \longrightarrow & \mathcal{F}^0(M)_{(n-1)} & \longrightarrow & \mathcal{F}_{n-1}^0(M_{(n-1)}) & \longrightarrow & 0,
\end{array}$$

in which $\nu_n$ is induced functorially by $\varrho_{n-1}^n$ and the central vertical map is the canonical projection. Upon passing to the limit over $n$ of these sequences, and recalling $\mathcal{F}^0(M)$ is pro-discrete (by (ii)), we therefore obtain an exact sequence (14), as required to complete the proof of (iii)(b). □

**Remark 4.6.** If $N$ is a principal $\mathbb{\Lambda}$-ideal that is generated by a non-zero divisor, then there exists a canonical exact sequence of $\mathbb{\Lambda}$-modules

$$0 \to N \xrightarrow{\subseteq} \varprojlim_{\varrho_{n-1}^n} \varrho_n(N) \xrightarrow{\xi_N} \varprojlim_{\varrho_{n-1}^n}{}^1 \mathrm{Ann}_{\mathbb{\Lambda}_n}\big(\varrho_n(N)\big) \to 0.$$

To see this, note $N$ is isomorphic to $\mathbb{\Lambda}$ and hence to the canonical limit $\varprojlim_n N_{(n)}$. The module $M = \mathbb{\Lambda}/N$ is therefore torsion over $R$ and has a presentation (13) in which $P = N$, $d = 1$ and every $\mathbb{\Lambda}_n$-module $N_{(n)}$ is free of rank 1. In addition, one has $\mathrm{Fit}^0_{\mathbb{\Lambda}}(\mathbb{\Lambda}/N) = N$ and $\mathrm{Fit}^0_{\mathbb{\Lambda}_n}((\mathbb{\Lambda}/N)_{(n)}) = \mathrm{Fit}^0_{\mathbb{\Lambda}_n}(\mathbb{\Lambda}_n/\varrho_n(N)) = \varrho_n(N)$ and so the inverse system $(\mathcal{T}_n(M), \nu_n)_n$ in Proposition 4.5(iii) is isomorphic to $(\mathrm{Ann}_{\mathbb{\Lambda}_n}(\varrho_n(N)), \varrho_{n-1}^n)_n$. Given these facts, the claimed exact sequence follows directly as a special case of (14).

The derived limit that occurs in the exact sequence (14) is usually non-zero. For example, if $H$ is trivial and, for any $n \in \mathbb{N}_0$, one sets $N := R \cdot \varpi_n$, then the displayed exact sequence in Remark 4.6 can be shown to recover the fundamental exact sequence (2). The following more general observation about derived limits will also be useful in the sequel.

**Lemma 4.7.** *Let $(Y_n)_{n\in\mathbb{N}}$ be an inverse system of finitely-generated abelian groups. Then $\varprojlim_n^1 Y_n$ is divisible and its torsion subgroup is supported on primes $\ell$ for which $\varprojlim_n(\mathbb{Z}_\ell \otimes_{\mathbb{Z}} Y_{n,\mathrm{tf}})$ is non-zero.*

*Proof.* Since each group $Y_{n,\mathrm{tor}}$ is finite, the Mittag-Leffler criterion implies that $\varprojlim_n^i Y_{n,\mathrm{tor}}$ vanishes for both $i = 1$ and $i = 2$. By passing to the limit over $n$ of the tautological short exact sequences $0 \to Y_{n,\mathrm{tor}} \to Y_n \to Y_{n,\mathrm{tf}} \to 0$, one deduces that the natural map $\varprojlim_n^1 Y_n \to \varprojlim_n^1 Y_{n,\mathrm{tf}}$ is bijective. Then, for each prime $\ell$ there is a short exact sequence

$$0 \to Y_{n,\mathrm{tf}} \to \mathbb{Z}_\ell \otimes_{\mathbb{Z}} Y_{n,\mathrm{tf}} \to (\mathbb{Z}_\ell/\mathbb{Z}) \otimes_{\mathbb{Z}} Y_{n,\mathrm{tf}} \to 0.$$

These sequences are compatible with the natural transition maps as $n$ varies. Hence, since each module $\mathbb{Z}_\ell \otimes_{\mathbb{Z}} Y_{n,\mathrm{tf}}$ is compact (and the relevant transition morphisms are continuous), upon passing to the inverse limit one obtains an exact sequence

$$0 \to \varprojlim_n Y_{n,\mathrm{tf}} \xrightarrow{\alpha} \varprojlim_n \big(\mathbb{Z}_\ell \otimes_{\mathbb{Z}} Y_{n,\mathrm{tf}}\big) \to \varprojlim_n \big((\mathbb{Z}_\ell/\mathbb{Z}) \otimes_{\mathbb{Z}} Y_{n,\mathrm{tf}}\big) \to \varprojlim_n^1 Y_n \to 0. \tag{18}$$

Now, since $\mathbb{Z}_\ell/\mathbb{Z}$ is uniquely $\ell$-divisible, the second term in this sequence is uniquely $\ell$-divisible. The exact sequence therefore implies that the group $\varprojlim_n^1 Y_n$ is $\ell$-divisible, respectively uniquely $\ell$-divisible if $\alpha$ is surjctive. This implies the claimed result. □

**Remark 4.8.** In view of Lemma 4.7, the short exact sequence (14) (and hence also that displayed in Remark 4.6) raises the question of whether finitely-generated $\mathbb{\Lambda}$-ideals are uniquely determined by their projections to $\mathbb{\Lambda}_n$ for each $n$. However, the chain of ideals constructed in Proposition 3.2(iii) shows that this is in general far from being true.

4.4. **Derived limit complexes.** In this subsection, we specialise the construction of derived limit complexes (as presented in [31]) to the setting of pro-discrete modules. The results obtained here will then be used in our study of Weil-étale cohomology in §6.

For a commutative ring $B$ we write $\mathsf{D}(B)$ for the derived category of $B$-modules. If $B$ is Noetherian, we further write $\mathsf{D}^{\mathrm{perf}}(B)$ for the full triangulated subcategory of $\mathsf{D}(B)$ comprising complexes that are isomorphic to a bounded complex of finitely generated projective $B$-modules and we recall that each object $X$ of $\mathsf{D}^{\mathrm{perf}}(B)$ defines a natural Euler characteristic $\chi_B(X)$ in the Grothendieck group $K_0(B)$.

We then assume to be given a family of complexes $(X_n^\bullet)_{n\in\mathbb{N}_0}$ of $\mathbb{\Lambda}$-modules satisfying the following hypothesis.

**Hypothesis 4.9.**

(i) *For each $n \in \mathbb{N}_0$, the action of $\mathbb{\Lambda}$ on $X_n^\bullet$ factors through $\mathbb{\Lambda}_n$.*
(ii) *For each $n \in \mathbb{N}$, there exists an isomorphism $\theta_n : \mathbb{\Lambda}_{n-1} \otimes^{\mathsf{L}}_{\mathbb{\Lambda}_n} X_n^\bullet \cong X_{n-1}^\bullet$ in $\mathsf{D}(\mathbb{\Lambda}_{n-1})$.*

In the context of (ii), we will also write $\theta_n$ for the induced composite morphism in $\mathsf{D}(\mathbb{\Lambda}_n)$

$$X_n^\bullet \to \mathbb{\Lambda}_{n-1} \otimes^{\mathsf{L}}_{\mathbb{\Lambda}_n} X_n^\bullet \cong X_{n-1}^\bullet.$$

**Definition 4.10.** The 'derived limit' of a family $(X_n^\bullet)_{n\in\mathbb{N}}$ of complexes satisfying Hypothesis 4.9 is a complex $\underleftarrow{\mathsf{Rlim}}_n X_n^\bullet$ that lies in an exact triangle in $\mathsf{D}(\mathbb{\Lambda})$ of the form

$$\underleftarrow{\mathsf{Rlim}}_n X_n^\bullet \to \prod\nolimits_{n\in\mathbb{N}} X_n^\bullet \xrightarrow{(1-\theta_n)_n} \prod\nolimits_{n\in\mathbb{N}} X_n^\bullet \to \cdot \tag{19}$$

(cf. [31, Def. 13.34.1]). We recall that such a complex always exists and is unique up to isomorphism in $\mathsf{D}(\mathbb{\Lambda})$ (cf. [31, Lem. 15.88.3, Lem. 15.88.5 and Rem. 15.88.6]).

The following result describes some properties of this construction that will be useful for us.

**Proposition 4.11.** *Assume there exist integers $a$ and $b$ with $a \leq b$ such that, for every $n \in \mathbb{N}_0$, the group $H^i(X_n^\bullet)$ vanishes for both $i < a$ and $i > b$. Then the following claims are valid.*

*(i) $H^i(\underleftarrow{\mathsf{Rlim}}_n X_n^\bullet)$ vanishes for both $i < a$ and $i > b$. Further, there exists a canonical isomorphism $H^a(\underleftarrow{\mathsf{Rlim}}_n X_n^\bullet) \cong \varprojlim_{n\in\mathbb{N}} H^a(X_n^\bullet)$ and, for $i > a$, a canonical short exact sequence*

$$0 \to \varprojlim_n^1 H^{i-1}(X_n^\bullet) \to H^i(\underleftarrow{\mathsf{Rlim}}_n X_n^\bullet) \to \varprojlim_n H^i(X_n^\bullet) \to 0,$$

*where the respective limits are taken with respect to the morphisms $H^{i-1}(\theta_n)$ and $H^i(\theta_n)$.*

*(ii) Suppose to be given two further families $(Y_n^\bullet, \theta_n')$ and $(Z_n^\bullet, \theta_n'')$ satisfying Hypothesis 4.9, together with a collection of compatible exact triangles $X_n^\bullet \to Y_n^\bullet \to Z_n^\bullet \to \cdot$ in $\mathsf{D}(\mathbb{\Lambda}_n)$. Then there is an exact triangle $\underleftarrow{\mathsf{Rlim}}_n X_n^\bullet \to \underleftarrow{\mathsf{Rlim}}_n Y_n^\bullet \to \underleftarrow{\mathsf{Rlim}}_n Z_n^\bullet \to \cdot$ in $\mathsf{D}(\mathbb{\Lambda})$.*

*(iii) For each $m \in \mathbb{N}_0$ there exists a canonical morphism $\mathbb{\Lambda}_m \otimes^{\mathsf{L}}_{\mathbb{\Lambda}} \underleftarrow{\mathsf{Rlim}}_n X^\bullet_n \to X^\bullet_m$ in $\mathsf{D}(\mathbb{\Lambda}_m)$.*

*(iv) Assume that $b = a + 1$ and that all of the following conditions are satisfied:*

(a) *for all $n$, the complex $X^\bullet_n$ belongs to $\mathsf{D}^{\mathrm{perf}}(\mathbb{\Lambda}_n)$;*

(b) *for all $n$, the module $H^a(X^\bullet_n)$ is $\mathbb{Z}$-free;*

(c) *for all $n$, the $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{\Lambda}_n$-modules spanned by $H^a(X^\bullet_n)$ and $H^{a+1}(X^\bullet_n)$ are isomorphic;*

(d) *$\underleftarrow{\lim}_n H^{a+1}(X^\bullet_n)$ is a finitely-generated $\mathbb{\Lambda}$-module.*

*Then $\underleftarrow{\mathsf{Rlim}}_n X^\bullet_n$ is isomorphic in $\mathsf{D}(\mathbb{\Lambda})$ to a complex of $\mathbb{\Lambda}$-modules $P \to \Pi$. Here $\Pi$ occurs in degree $a + 1$ and is a free-module of finite rank, $t$ say, and $P$ is finitely-generated and also, unless $p$ is exceptional, projective. Further, in every case, for each $m \in \mathbb{N}_0$, the $\mathbb{\Lambda}_m$-module $P_{(m)}$ is locally-free of rank $t$, respectively free of rank $t$ if $\chi_{\mathbb{\Lambda}_m}(X^\bullet_m) = 0$, the morphism in (iii) is an isomorphism and the induced map $H^{a+1}(\underleftarrow{\mathsf{Rlim}}_n X^\bullet_n)_{(m)} \to H^{a+1}(X^\bullet_m)$ is bijective.*

*Proof.* The long exact sequence of cohomology of the exact triangle (19) gives in each degree $k$ a short exact sequence

$$0 \to \underleftarrow{\lim}^1_n H^{k-1}(X^\bullet_n) \to H^k(\underleftarrow{\mathsf{Rlim}}_n X^\bullet_n) \to \underleftarrow{\lim}_n H^k(X^\bullet_n) \to 0, \tag{20}$$

in which each limit is taken with respect to the transition morphisms induced by the respective maps $H^i(\theta_n)$. The isomorphism in the second assertion of (i), together with the acyclicity of $\underleftarrow{\mathsf{Rlim}}_n X^\bullet_n$ in all degrees less than $a$ and greater than $b + 1$, follow directly from these sequences. The acyclicity in degree $b + 1$ follows in the same way since each morphism $H^b(\theta_n) : H^b(X^\bullet_n) \to H^b(X^\bullet_{n-1})$ is surjective (as $X^\bullet_n$ is acyclic in degrees greater than $b$).

To prove (ii) we note that restriction of scalars along $\mathbb{\Lambda} \to \mathbb{\Lambda}_n$ is an exact functor between the respective module categories and hence that each of the given triangles is exact in $\mathsf{D}(\mathbb{\Lambda})$. In particular, since the category of $\mathbb{\Lambda}$-modules has exact products, the given assumptions imply the existence of a commutative diagram of exact triangles in $\mathsf{D}(\mathbb{\Lambda})$ of the form

$$\begin{array}{ccccccc}
 & & \uparrow & & \uparrow & & \\
\underleftarrow{\mathsf{Rlim}}_n Z^\bullet_n & \longrightarrow & \prod_{n\in\mathbb{N}} Z^\bullet_n & \xrightarrow{(1-\theta''_n)_n} & \prod_{n\in\mathbb{N}} Z^\bullet_n & \longrightarrow & \cdot \\
 & & \uparrow & & \uparrow & & \\
\underleftarrow{\mathsf{Rlim}}_n Y^\bullet_n & \longrightarrow & \prod_{n\in\mathbb{N}} Y^\bullet_n & \xrightarrow{(1-\theta'_n)_n} & \prod_{n\in\mathbb{N}} Y^\bullet_n & \longrightarrow & \cdot \\
 & & \uparrow & & \uparrow & & \\
\underleftarrow{\mathsf{Rlim}}_n X^\bullet_n & \longrightarrow & \prod_{n\in\mathbb{N}} X^\bullet_n & \xrightarrow{(1-\theta_n)_n} & \prod_{n\in\mathbb{N}} X^\bullet_n & \longrightarrow & \cdot
\end{array}$$

in which each row is the corresponding case of (19). By applying the Octahedral axiom to this diagram, one obtains an exact triangle as in (ii).

For each $m > n$ we write $\theta_{m,n} : X^\bullet_m \to X^\bullet_n$ for the composite morphism $\theta_{n+1} \circ \cdots \circ \theta_{m-1} \circ \theta_m$. Then, to prove (iii), we use the commutative square in $\mathsf{D}(\mathbb{\Lambda})$

$$\begin{array}{ccc}
\mathbb{\Lambda}_n \otimes^{\mathsf{L}}_{\mathbb{\Lambda}} \prod_{m\in\mathbb{N}} X^\bullet_m & \xrightarrow{\mathrm{id}\otimes(1-\theta_m)_m} & \mathbb{\Lambda}_n \otimes^{\mathsf{L}}_{\mathbb{\Lambda}} \prod_{m\in\mathbb{N}} X^\bullet_m \\
\downarrow{\scriptstyle (\theta_{m,n})_{m>n}} & & \downarrow{\scriptstyle (\theta_{m,n})_{m>n}} \\
\prod_{m>n} X^\bullet_n & \xrightarrow{(1-\mathrm{id}_m)_m} & \prod_{m>n} X^\bullet_n
\end{array}$$

in which $\mathrm{id}_m$ denotes the identity morphism from $X_n^\bullet$ in the $m$-th component of the direct product to $X_n^\bullet$ in the $(m-1)$-st component of the product. This diagram combines with the exact triangle (19) to induce a morphism from $\mathbb{\Lambda}_n \otimes^{\mathsf{L}}_{\mathbb{\Lambda}} \underleftarrow{\mathsf{R}\lim}_m X_m^\bullet$ to the derived limit of the constant system $(X_n^\bullet, \mathrm{id})_{m\in\mathbb{N}_0}$. This in turn induces a morphism of the required sort since the latter system is $\underleftarrow{\lim}_n$-acyclic and so [31, Lem. 15.88.1(5) and Lem. 15.88.3] combine to imply its derived limit is represented by $X_n^\bullet$.

To prove (iv) we claim first that the $R$-module $\underleftarrow{\lim}_n H^{a+1}(X_n^\bullet)$ is pro-discrete. Indeed, this is true since the given isomorphism $\theta_n$ combines with the fact that $X_n^\bullet$ and $X_{n-1}^\bullet$ are each acyclic in degrees greater than $a+1$ to induce an isomorphism of $R_{n-1}$-modules

$$H^{a+1}(X_n^\bullet)_{(n-1)} \cong H^{a+1}(X_{n-1}^\bullet). \tag{21}$$

We next note that condition (d) allows us to fix a free $\mathbb{\Lambda}$-module $\Pi$ of finite rank $t$ (which we assume, as we may, is at least 2), together with a surjective homomorphism

$$\pi : \Pi \to \underleftarrow{\lim}_n H^{a+1}(X_n^\bullet).$$

Then, for each $n \in \mathbb{N}_0$ we write $\Pi_n$ for the free, rank $t$, $\mathbb{\Lambda}_n$-module $\mathbb{\Lambda}_n \otimes_{\mathbb{\Lambda}} \Pi$ and we note that, as $\underleftarrow{\lim}_n H^{a+1}(X_n^\bullet)$ is pro-discrete, the map $\pi_n : \Pi_n \to H^{a+1}(X_n^\bullet)$ induced by $\pi$ is surjective (cf. [7, Rem. 3.3]). Now, by a standard argument (as in [8, §5.4]), the conditions (a), (b) and (c) combine to imply $X_n^\bullet$ is isomorphic in $\mathsf{D}(\mathbb{\Lambda}_n)$ to a complex $\Pi_n^\bullet$ of the form $P_n \xrightarrow{\phi_n} \Pi_n$, where $P_n$ is a finitely-generated locally-free (and hence projective) $\mathbb{\Lambda}_n$-module of rank $t$ placed in degree $a$, and $\mathrm{cok}(\phi_n)$ is identified with $H^{a+1}(X_n^\bullet)$ via $\pi_n$. In addition, if $\chi_{\mathbb{\Lambda}_n}(X_n^\bullet) = 0$, then (since $t \geq 2 > \dim(\mathbb{\Lambda}_n)$) the Bass Cancellation Theorem [13, Th. (41.20)] implies $P_n$ is isomorphic to $\Pi_n$ and so is free of rank $t$.

Given these representatives, the composite morphism in $\mathsf{D}(\mathbb{\Lambda}_n)$

$$\Pi_n^\bullet \cong X_n^\bullet \xrightarrow{\theta_n} X_{n-1}^\bullet \cong \Pi_{n-1}^\bullet$$

is realised by a concrete morphism of complexes $\theta_n^\bullet = (\theta_n^a, \theta_n^{a+1})$ and, by changing $\theta_n^\bullet$ by a homotopy (if necessary), we can assume that $\theta_n^{a+1}$ is the natural projection map $\rho_n : \Pi_n \to \Pi_{n-1}$. We now consider the following commutative diagram of $\mathbb{\Lambda}_{n-1}$-modules

$$\begin{array}{ccccc}
(P_n)_{(n-1)} & \xrightarrow{(\theta_n^a)_{(n-1)}} & P_{n-1} & & \\
\| & & \downarrow{\scriptstyle x\mapsto(x,0)} & & \\
(P_n)_{(n-1)} & \xrightarrow{((\theta_n^a)_{(n-1)},(\phi_n)_{(n-1)})} & P_{n-1}\oplus(\Pi_n)_{(n-1)} & \xrightarrow{(\phi_{n-1},-(\theta_n^{a+1})_{(n-1)})} & \Pi_{n-1} \\
 & & {\scriptstyle (y,z)\mapsto -z}\downarrow & & \| \\
 & & (\Pi_n)_{(n-1)} & \xrightarrow{(\theta_n^{a+1})_{(n-1)}} & \Pi_{n-1}.
\end{array}$$

The central row of this diagram is the mapping cone of the morphism $\mathbb{\Lambda}_{n-1} \otimes_{\mathbb{\Lambda}_n} \Pi_n^\bullet \to \Pi_{n-1}^\bullet$ induced by $\theta_n^\bullet$. In particular, since $\theta_n^\bullet$ represents $\theta_n$, this mapping cone is acyclic and so the central row is exact. Since the morphism $(\theta_n^{a+1})_{(n-1)}$ on the bottom row is induced by $\rho_n$ and so is bijective, and the central column is a short exact sequence, a diagram chase shows that the morphism $(\theta_n^a)_{(n-1)}$ in the upper row is also bijective. It follows that the family $(P_n, \theta_n^a)_n$ is a pro-discrete system and so Propositions 2.3(ii) and 4.1 combine to imply that its limit $P := \underleftarrow{\lim}_n P_n$ is a finitely-generated, pro-discrete $\mathbb{\Lambda}$-module that is also, unless $p$ is exceptional, projective. From Proposition 2.4(i) we also know that the canonical map $P_{(n)} \to P_n$ is bijective for every $n$.

Writing $\phi_\infty$ for the limit homomorphism $\underleftarrow{\lim}_n \phi_n$, we next claim that the complex $\Pi^\bullet$ given by $P \xrightarrow{\phi_\infty} \Pi$, where $P$ is placed in degree $a$, is isomorphic to $\underleftarrow{\mathsf{R}\lim}_n X_n^\bullet$ in $\mathsf{D}(\mathbb{\Lambda})$.

To show this we note that, since $\theta_n^\bullet$ represents $\theta_n$ in $\mathsf{D}(\mathbb{\Lambda}_n)$, the derived limit $\underleftarrow{\mathsf{R}\lim}_n \Pi_n^\bullet$ of the family $(\Pi_n^\bullet, \theta_n^\bullet)$ is isomorphic in $\mathsf{D}(\mathbb{\Lambda})$ to $\underleftarrow{\mathsf{R}\lim}_n X_n^\bullet$. It is therefore enough to show that $\Pi^\bullet$ is isomorphic in $\mathsf{D}(\mathbb{\Lambda})$ to $\underleftarrow{\mathsf{R}\lim}_n \Pi_n^\bullet$.

Write $\mathsf{Mod}^{\mathbb{N}}(\mathbb{\Lambda})$ for the abelian category of inverse systems $(M_n, \psi_n)$ of $\mathbb{\Lambda}$-modules where each $M_n$ is naturally a $\mathbb{\Lambda}_n$-module and $\psi_n : M_n \to M_{n-1}$ is a homomorphism of $\mathbb{\Lambda}$-modules. Then, since $\mathsf{Mod}^{\mathbb{N}}(\mathbb{\Lambda})$ has enough injectives, the natural inverse limit functor $\underleftarrow{\lim}_n : \mathsf{Mod}^{\mathbb{N}}(\mathbb{\Lambda}) \to \mathsf{Mod}(\mathbb{\Lambda})$ induces a functor $\underleftarrow{\mathsf{R}\lim}'_n : \mathsf{D}(\mathsf{Mod}^{\mathbb{N}}(\mathbb{\Lambda})) \to \mathsf{D}(\mathbb{\Lambda})$ on the corresponding derived categories. The key point now is that the family $(\Pi_n^\bullet, \theta_n^\bullet)$ defines an object of $\mathsf{D}(\mathsf{Mod}^{\mathbb{N}}(\mathbb{\Lambda}))$ with the property that both occurring systems $(P_n, \theta_n^a)$ and $(\Pi_n, \theta_n^{a+1})$ are $\underleftarrow{\lim}_n$-acyclic since every transition map $\theta_n^a$ and $\theta_n^{a+1}$ is surjective. It follows from [31, Lem. 15.88.1(5)] that $\underleftarrow{\mathsf{R}\lim}'_n \Pi_n^\bullet$ is represented by the complex $\Pi^\bullet$ and hence, by [31, Lem. 15.88.3], that $\underleftarrow{\mathsf{R}\lim}_n \Pi_n^\bullet$ is naturally isomorphic to $\Pi^\bullet$ in $\mathsf{D}(\mathbb{\Lambda})$, as required. This completes the proof of all but the final assertion of (iv). The final assertion of (iv) is then true since, in this case, the morphism in (iii) coincides with the composite morphism

$$\underleftarrow{\mathsf{R}\lim}_n X_n^\bullet \cong \Pi^\bullet \xrightarrow{x \mapsto 1 \otimes x} \mathbb{\Lambda}_m \otimes_{\mathbb{\Lambda}} \Pi^\bullet = \Pi_m^\bullet \cong X_m^\bullet,$$

and so the induced map of $\mathbb{\Lambda}_m$-modules $H^{a+1}(\underleftarrow{\mathsf{R}\lim}_n X_n^\bullet)_{(m)} \to H^{a+1}(X_m^\bullet)$ is bijective as $\mathrm{cok}(\phi_\infty)_{(m)}$ identifies with $\mathrm{cok}(\phi_m)$. □

## 5. Divisor class groups

For any global function field $E$ we write $C_E$ for the unique geometrically irreducible smooth projective curve that has function field $E$, and $E^{\mathrm{con}}$ for the constant $\mathbb{Z}_p$-extension of $E$. We also write $\mathrm{d}_E(v)$ for the degree (over $E$) of each place $v$ of $E$.

In this section we fix such a field $k$ of characteristic $p$ together with a finite, possibly empty, set of places $\Sigma$ of $k$. For a finite extension $F$ of $k$, we write $\Sigma_F$ for the set of places of $F$ that lie above those in $\Sigma$. If $\Sigma \neq \emptyset$, we write $\mathcal{O}_{F,\Sigma}$ for the subring of $F$ comprising elements integral at all places outside $\Sigma_F$, $B_\Sigma(F)$ for $\bigoplus_{w \in \Sigma_F} \mathbb{Z}$ and $B_\Sigma^0(F)$ for the kernel of the map $B_\Sigma(F) \to \mathbb{Z}$ sending each $(n_w)_w$ to $\sum_w n_w$. We also define a finite abelian group

$$A_\Sigma(F) := \begin{cases} \mathrm{Pic}(\mathcal{O}_{F,\Sigma}), & \text{if } \Sigma \neq \emptyset \\ \mathrm{Pic}^0(C_F), & \text{if } \Sigma = \emptyset \end{cases}$$

and, for each $\ell \in \mathcal{P}$, set $A_\Sigma^\ell(F) := A_\Sigma(F)[\ell^\infty]$. We then write $A'_\Sigma(F)$ for $\bigoplus_{\ell \in \mathcal{P} \setminus \{p\}} A_\Sigma^\ell(F)$, so that $A_\Sigma(F) = A_\Sigma^p(F) \oplus A'_\Sigma(F)$.

We now fix a $\mathbb{Z}_p$-extension $\mathcal{K}$ of $k$ that is ramified at only finitely many places. For $n \in \mathbb{N}_0$, we write $\mathcal{K}_n$ for the extension of $k$ of degree $p^n$ inside $\mathcal{K}$. Then, for $n \in \mathbb{N}$, the field-theoretic norm map $\mathrm{N}_n : \mathcal{K}_n^\times \to \mathcal{K}_{n-1}^\times$ induces a map $A_\Sigma(\mathcal{K}_n) \to A_\Sigma(\mathcal{K}_{n-1})$ (that we also denote by $\mathrm{N}_n$) and we set

$$A_\Sigma(\mathcal{K}) := \underleftarrow{\lim}_{\mathrm{N}_n} A_\Sigma(\mathcal{K}_n) \quad \text{and} \quad A_\Sigma^\ell(\mathcal{K}) := \underleftarrow{\lim}_{\mathrm{N}_n} A_\Sigma^\ell(\mathcal{K}_n) \quad \text{for } \ell \in \mathcal{P}.$$

We fix a topological generator of $G_{\mathcal{K}/k} := \mathrm{Gal}(\mathcal{K}/k)$, thereby regarding $A_\Sigma(\mathcal{K})$ and each $A_\Sigma^\ell(\mathcal{K})$ as an $R$-module, and our aim is then to use the theory of pro-discrete $R$-modules to study $A_\Sigma(\mathcal{K})$.

To state the main result of this section, we use the notion of a pseudo-coherent module, as recalled just before Proposition 4.5. In addition, for each $q \in \mathcal{P}$, we define the '$q$-rank' $\mathrm{r}_q(X)$ of a finite abelian group $X$ via the equality $q^{\mathrm{r}_q(X)} = |X/q|$ and, for $n \in \mathbb{N}_0$, write $t_n(q)$ for the number of $q$-adic places of the field $\mathbb{Q}(e^{2\pi i/p^n})$. Finally, we recall the ring homomorphisms $\tau_n^n : R_n \to \mathcal{O}_n$ defined just before (8) and, as in Proposition 3.8, write $\mathrm{MaxSpec}_{(q)}(\mathcal{O}_n)$ for $q \in \mathcal{P}$ for the subspace of $\mathrm{MaxSpec}(\mathcal{O}_n)$ comprising prime ideals above $q$.

**Theorem 5.1.** *For data $\mathcal{K}/k$ and $\Sigma$ as above, the following claims are valid.*

(i) *$A_\Sigma(\mathcal{K})$ is both pro-discrete and torsion as an $R$-module.*
(ii) *$\mu_R\big(A^p_\Sigma(\mathcal{K})\big)$ is finite if and only if $A^p_\Sigma(\mathcal{K})^{\omega_n=0}$ is finite for every $n \in \mathbb{N}_0$.*
(iii) *$\mu_R\big(A^\ell_\Sigma(\mathcal{K})\big)$ is finite for $\ell \in \mathcal{P} \setminus \{p\}$ if and only if $\mathrm{r}_\ell(A_\Sigma(\mathcal{K}_n)) = O(p^n)$ as $n \to \infty$.*
(iv) *$A_\Sigma(\mathcal{K})$ is pseudo-coherent as an $R$-module if and only if $\mu_R\big(A_\Sigma(\mathcal{K})\big)$ is finite.*
(v) *$A_\Sigma(\mathcal{K})$ is a pro-discrete, torsion, pseudo-coherent $R$-module if and only if both of the following conditions are satisfied:*
  (a) *$A^p_\Sigma(\mathcal{K})^{\omega_n=0}$ is finite for every $n \in \mathbb{N}_0$;*
  (b) *$(t_n(\ell)/p^n)\cdot \mathrm{r}_\ell((\mathcal{O}_n/\mathcal{L}) \otimes_{R_n,\tau^n_n} A_\Sigma(\mathcal{K}_n))$ is bounded independently of $\ell \in \mathcal{P} \setminus \{p\}$, $n \in \mathbb{N}$ and $\mathcal{L} \in \mathrm{MaxSpec}_{(\ell)}(\mathcal{O}_n)$.*

  *In addition, condition (b) is satisfied if $(t_n(\ell)/p^n)\cdot \mathrm{r}_\ell(A_\Sigma(\mathcal{K}_n))$ is bounded independently of $\ell \in \mathcal{P} \setminus \{p\}$ and $n \in \mathbb{N}$.*
(vi) *Both of the conditions in (v) are satisfied if $\mathcal{K} = k^{\mathrm{con}}$.*

*Proof.* Since $A^\ell_\Sigma(\mathcal{K})$ is the maximal pro-$\ell$-subgroup of $A_\Sigma(\mathcal{K})$, one has $A_\Sigma(\mathcal{K}) = A^p_\Sigma(\mathcal{K}) \oplus A'_\Sigma(\mathcal{K})$, with $A'_\Sigma(\mathcal{K}) := \varprojlim_{\mathrm{N}_n} A'_\Sigma(\mathcal{K}_n) = \prod_{\ell\in\mathcal{P}\setminus\{p\}} A^\ell_\Sigma(\mathcal{K})$. It follows that $A_\Sigma(\mathcal{K})$ is pro-discrete if and only if the modules $A^p_\Sigma(\mathcal{K})$ and $A'_\Sigma(\mathcal{K})$ are both pro-discrete.

To analyse $A^p_\Sigma(\mathcal{K})$, we use the following notation for $n \in \mathbb{N}_0$: we write $\delta^\Sigma_n : B_\Sigma(\mathcal{K}_n) \to \mathbb{Z}$ for the homomorphism sending each $w \in \Sigma_{\mathcal{K}_n}$ to $\mathrm{d}_k(w)$, $m^\Sigma_n$ for the natural number that generates $\mathrm{im}(\delta^\Sigma_n)$; $\Delta^\Sigma_n$ for the divisor map $\mathcal{O}^\times_{\mathcal{K}_n,\Sigma} \to \ker(\delta^\Sigma_n)$ and $c_n$ for the dimension of the field of constants of $\mathcal{K}_n$ over the field of constants of $k$ (so that $c_n$ divides $m^\Sigma_n$). Then, for each $n \in \mathbb{N}$, there are natural exact commutative diagrams of $R_n$-modules

$$\begin{array}{ccccccccc}
0 \to & \ker(\delta^\Sigma_n) & \xrightarrow{\subset} & B_\Sigma(\mathcal{K}_n) & \xrightarrow{\delta^\Sigma_n} & \mathbb{Z} & \to & \mathbb{Z}/(\mathbb{Z}\cdot m^\Sigma_n) & \to 0 \\
& \downarrow{\scriptstyle \alpha'_n} & & \downarrow{\scriptstyle \alpha_n} & & \| & & \downarrow{\scriptstyle \beta_n} & \\
0 \to & \ker(\delta^\Sigma_{n-1}) & \xrightarrow{\subset} & B_\Sigma(\mathcal{K}_{n-1}) & \xrightarrow{\delta^\Sigma_{n-1}} & \mathbb{Z} & \to & \mathbb{Z}/(\mathbb{Z}\cdot m^\Sigma_{n-1}) & \to 0,
\end{array}$$

$$\begin{array}{ccccccccccc}
0 \to & \mathcal{O}^\times_{\mathcal{K}_n,\Sigma} & \xrightarrow{\Delta^\Sigma_n} & \ker(\delta^\Sigma_n) & \to & A_\emptyset(\mathcal{K}_n) & \to & A_\Sigma(\mathcal{K}_n) & \to & (\mathbb{Z}\cdot c_n)/(\mathbb{Z}\cdot m^\Sigma_n) & \to 0 \\
& \downarrow{\scriptstyle \mathrm{N}_n} & & \downarrow{\scriptstyle \alpha'_n} & & \downarrow{\scriptstyle \mathrm{N}_n} & & \downarrow{\scriptstyle \mathrm{N}_n} & & \downarrow{\scriptstyle \beta'_n} & \\
0 \to & \mathcal{O}^\times_{\mathcal{K}_{n-1},\Sigma} & \xrightarrow{\Delta^\Sigma_{n-1}} & \ker(\delta^\Sigma_{n-1}) & \to & A_\emptyset(\mathcal{K}_{n-1}) & \to & A_\Sigma(\mathcal{K}_{n-1}) & \to & (\mathbb{Z}\cdot c_{n-1})/(\mathbb{Z}\cdot m^\Sigma_{n-1}) & \to 0.
\end{array} \tag{22}$$

Here $G_{\mathcal{K}/k}$ (and hence $R_n$) acts trivially on $\mathbb{Z}$ and on each quotient $(\mathbb{Z}\cdot c_n)/(\mathbb{Z}\cdot m^\Sigma_n)$ and $\mathbb{Z}/(\mathbb{Z}\cdot m^\Sigma_n)$, the maps $\alpha_n$ and $\alpha'_n$ send each place $w \in \Sigma_{\mathcal{K}_n}$ to $f_v \cdot v$, where $v$ is the restriction of $w$ to $\mathcal{K}_{n-1}$ and $f_v$ its residue degree in $\mathcal{K}_n/\mathcal{K}_{n-1}$ and the maps $\beta_n$ and $\beta'_n$ are induced by the inclusions $\mathbb{Z}\cdot c_n \subseteq \mathbb{Z}\cdot c_{n-1}$ and $\mathbb{Z}\cdot m^\Sigma_n \subseteq \mathbb{Z}\cdot m^\Sigma_{n-1}$. (The exact rows of the second diagram can be derived, for example, by applying the Snake Lemma to the diagram of [24, (11)] and then using the isomorphism of [24, (12)], and the commutativity of both diagrams is easy to check.)

By tensoring these diagrams with $\mathbb{Z}_p$ and passing to the limit over $n$, exactness is preserved (as finitely-generated $\mathbb{Z}_p$-modules are compact) and so one derives an inclusion $\varprojlim_{\alpha'_n}(\mathbb{Z}_p \otimes_\mathbb{Z} \ker(\delta^\Sigma_n)) \subseteq \varprojlim_{\alpha_n}(\mathbb{Z}_p \otimes_\mathbb{Z} B_\Sigma(\mathcal{K}_n))$ and an exact sequence of $R^p$-modules

$$\varprojlim_{\alpha'_n}(\mathbb{Z}_p \otimes_\mathbb{Z} \ker(\delta^\Sigma_n)) \to A^p_\emptyset(\mathcal{K}) \to A^p_\Sigma(\mathcal{K}) \to Q_p \tag{23}$$

in which $Q_p$ denotes the (possibly finite) quotient $\varprojlim_{\beta'_n}(\mathbb{Z}_p\cdot c_n)/(\mathbb{Z}_p\cdot m^\Sigma_n)$ of $\mathbb{Z}_p$ endowed with trivial action of $R^p$. At this point we recall that, since $\mathcal{K}/k$ is ramified at only finitely many places, a result of Gold and Kisilevsky [17, Th. 2] implies $A^p_\emptyset(\mathcal{K})$ is a finitely-generated, torsion $R^p$-module. The

last displayed seqeunce therefore implies $A^p_\Sigma(\mathcal{K})$ is a finitely-generated, torsion $R^p$-module and, by Proposition 4.3(ii), it then follows directly that $A_\Sigma(\mathcal{K})$ is a torsion $R$-module, as claimed in (i).

In addition, since $A^p_\Sigma(\mathcal{K})_{(n)}$ is the quotient of $A^p_\Sigma(\mathcal{K})$ by $\varpi_n(A^p_\Sigma(\mathcal{K}))$ (by [7, Lem. 3.1(i)]), a general property of finitely-generated $R^p$-modules implies the natural map $A^p_\Sigma(\mathcal{K}) \to \varprojlim_n A^p_\Sigma(\mathcal{K})_{(n)}$ is bijective, thereby showing that $A^p_\Sigma(\mathcal{K})$ is pro-discrete.

For $n \in \mathbb{N}$, we write $\eta_{n,\Sigma} : A'_\Sigma(\mathcal{K}_n) \to A'_\Sigma(\mathcal{K}_{n-1})$ and $\overline{\eta}_{n,\Sigma} : R_{n-1} \otimes_{R_n} A'_\Sigma(\mathcal{K}_n) \to A'_\Sigma(\mathcal{K}_{n-1})$ for the maps induced by $\mathrm{N}_n$. Then, to prove $A'_\Sigma(\mathcal{K})$ is pro-discrete, it is enough to show each $\overline{\eta}_{n,\Sigma}$ is bijective. Writing $\iota_n$ for the inflation map $A'_\Sigma(\mathcal{K}_{n-1}) \to A'_\Sigma(\mathcal{K}_n)$, the composite $\eta_{n,\Sigma} \circ \iota_n$ is equal to multiplication-by-$p$ and hence, since $|A'_\Sigma(\mathcal{K}_{n-1})|$ is prime-to-$p$, we deduce $\iota_n$ is injective and $\eta_{n,\Sigma}$, and hence also $\overline{\eta}_{n,\Sigma}$, is surjective. Thus, if we use $\iota_n$ to regard $A'_\Sigma(\mathcal{K}_{n-1})$ as a submodule of $A'_\Sigma(\mathcal{K}_n)$, then $\eta_{n,\Sigma}$ is induced by the action of $T_n := \sum_{g \in G_{\mathcal{K}_n/\mathcal{K}_{n-1}}} g \in R_n$. In particular, since $|A'_\Sigma(\mathcal{K}_n)|$ is prime-to-$p$ (and hence cohomologically-trivial as a $G_{\mathcal{K}_n/\mathcal{K}_{n-1}}$-module), one has $\ker(\eta_{n,\Sigma}) = \varpi_{n-1}(A'_\Sigma(\mathcal{K}_n))$ and so $\overline{\eta}_{n,\Sigma}$ is bijective, as required. This completes the proof of (i).

Since the above argument shows $A^p_\Sigma(\mathcal{K})$ and $A^\ell_\Sigma(\mathcal{K})$ for $\ell \in \mathcal{P} \setminus \{p\}$ to be the limits of the pro-discrete systems $(A^p_\Sigma(\mathcal{K})_{(n)})_n$ and $(A^\ell_\Sigma(\mathcal{K}_n))_n$, Proposition 2.3(i) reduces the proof of (ii) and (iii) to showing that the stated conditions are equivalent to the respective quantities $\mu_{R_n}(A^p_\Sigma(\mathcal{K})_{(n)})$ and $\mu_{R_n}(A^\ell_\Sigma(\mathcal{K}_n))$ being finite and bounded independently of $n$.

To prove (ii) we note that, since $A^p_\Sigma(\mathcal{K})$ is finitely-generated over $R^p$, each module $A^p_\Sigma(\mathcal{K})_{(n)}$ is finitely-generated over $R^p_{(n)} = \mathbb{Z}_p \otimes_{\mathbb{Z}} R_n$. Hence, since $\mathbb{Z}_p$ is not finitely-generated over $\mathbb{Z}$, the module $A^p_\Sigma(\mathcal{K})_{(n)}$ is finitely-generated over $R_n$ if and only if it is finite and, if this is the case, then

$$\mu_{R_n}(A^p_\Sigma(\mathcal{K})_{(n)}) = \mu_{R^p_{(n)}}(A^p_\Sigma(\mathcal{K})_{(n)}) \leq \mu_{R^p}(A^p_\Sigma(\mathcal{K})).$$

Claim (ii) is therefore true since the structure theory of finitely-generated $R^p$-modules implies that $A^p_\Sigma(\mathcal{K})_{(n)}$ is finite if and only if $A^p_\Sigma(\mathcal{K})^{\varpi_n=0}$ is finite.

To prove (iii), we note the decomposition (8) induces an isomorphism of $\mathbb{Z}[1/p]$-algebras

$$R_n[1/p] \cong R_{n-1}[1/p] \times \mathcal{O}_n[1/p], \quad x \mapsto (\varrho^n_{n-1}(x), \tau^n_n(x)),$$

and that the above argument establishes a natural isomorphism of $R_n$-modules

$$A^\ell_\Sigma(\mathcal{K}_n) \cong A^\ell_\Sigma(\mathcal{K}_{n-1}) \oplus \ker(\eta_{n,\Sigma})[\ell^\infty]. \tag{24}$$

In particular, since $R_n$ acts on $\ker(\eta_{n,\Sigma})[\ell^\infty]$ via $\tau^n_n$, these decompositions induce an isomorphism between $\ker(\eta_{n,\Sigma})[\ell^\infty]$ and $Z^\ell_n := \mathcal{O}_n \otimes_{R_n,\tau^n_n} A^\ell_\Sigma(\mathcal{K}_n)$ and hence imply that

$$\mu_{R_n}\big(A^\ell_\Sigma(\mathcal{K}_n)\big) = \max\{\mu_{R_{n-1}}\big(A^\ell_\Sigma(\mathcal{K}_{n-1})\big), \mu_{\mathcal{O}_n}\big(Z^\ell_n\big)\}.$$

Since this equality implies (by induction on $n$) that

$$\mu_{R_n}\big(A^\ell_\Sigma(\mathcal{K}_n)\big) = \max\{\mu_{\mathcal{O}_m}\big(Z^\ell_m\big) : m \in \{0\} \cup [n]\},$$

it follows that $\mu_R(A^\ell_\Sigma(\mathcal{K}))$ is finite if and only if $\mu_{\mathcal{O}_m}\big(Z^\ell_m\big)$ is bounded independently of $m$. To explicate this condition, we note the Chinese Remainder Theorem combines with the structure theorem for finitely-generated torsion $\mathcal{O}_m$-modules to imply

$$\mu_{\mathcal{O}_m}\big(Z^\ell_m\big) = d_{m,\Sigma.\ell} := \max\{d_{m,\Sigma,\mathcal{L}} : \mathcal{L} \in \mathrm{MaxSpec}_{(\ell)}(\mathcal{O}_m)\}.$$

Here $d_{m,\Sigma,\mathcal{L}}$ denotes the number of invariant factors with support $\mathcal{L}$ in the decomposition of $Z^\ell_m$, and so can be computed as

$$\begin{aligned} d_{m,\Sigma,\mathcal{L}} &= \dim_{\mathcal{O}_m/\mathcal{L}}\big((\mathcal{O}_m/\mathcal{L}) \otimes_{R_m,\tau_m^m} A_\Sigma^\ell(\mathcal{K}_m)\big) \\ &= \mathrm{r}_\ell\big(\mathcal{O}_m/\mathcal{L}\big)^{-1} \cdot \mathrm{r}_\ell\big((\mathcal{O}_m/\mathcal{L}) \otimes_{R_m,\tau_m^m} A_\Sigma^\ell(\mathcal{K}_m)\big) \\ &= (p^{m-1}(p-1)/t_m(\ell))^{-1} \cdot \mathrm{r}_\ell\big((\mathcal{O}_m/\mathcal{L}) \otimes_{R_m,\tau_m^m} A_\Sigma^\ell(\mathcal{K}_m)\big). \end{aligned} \tag{25}$$

It follows that $\mu_R(A_\Sigma^\ell(\mathcal{K}))$ is finite if and only if $d_{m,\Sigma,\mathcal{L}}$ is bounded independently of $m$ and $\mathcal{L}$ (a similar general point is observed in [3, Prop. 4.4]). Also, since $t_m(\ell) = |\mathrm{MaxSpec}_{(\ell)}(\mathcal{O}_m)|$, one has

$$d_{m,\Sigma,\ell} \cdot p^{m-1}(p-1) = t_m(\ell) d_{m,\Sigma,\ell} \cdot p^{m-1}(p-1)/t_m(\ell) \geq \mathrm{r}_\ell\big(Z_m^\ell\big) \geq d_{m,\Sigma,\ell} \cdot p^{m-1}(p-1)/t_m(\ell). \tag{26}$$

Writing $t(\ell)$ for the (finite) number of $\ell$-adic places of the field $\bigcup_{n\in\mathbb{N}} \mathbb{Q}(e^{2\pi i/p^n})$ (so that each $t_m(\ell)$ divides $t(\ell)$), one therefore has

$$\begin{aligned} \max\{d_{m,\Sigma,\ell} : 0 \leq m \leq n\} \cdot p^n &\geq d_{0,\Sigma,\ell} + \sum\nolimits_{m\in[n]} t_m(\ell) d_{m,\Sigma,\ell} \cdot p^{m-1}(p-1)/t_m(\ell) \\ &\geq \sum\nolimits_{m\in\{0\}\cup[n]} \mathrm{r}_\ell\big(Z_m^\ell\big) \\ &= \mathrm{r}_\ell(A_\Sigma(\mathcal{K}_n)) \\ &\geq d_{n,\Sigma,\ell} \cdot p^{n-1}(p-1)/t_n(\ell) \\ &\geq t(\ell)^{-1} \cdot d_{n,\Sigma,\ell} \cdot p^{n-1}(p-1). \end{aligned} \tag{27}$$

These inequalities imply $\{d_{n,\Sigma,\ell} : n \in \mathbb{N}\}$ is bounded if and only if $\mathrm{r}_\ell(A_\Sigma(\mathcal{K}_n)) = O(p^n)$ as $n \to \infty$, as required to prove (iii).

To prove (iv), it suffices to assume that $A_\Sigma(\mathcal{K})$ is finitely-generated over $R$ and deduce it is pseudo-coherent (or, equivalently, finitely $\infty$-presented). Recalling [7, Prop. 5.1(ii)], it is thus enough for us to show $A_\Sigma(\mathcal{K})$ is finitely 2-presented over $R$. As a first step, we note that, since $A_\Sigma(\mathcal{K})$ is pro-discrete, the last assertion of Proposition 2.4(ii) (with $M = A_\Sigma(\mathcal{K})$) implies it is finitely-presented if and only if the quantity

$$\mu_R(A_\Sigma(\mathcal{K})^{\varpi_n=0}) = \max\big\{\mu_R(A_\Sigma^p(\mathcal{K})^{\varpi_n=0}), \mu_R(A'_\Sigma(\mathcal{K})^{\varpi_n=0})\big\}$$

is finite and bounded independently of $n$. In this regard, we note [7, Lem. 3.7] (with $M = A'_\Sigma(\mathcal{K})$) implies the $R$-module $A'_\Sigma(\mathcal{K})^{\varpi_n=0}$ is isomorphic to $A'_\Sigma(\mathcal{K})_{(n)}$, and so $\mu_R(A'_\Sigma(\mathcal{K})^{\varpi_n=0}) \leq \mu_R(A'_\Sigma(\mathcal{K}))$. To consider $\mu_R(A_\Sigma^p(\mathcal{K})^{\varpi_n=0})$, we fix (as we may, by the structure theory of $R^p$-modules) an exact sequence of $R^p$-modules

$$0 \to M_1 \to A_\Sigma^p(\mathcal{K}) \to M_2 \oplus M_2'$$

in which $M_1$ is finite, $M_2$ is a finitely-generated free $\mathbb{Z}_p$-module and $M_2'$ is a finite direct sum of modules of the form $R^p/(p^i)$. In particular, since $(M_2')^{\varpi_n=0} = (0)$ and $A_\Sigma^p(\mathcal{K})^{\varpi_n=0}$ is finite (by (ii) and the current assumption that $A_\Sigma(\mathcal{K})$ is finitely-generated over $R$), the above sequence induces an isomorphism $M_1^{\varpi_n=0} \cong A_\Sigma^p(\mathcal{K})^{\varpi_n=0}$. Since $M_1$ is finite, it is therefore clear that the quantities $\mu_R(A_\Sigma^p(\mathcal{K})^{\varpi_n=0})$ are finite and bounded independently of $n$, as required.

We now know that $A_\Sigma(\mathcal{K})$ is finitely-presented, and so can fix an exact sequence of $R$-modules

$$R^{t_2} \xrightarrow{\psi} R^{t_1} \to A_\Sigma(\mathcal{K}) \to 0$$

(for suitable $t_1$ and $t_2$ in $\mathbb{N}$). We must prove that the finitely-generated $R$-module $\mathrm{im}(\psi)$ is finitely-presented. To do this we note that, for each $n$, the above sequence induces isomorphisms

$$(R^{t_1}/\,\mathrm{im}(\psi))^{\varpi_n=0}[p^\infty] \cong A_\Sigma(\mathcal{K})^{\varpi_n=0}[p^\infty] = A_\Sigma^p(\mathcal{K})^{\varpi_n=0}[p^\infty] \cong M_1^{\varpi_n=0},$$

with $M_1$ as fixed above. In particular, since $M_1$ is finite, we can apply the criterion of Proposition 2.4(iii) to $M = \mathrm{im}(\psi)$ in order to deduce that $\mathrm{im}(\psi)$ is finitely-presented, as required to prove (iv).

Turning to (v), we note Proposition 2.3(i) implies that the pro-discrete $R$-module $A_\Sigma(\mathcal{K}) = A^p_\Sigma(\mathcal{K}) \oplus \prod_{\ell\in\mathcal{P}\setminus\{p\}} A^\ell_\Sigma(\mathcal{K})$ is finitely-generated if and only if $\mu_R\big(A^p_\Sigma(\mathcal{K})\big)$ is finite and, in addition, the quantities $\mu_R\big(A^\ell_\Sigma(\mathcal{K})\big)$ for $\ell \in \mathcal{P}\setminus\{p\}$ are both finite and bounded independently of $\ell$. Given this, and the results of (i), (ii), (iii) and (iv), the proof of (v) is thus reduced to the following observations. Firstly, the formula (25) (and the discussion preceeding it) implies the explicit condition in (v)(b) is equivalent to requiring $\mu_R(A^\ell_\Sigma(\mathcal{K}))$ is finite and bounded independently of $\ell \in \mathcal{P}\setminus\{p\}$. Secondly, to verify the final assertion of (v), it is enough to show that if $(t_n(\ell)/p^n)\cdot \mathrm{r}_\ell(A_\Sigma(\mathcal{K}_n))$ is bounded independently of $n$ and $\ell \in \mathcal{P}\setminus\{p\}$, then so is the multiplicity $d_{n,\Sigma,\ell}$. But this follows directly from the fact that the penultimate inequality in (27), implies $(t_n(\ell)/p^n)\cdot \mathrm{r}_\ell(A_\Sigma(\mathcal{K}_n)) \geq d_{n,\Sigma,\ell}\cdot(1-1/p)$.

To prove (vi) we assume $\mathcal{K} = k^{\mathrm{con}}$. In this case, every place in $\Sigma$ has an open decomposition group in $G_{\mathcal{K}/k}$ and so, for all sufficiently large $n$, the terms $c_n$ and $m^\Sigma_n$ occurring in (22) satisfy $c_{n+1} = pc_n$ and $m^\Sigma_{n+1} = pm^\Sigma_n$. This implies, in particular, that the limit module $Q_p$ in (23) is finite. One checks similarly in this case that the limit $\varprojlim_{\alpha'_n}(\mathbb{Z}_p \otimes_{\mathbb{Z}} B_\Sigma(\mathcal{K}_n))$, and hence also its submodule $\varprojlim_{\alpha'_n}(\mathbb{Z}_p \otimes_{\mathbb{Z}} \ker(\delta^\Sigma_n))$ vanishes, and so (23) gives an exact sequence

$$0 \to A^p_\emptyset(\mathcal{K}) \to A^p_\Sigma(\mathcal{K}) \to Q_p$$

in which $Q_p$ is finite. To show that each module $A^p_\Sigma(\mathcal{K})^{\varpi_n=0}$ is finite (as required by (v)(a)), we are therefore reduced to considering the special case that $\Sigma = \emptyset$. To deal with this case, we write $\kappa$ for the constant field of $k$. Then $k$ is the function field of the curve $C_k$ over $\kappa$ and, if $J$ is the Jacobian of $C_k$, then for each $n \in \mathbb{N}_0$ one has $A_\emptyset(\mathcal{K}_n) = J(\kappa_n)$, where $\kappa_n$ is the (unique) field extension of $\kappa$ of degree $n$. It follows that $A_\emptyset(\mathcal{K}_n) = A_\emptyset(\mathcal{K}_m)^{\varpi_n=0}$ for every $m > n$, and hence that

$$A^p_\emptyset(\mathcal{K})^{\varpi_n=0} = \big(\varprojlim_{m>n} A^p_\emptyset(\mathcal{K}_n)\big)^{\varpi_n=0} = \varprojlim_{m>n} A^p_\emptyset(\mathcal{K}_m)^{\varpi_n=0} = \varprojlim_{m>n} A^p_\emptyset(\mathcal{K}_n) = (0),$$

where the last equalty is valid since all transition morphisms in the third limit are the multiplication-by-$p$ map. The condition on $A^p_\emptyset(\mathcal{K})$ stated in (v)(a) is therefore clearly satisfied.

We next verify condition (v)(b) by using the final assertion of (v). For this, we fix $\ell \in \mathcal{P}\setminus\{p\}$. Then by tensoring the upper row of (22) with $\mathbb{Z}_\ell$ one obtains an exact sequence

$$A^\ell_\emptyset(\mathcal{K}_n) \to A^\ell_\Sigma(\mathcal{K}_n) \to (\mathbb{Z}_\ell\cdot c_n)/(\mathbb{Z}_\ell\cdot m^\Sigma_n)$$

and hence an inequality

$$\mathrm{r}_\ell\big(A_\Sigma(\mathcal{K}_n)\big) \leq \mathrm{r}_\ell\big(A_\emptyset(\mathcal{K}_n)\big) + \mathrm{r}_\ell\big((\mathbb{Z}_\ell\cdot c_n)/(\mathbb{Z}_\ell\cdot m^\Sigma_n)\big) \leq \mathrm{r}_\ell\big(A_\emptyset(\mathcal{K}_n)\big) + 1.$$

In addition, writing $\mathrm{gen}(C_k)$ for the genus of $C_k$, for every $n \in \mathbb{N}$ one has

$$\mathrm{r}_\ell\big(A_\emptyset(\mathcal{K}_n)\big) = \mathrm{r}_\ell\big(J(\kappa_n)[\ell^\infty]\big) \leq 2\cdot\mathrm{gen}(C_k)$$

(by [25, Th. 11.12]), and hence also $\mathrm{r}_\ell\big(A_\Sigma(\mathcal{K}_n)\big) \leq 2\cdot\mathrm{gen}(C_k)+1$ (in fact, a similar observation about $\ell$-ranks in this context is made by Washington in [34, §II]). In particular, since $t_n(\ell) < p^n$ for every $n$, one therefore has

$$(t_n(\ell)/p^n)\cdot\mathrm{r}_\ell\big(A_\Sigma(\mathcal{K}_n)\big) < 2\cdot\mathrm{gen}(C_k)+1.$$

It follows that, in this case, the condition on $\mathrm{r}_\ell\big(A_\Sigma(\mathcal{K}_n)\big)$ stated at the end of (v) is satisfied with the universal bound taken to be $2\cdot\mathrm{gen}(C_k)+1$.

This completes the proof of Theorem 5.1. □

**Remark 5.2.** The condition that $A^p_\Sigma(\mathcal{K})^{\varpi_n=0}$ is finite for every $n$ is equivalent to requiring that the characteristic polynomial of $A^p_\Sigma(\mathcal{K})$ as a (finitely-generated, torsion) $R^p$-module is not divisible by any cyclotomic polynomial $\Phi_i(\gamma)$ (see the proof of Proposition 3.4).

**Remark 5.3.** The finite generation of $A_\Sigma(\mathcal{K})$ as an $R$-module does not imply that its submodule $\bigoplus_{\ell\in\mathcal{P}} A^\ell_\Sigma(\mathcal{K})$ is also finitely-generated. In particular, the latter module cannot be finitely-generated if $A^\ell_\Sigma(\mathcal{K})$ is non-zero for infinitely many primes $\ell$ and this is always the case for $\mathcal{K} = k^{\mathrm{con}}$ and $\Sigma = \emptyset$ (cf. Rosen [25, Cor. to Th. 11.6]).

## 6. Derived Weil-étale cohomology

In the main result (Theorem 6.5) of this section, we prove a precise link between the values at 0 of Dirichlet $L$-series and the $\mathbb{Z}[\![\mathbb{Z}_p]\!]$-module structure of the cohomology in degree 1 of the derived limits over $\mathbb{Z}_p$-extensions of the Weil-étale cohomology complexes of $\mathbb{G}_m$.

We continue to use the general notation introduced in §5 and also now assume the given finite extension $F/k$ to be Galois.

6.1. **Definitions and basic properties.** For any scheme $Y$ of finite type over the constant field $\kappa$ of $k$, we write $Y_{\mathrm{W}}$ for the Weil-étale topos on $Y$ defined by Lichtenbaum in [22, §2]. By a slight abuse of notation, we then denote by $\mathbb{G}_m$ the Weil-étale sheaf on $Y$ that is obtained by restricting the étale sheaf $\mathbb{G}_m$ on $Y$ to the corresponding Weil-étale site.

We assume $\Sigma \neq \emptyset$ and also fix a finite set of places $\Sigma'$ of $k$ disjoint from $\Sigma$. We write $C_F^\Sigma$ for the affine curve $\mathrm{Spec}(\mathcal{O}_{F,\Sigma})$ and $j_F^\Sigma : C_F^\Sigma \to C_F$ for the corresponding open immersion of $\kappa$-schemes. We identify each $w \in \Sigma'_F$ with the corresponding (closed) point of $C_F^\Sigma$, write $\kappa(w)$ for its residue field and set $Z_w := \mathrm{Spec}(\kappa(w))$. We note the associated closed immersion $i_w : Z_w \to C_F^\Sigma$ induces a canonical morphism $\mathbb{G}_m \to i_{w,*}i_w^*\mathbb{G}_m$ in $(C_F^\Sigma)_{\mathrm{W}}$ and hence also an induced morphism in $\mathsf{D}(\mathbb{Z}[G_{F/k}])$

$$\varrho_w : \mathsf{R}\Gamma((C_F^\Sigma)_{\mathrm{W}}, \mathbb{G}_m) \to \mathsf{R}\Gamma((C_F^\Sigma)_{\mathrm{W}}, i_{w,*}i_w^*\mathbb{G}_m) \cong \mathsf{R}\Gamma(\kappa(w)_{\mathrm{W}}, \mathbb{G}_m).$$

Following Kurihara et al [8, §2.2], we then define (up to isomorphism in $D(\mathbb{Z}[G_{F/k}])$) the $\Sigma'$-modified Weil-étale cohomology complex $\mathsf{R}\Gamma^{\Sigma'}((C_F^\Sigma)_{\mathrm{W}}, \mathbb{G}_m)$ of $\mathbb{G}_m$ on $(C_F^\Sigma)_{\mathrm{W}}$ via the exact triangle in $D(\mathbb{Z}[G_{F/k}])$

$$\mathsf{R}\Gamma^{\Sigma'}((C_F^\Sigma)_{\mathrm{W}}, \mathbb{G}_m) \to \mathsf{R}\Gamma((C_F^\Sigma)_{\mathrm{W}}, \mathbb{G}_m) \xrightarrow{(\varrho_w)_w} \bigoplus_{w\in\Sigma'_F} \mathsf{R}\Gamma(\kappa(w)_{\mathrm{W}}, \mathbb{G}_m) \to \cdot \tag{28}$$

In each degree $i$, we set $H^{i,\Sigma'}((C_F^\Sigma)_{\mathrm{W}}, \mathbb{G}_m) := H^i(\mathsf{R}\Gamma^{\Sigma'}((C_F^\Sigma)_{\mathrm{W}}, \mathbb{G}_m))$.

We now fix a $\mathbb{Z}_p$-extension $\mathcal{K}$ of $k$ in which only finitely many places ramify and, for $n \in \mathbb{N}_0$, write $\mathcal{K}_n$ for the unique extension of $k$ in $\mathcal{K}$ of degree $p^n$. We then impose the following conditions on the data $F, \Sigma$ and $\Sigma'$.

**Hypothesis 6.1.**

(i) *$F$ is a finite abelian extension of $k$ that is disjoint from $\mathcal{K}$. We set $H := G_{F/k}$.*
(ii) *$\Sigma$ is finite, non-empty and contains all places that ramify in $\mathcal{F} := F\cdot\mathcal{K}$ but no place that splits completely in $\mathcal{K}$.*
(iii) *$\Sigma'$ is finite and non-empty.*

In the sequel we regard $F$ (and hence $\mathcal{F}$), $\Sigma$ and $\Sigma'$ as fixed and, for each $n \in \mathbb{N}_0$, set

$$\mathcal{F}_n := F\cdot\mathcal{K}_n,\ G_n := G_{\mathcal{F}_n/k},\ U_n := \mathcal{O}^{\times,\Sigma'}_{\mathcal{F}_n,\Sigma},\ A_n := A^{\Sigma'}_\Sigma(\mathcal{F}_n),\ B_n := B^0_\Sigma(\mathcal{F}_n),\ C^\bullet_n := \mathsf{R}\Gamma^{\Sigma'}((C^\Sigma_{\mathcal{F}_n})_{\mathrm{W}}, \mathbb{G}_m).$$

Here $\mathcal{O}^{\times,\Sigma'}_{\mathcal{F}_n,\Sigma}$ denotes the (finite index) $G_n$-submodule of $\mathcal{O}^\times_{\mathcal{F}_n,\Sigma}$ comprising elements that are congruent to 1 modulo all places in $\Sigma'_{\mathcal{F}_n}$ and $A^{\Sigma'}_\Sigma(\mathcal{F}_n)$ the ray class group of $\mathcal{O}_{\mathcal{F}_n,\Sigma}$ modulo $\Sigma'_{\mathcal{F}_n}$.

We identify $G_{\mathcal{K}/k}$ with $\mathbb{Z}_p$ (as in §5). Then, for each $n \in \mathbb{N}_0$, Hypothesis 6.1(i) identifies $G_n$ with $H \times (\mathbb{Z}/p^n)$ and Hypothesis 6.1(ii) ensures $\Sigma$ contains all places ramifying in $\mathcal{F}_n/k$. It follows that each complex $C^\bullet_n$ belongs to $\mathsf{D}^{\mathrm{perf}}(\mathbb{Z}[G_n])$ and there exists a 'projection formula' isomorphism

$$\rho_n : \mathbb{Z}[G_{n-1}] \otimes^{\mathsf{L}}_{\mathbb{Z}[G_n]} C^\bullet_n \cong C^\bullet_{n-1}$$

in $\mathsf{D}^{\mathrm{perf}}(\mathbb{Z}[G_{n-1}])$. (The last two claims are proved in [6, §2.2, Lem. 1] for $\Sigma' = \emptyset$ and in [8, §2.2] for $\Sigma' \neq \emptyset$.) It is then clear that the associated family $(C_n^\bullet, \rho_n)_n$ satisfies Hypothesis 4.9 and so, recalling $\mathbb{\Lambda}$ denotes $R[H] = \varprojlim_n \mathbb{Z}[G_n]$, one obtains (via Definition 4.10) an object of $\mathsf{D}(\mathbb{\Lambda})$

$$\mathsf{R}\Gamma^{\Sigma'}((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) := \underset{\longleftarrow}{\mathsf{R}\mathrm{lim}}_{\rho_n} C_n^\bullet$$

that is well-defined up to isomorphism.

We can now define the modules that will be the primary focus of our study.

**Definition 6.2.** In each degree $i$, the 'Iwasawa Weil-étale cohomology' and 'derived Iwasawa Weil-étale cohomology' of $\mathbb{G}_m$ relative to $(\mathcal{F}, \Sigma, \Sigma')$ are the respective $\mathbb{\Lambda}$-modules

$$H^{i,\Sigma'}_{\mathrm{Iw}}((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) := \varprojlim_n H^{i,\Sigma'}((C_{\mathcal{F}_n}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) \quad \text{and} \quad H^{i,\Sigma'}_{\mathrm{dIw}}((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) := H^i(\mathsf{R}\Gamma^{\Sigma'}((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m)).$$

Here the given limit is taken with respect to the maps

$$H^{i,\Sigma'}((C_{\mathcal{F}_n}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) \xrightarrow{H^i(\rho_n')} H^i(\mathbb{Z}[G_{n-1}] \otimes^{\mathsf{L}}_{\mathbb{Z}[G_n]} C_n^\bullet) \xrightarrow{H^i(\rho_n)} H^{i,\Sigma'}((C_{\mathcal{F}_{n-1}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m),$$

with $\rho_n'$ the obvious morphism $C_n^\bullet \to \mathbb{Z}[G_{n-1}] \otimes^{\mathsf{L}}_{\mathbb{Z}[G_n]} C_n^\bullet$.

To compute these modules, we use the inverse limits

$$A(\mathcal{F}) = A_{\Sigma}^{\Sigma'}(\mathcal{F}) := \varprojlim_{\mathrm{N}_n} A_n, \;\; U(\mathcal{F}) = U_{\Sigma}^{\Sigma'}(\mathcal{F}) := \varprojlim_{\mathrm{N}_n} U_n \quad \text{and} \quad B^0(\mathcal{F}) = B^0_{\Sigma}(\mathcal{F}) := \varprojlim_{\mu_n} B^0_n.$$

Here the transition morphisms $\mu_n : B^0_n \to B^0_{n-1}$ are induced by the maps $\Sigma_{\mathcal{F}_n} \to \Sigma_{\mathcal{F}_{n-1}}$ that are obtained by restricting places of $\mathcal{F}_n$ to $\mathcal{F}_{n-1}$.

**Lemma 6.3.** *Fix* $* \in \{\mathrm{Iw}, \mathrm{dIw}\}$. *Then* $H^{i,\Sigma'}_*((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) = (0)$ *for* $i \notin \{0,1\}$ *and* $H^{0,\Sigma'}_*((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m)$ *identifies with* $U(\mathcal{F})$. *Further, there exist canonical short exact sequences of* $R$*-modules*

$$0 \to \varprojlim\nolimits^1_{\mathrm{N}_n} U_n \to H^{1,\Sigma'}_{\mathrm{dIw}}((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) \xrightarrow{\kappa'} H^{1,\Sigma'}_{\mathrm{Iw}}((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) \to 0,$$
$$0 \to A(\mathcal{F}) \xrightarrow{\kappa} H^{1,\Sigma'}_{\mathrm{Iw}}((C_{\mathcal{F}}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) \to B^0(\mathcal{F}) \to 0.$$

*Proof.* The computations of Lichtenbaum [22, Th. 6.1] (or of [6, §2.2, Lem. 1]) for $\Sigma' = \emptyset$ and of Kurihara et al [8, Rems. 2.5(iii) and 2.7] for $\Sigma' \neq \emptyset$, show that $H^{i,\Sigma'}((C_{\mathcal{F}_n}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) = (0)$ for $i \notin \{0,1\}$ and also $H^{0,\Sigma'}((C_{\mathcal{F}_n}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m) = U_n$, $H^{1,\Sigma'}((C_{\mathcal{F}_n}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m)_{\mathrm{tor}} = A_n$ and $H^{1,\Sigma'}((C_{\mathcal{F}_n}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m)_{\mathrm{tf}} = B^0_n$. Further, with respect to these identifications, the maps induced on cohomology by the composite morphism $\rho_n \circ \rho_n'$ (that occurs in Definition 6.2) agree in the first two cases with those induced by the norm $\mathrm{N}_n$ and in the third case is the restriction map $\mu_n : B^0_n \to B^0_{n-1}$.

Given these descriptions, all assertions except for the existence of the second exact sequence now follow directly from the general result of Proposition 4.11(i). For the second exact sequence one then only needs to note that, since each group $A_n$ is finite, the derived limit $\varprojlim^1_{\mathrm{N}_n} A_n$ vanishes. □

In the sequel, for each $n \in \mathbb{N}$, we use the isomorphisms (24) to identify $A_{n-1} \otimes_{\mathbb{Z}} \mathbb{Z}[1/|G_n|]$ with $\mathrm{N}_n(A_n \otimes_{\mathbb{Z}} \mathbb{Z}[1/|G_n|])$. We then assume to be given, for each $n \in \mathbb{N}_0$, a $\mathbb{\Lambda}_n$-submodule $X_n$ of $A_n \otimes_{\mathbb{Z}} \mathbb{Z}[1/|G_n|]$ with $X_{n-1} = \mathrm{N}_n(X_n)$ for $n > 0$. Then, writing $\mathrm{N}_n'$ for the map $(X_n)_{(n-1)} \to X_{n-1}$ induced by $\mathrm{N}_n$, the argument of Theorem 5.1(i) implies that $(X_n, \mathrm{N}_n')_{n \in \mathbb{N}_0}$ is a pro-discrete system and so gives rise to a pro-discrete $\mathbb{\Lambda}$-submodule $X := \varprojlim_{\mathrm{N}_n'} X_n$ of $A(\mathcal{F})$.

Since the orders of $X_n$ and $G_n$ are coprime, the canonical convergent spectral sequence

$$\prod\nolimits_{i \in \mathbb{Z}} \mathsf{Ext}^a_{\mathbb{\Lambda}_n}(H^i(X_n[-1]), H^{b+i}(C_n^\bullet)) \implies H^{a+b}(\mathsf{R}\mathrm{Hom}_{\mathbb{\Lambda}_n}(X_n[-1], C_n^\bullet)) \tag{29}$$

(cf. [33, III, 4.6.10]) collapses to imply bijectivity of the map

$$\mathsf{R}\mathrm{Hom}_{\mathsf{D}(\mathbb{\Lambda}_n)}(X_n[-1], C_n^\bullet) \xrightarrow{\phi \mapsto H^1(\phi)} \mathrm{Hom}_{\mathbb{\Lambda}_n}(X_n, H^{1,\Sigma'}((C_{\mathcal{F}_n}^{\Sigma})_{\mathrm{W}}, \mathbb{G}_m)).$$

In particular, there exists a unique morphism $\iota_{X_n} : X_n[-1] \to C^\bullet_n$ in $\mathsf{D}(\mathbb{\Lambda}_n)$ for which $H^1(\iota_{X_n})$ is the inclusion $X_n \subseteq A_n \subseteq H^{1,\Sigma'}((C^\Sigma_{\mathcal{F}_n})_{\mathrm{W}}, \mathbb{G}_m)$. Writing $C^\bullet_{X_n}$ for mapping cone of $\iota_{X_n}$, we then obtain a morphism of exact triangles in $\mathsf{D}(\mathbb{\Lambda}_{n-1})$

$$\begin{array}{ccccccc} \mathbb{\Lambda}_{n-1} \otimes^{\mathsf{L}}_{\mathbb{\Lambda}_n} X_n[-1] & \xrightarrow{1\otimes_{\mathbb{\Lambda}_n} \iota_{X_n}} & \mathbb{\Lambda}_{n-1} \otimes^{\mathsf{L}}_{\mathbb{\Lambda}} C^\bullet_n & \longrightarrow & \mathbb{\Lambda}_{n-1} \otimes^{\mathsf{L}}_{\mathbb{\Lambda}_n} C^\bullet_{X_n} & \longrightarrow & \cdot \\ \rho_{X,n}\big\downarrow & & \rho_n\big\downarrow & & \theta'_{X,n}\big\downarrow & & \\ X_{n-1}[-1] & \xrightarrow{\iota_{X_{n-1}}} & C^\bullet_{n-1} & \longrightarrow & C^\bullet_{X_{n-1}} & \longrightarrow & \cdot \end{array} \tag{30}$$

Here the upper row is obtained by applying the functor $\mathbb{\Lambda}_{n-1} \otimes^{\mathsf{L}}_{\mathbb{\Lambda}_n} -$ to the exact triangle arising from the definition of $C^\bullet_{X_n}$ as a mapping cone, the lower row is the corresponding tautological exact triangle with $n$ replaced by $n-1$ and $\rho_{X,n}$ is the isomorphism in $\mathsf{D}(\mathbb{\Lambda}_{n-1})$ induced by the isomorphism $\mathrm{N}'_n : (X_n)_{(n-1)} \cong X_{n-1}$. In particular, since $\rho_n$ is also an isomorphism in $\mathsf{D}(\mathbb{\Lambda}_{n-1})$ and the first sequre commutes, the Octahedral axiom implies the diagram can be completed to give a morphism of exact triangles in which $\rho'_{X,n}$ is an isomorphism in $\mathsf{D}(\mathbb{\Lambda}_{n-1})$. It follows that the family $(C^\bullet_{X_n}, \rho'_{X,n})_{n\in\mathbb{N}_0}$ satisfies Hypothesis 4.9 and so we obtain an object of $\mathsf{D}(\mathbb{\Lambda})$ (that is well-defined up to isomorphism) by setting

$$C^\bullet_X := \underleftarrow{\mathsf{Rlim}}_{\rho'_{X,n}} C^\bullet_{X_n}.$$

**Lemma 6.4.** *If $i \notin \{0,1\}$, then $H^i(C^\bullet_X) = (0)$. In addition, $H^0(C^\bullet_X)$ identifies with $U(\mathcal{F})$ and there exist canonical short exact sequences of $\mathbb{\Lambda}$-modules*

$$0 \to X \xrightarrow{\iota_X} H^{1,\Sigma'}_{\mathrm{dIw}}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m) \to H^1(C^\bullet_X) \to 0$$
$$0 \to \underleftarrow{\lim}^1_{\mathrm{N}_n} U_n \to H^1(C^\bullet_X) \to \underleftarrow{\lim}_n H^1(C^\bullet_{X_n}) \to 0$$
$$0 \to A(\mathcal{F})/X \xrightarrow{\subseteq} \underleftarrow{\lim}_n H^1(C^\bullet_{X_n}) \to B^0(\mathcal{F}) \to 0.$$

*Proof.* With each $\rho_{X,n}$ as in (30), the explicit definition (19) of the derived limit $\underleftarrow{\mathsf{Rlim}}_{\rho_{X,n}} X_n[-1]$ implies it identifies with the complex $X[-1]$. Hence, upon combining the morphisms of exact triangles that correspond to (30) (for each $n \in \mathbb{N}$) with the general result of Proposition 4.11(ii), one obtains an exact triangle in $\mathsf{D}(\mathbb{\Lambda})$

$$X[-1] \xrightarrow{\hat{\iota}_X} \mathsf{R}\Gamma^{\Sigma'}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m) \to C^\bullet_X \to \cdot$$

for which $\kappa' \circ H^1(\hat{\iota}_X)$ is the injective map $X \to H^{1,\Sigma'}_{\mathrm{Iw}}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m)$ induced by $\kappa$ (for the maps $\kappa$ and $\kappa'$ in Lemma 6.3). Set $\iota_X := H^1(\hat{\iota}_X)$. Then, except for the existence of the second and third claimed exact sequences, all assertions now follow by combining the long exact cohomology sequence of the above exact triangle together with the descriptions of the cohomology of $\mathsf{R}\Gamma^{\Sigma'}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m)$ given in Lemma 6.3.

Next we note that the long exact cohomology sequence of the lower row of (30) (with $n-1$ replaced by $n$) implies $C^\bullet_{X_n}$ is acyclic outside degrees 0 and 1, that $H^0(C^\bullet_{X_n}) = H^0(C^\bullet_n) = U_n$ and, after taking account of the description of $H^1(C^\bullet_n)$ given in the proof of Lemma 6.3, that there exists a canonical short exact sequence $0 \to A_n/X_n \to H^1(C^\bullet_{X_n}) \to B^0_n \to 0$.

The third claimed exact sequence is now obtained by passing to the inverse limit over $n$ of the latter sequences and noting that, since each module $A_n$ and $X_n$ is finite, exactness is preserved and $\underleftarrow{\lim}_n (A_n/X_n) = A(\mathcal{F})/X$. Finally, the existence of the second claimed exact sequence follows from the general result of Proposition 4.11(i) and the fact that $H^0(C^\bullet_n) = U_n$ for all $n$. □

6.2. **Iwasawa theory.** For each pro-discrete module $X$ as above, we now investigate the $\mathbb{\Lambda}$-modules

$$M_X := H^1(C_X^\bullet) \cong H^{1,\Sigma'}_{\mathrm{dIw}}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m)/\operatorname{im}(\iota_X) \text{ and } M'_X := H^{1,\Sigma'}_{\mathrm{Iw}}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m)/\kappa(X),$$

where the isomorphism is induced by the first exact sequence in Lemma 6.4 and the map $\kappa$ is from Lemma 6.3.

To help the reader, we recall that an ideal $I_n$ of $\mathbb{\Lambda}_n = \mathbb{Z}[G_n]$ is said to be 'invertible' if it is locally-free of rank 1 as a $\mathbb{\Lambda}_n$-module and that, for any such ideal, the set

$$I_n^{-1} := \{x \in \mathbb{Q}[G_n] : x \cdot I_n \subseteq \mathbb{\Lambda}_n\}$$

is a locally-free $\mathbb{\Lambda}_n$-module of rank 1 such that $I_n \cdot I_n^{-1} = \mathbb{\Lambda}_n$ (cf. [13, §35]).

By a slight abuse of notation, in the sequel we will not distinguish between the natural projection map $\varrho^n_{n-1} : \mathbb{\Lambda}_n \to \mathbb{\Lambda}_{n-1}$ and its $\mathbb{Q}$-linear extension $\mathbb{Q}[G_n] \to \mathbb{Q}[G_{n-1}]$.

6.2.1. *Statement of the main results.* For each $n \in \mathbb{N}_0$, we write $\theta_n = \theta^{\Sigma'}_{\mathcal{F}_n,\Sigma}$ for the $(\Sigma, \Sigma')$-relative Stickelberger element of $\mathcal{F}_n/k$. We recall $\theta_n$ is an element of $\mathbb{\Lambda}_n$ that interpolates the values at 0 of the $\Sigma'$-modified $\Sigma$-truncated Dirichlet $L$-series attached to complex characters of $G_{\mathcal{F}_n/k}$ (for details see [8, §5.1]). In particular, the inflation invariance of $L$-series implies that $\varrho^n_{n-1}(\theta_n) = \theta_{n-1}$ for all $n \in \mathbb{N}$, and so one obtains a well-defined Iwasawa-theoretic Stickelberger element by setting

$$\theta_{\mathcal{F}} = \theta^{\Sigma'}_{\mathcal{F},\Sigma} := (\theta_n)_{n\in\mathbb{N}} \in \varprojlim_{\varrho^n_{n-1}} \mathbb{\Lambda}_n = \mathbb{\Lambda}.$$

Our main result will explicitly link this element to the structures of the $\mathbb{\Lambda}$-modules $M_X$. To state it, we note that, since each $X_n$ is finite and of order prime to $|G_n|$, the $\mathbb{\Lambda}_n$-ideal $\mathrm{Fitt}^0_{\mathbb{\Lambda}_n}(X_n)$ is invertible. We further recall that, since $(X_n, \mathrm{N}'_n)_n$ is a pro-discrete system, the argument of Proposition 4.3(ii) implies $\varrho^n_{n-1}(\mathrm{Fitt}^0_{\mathbb{\Lambda}_n}(X_n)) = \mathrm{Fitt}^0_{\mathbb{\Lambda}_{n-1}}(X_{n-1})$ for all $n \in \mathbb{N}$. In particular, each of the families

$$\big(\mathrm{Fitt}^0_{\mathbb{\Lambda}_n}(X_n)^{-1}, \varrho^n_{n-1}\big)_{n\in\mathbb{N}}, \ \big(\theta_n \cdot \mathrm{Fitt}^0_{\mathbb{\Lambda}_n}(X_n)^{-1}, \varrho^n_{n-1}\big)_{n\in\mathbb{N}} \text{ and } \big(\mathrm{Ann}_{\mathbb{\Lambda}_n}(\theta_n), \varrho^n_{n-1}\big)_{n\in\mathbb{N}}$$

is an inverse system.

**Theorem 6.5.** *Assume Hypothesis 6.1 and also that $M'_X$ is finitely-generated over $R$. Then $M_X$ is finitely-presented over $\mathbb{\Lambda}$ and there exists a canonical exact sequence of $\mathbb{\Lambda}$-modules*

$$0 \to \mathrm{Fit}^0_{\mathbb{\Lambda}}(M_X) \xrightarrow{\subseteq} \varprojlim_{\varrho^n_{n-1}} (\theta_n \cdot \mathrm{Fit}^0_{\mathbb{\Lambda}_n}(X_n)^{-1}) \to \varprojlim^1_{\nu_n} \mathcal{T}_n(M_X) \to 0,$$

*where the inverse system $(\mathcal{T}_n(M_X), \nu_n)_n$ is as defined in Proposition 4.5(iii). In particular, one has*

$$(\theta_{\mathcal{F}})^2 \in \mathrm{Fit}^0_{\mathbb{\Lambda}}(M_X) \tag{31}$$

*and so $M_X$ is torsion over $\mathbb{\Lambda}$. In addition, the module $\varprojlim^1_{\nu_n} \mathcal{T}_n(M_X)$ is divisible and $p$-torsion-free and vanishes only if both $|\Sigma| = 1$ and the unique place in $\Sigma$ has full decomposition group in $G_{\mathcal{F}/k}$.*

This result is proved in §6.2.2 and the assumed finite-generation of $H^{1,\Sigma'}_{\mathrm{Iw}}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m)/\kappa(X)$ is then discussed in §6.2.4.

We next record a consequence of the proof of Theorem 6.5 concerning finite-presentability of the module $M'_X$. In particular, it shows that, even if $M'_X$ is finitely-generated over $R$, it is very unlikely to be finitely-presented and so cannot be used in any general main-conjecture-type formalism relative to $R$. (In fact, this observation motivated our introduction of derived Weil-étale cohomology.)

**Corollary 6.6.** *Assume Hypothesis 6.1 and also that $M'_X$ is finitely-generated over $R$. Then the following conditions are equivalent.*

(i) *$M'_X$ is finitely-presented over $R$.*
(ii) *The canonical map $H^{1,\Sigma'}_{\mathrm{dIw}}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m) \to H^{1,\Sigma'}_{\mathrm{Iw}}((C^\Sigma_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m)$ is an isomorphism.*
(iii) *The inverse system $(U_n, \mathrm{N}_n)_n$ is Mittag-Leffler.*

(iv) $|\Sigma| = 1$ *(so that $\mathcal{F}/k$ is ramified at at most one place) and the unique place in $\Sigma$ has full decomposition subgroup in $G_{\mathcal{F}/k}$.*

This result is proved in §6.2.3.

6.2.2. *The proof of Theorem 6.5.* As a first step, we show that the family $(C^\bullet_{X_n})_n$ satisfies the conditions of Proposition 4.11(iv) with $a = 0$.

The validity of condition (a) in this case follows by combining the exact triangle given by the lower row of (30) with the facts $C^\bullet_{n-1}$ belongs to $\mathsf{D}^{\mathrm{perf}}(\mathbb{A}_{n-1})$ and $X_{n-1}$ is a finite cohomologically-trivial $G_{n-1}$-module (as its order is prime to $|G_{n-1}|$). To check the remaining conditions, we use the explicit descriptions of the groups $H^i(C^\bullet_{X_n})$ given in the proof of Lemma 6.4. In particular, to verify (b), we need to show $U_n$ is torsion-free. However, if $u \in U_n \setminus \{1\}$ satisfies $u^m = 1$ for some $m \in \mathbb{N}$, then, modulo any place in $\Sigma'$, there is a congruence

$$m \equiv \sum\nolimits_{a=0}^{a=m-1} u^a = (u^m - 1)/(u - 1) = 0$$

which contradicts the fact that $m$ is prime to the characteristic $p$ of $\mathcal{F}_n$. Similarly, (c) is valid since the Riemann-Roch Theorem implies the map $U_n \to B^0_n$ sending $u$ to $(\mathrm{ord}_w(u)\,\mathrm{d}_{\mathcal{F}_n}(w))_{w\in\Sigma_{\mathcal{F}_n}}$ induces a composite isomorphism of $\mathbb{Q}[G_n]$-modules

$$\mathbb{Q} \otimes_{\mathbb{Z}} H^0(C^\bullet_{X_n}) = \mathbb{Q} \otimes_{\mathbb{Z}} U_n \cong \mathbb{Q} \otimes_{\mathbb{Z}} B^0_n \cong \mathbb{Q} \otimes_{\mathbb{Z}} H^1(C^\bullet_{X_n}) \tag{32}$$

(in which the second isomorphism is induced by the fact $A_n/X_n$ is finite). Finally, we note that the long exact cohomology sequence of the exact triangle in the lower row of (30) (with $n-1$ replaced by $n$) gives an exact sequence of $\mathbb{A}_n$-modules $0 \to X_n \to H^{1,\Sigma'}((C^\Sigma_{\mathcal{F}_n})_{\mathrm{W}}, \mathbb{G}_m) \to H^1(C^\bullet_{X_n}) \to 0$ that is compatible with the natural transition morphisms as $n$ varies. Hence, since each $X_n$ is finite, by taking the inverse limit of these sequences one derives an isomorphism of $\mathbb{A}$-modules

$$\varprojlim\nolimits_n H^1(C^\bullet_{X_n}) \cong M'_X. \tag{33}$$

In particular, the given assumptions imply that $\varprojlim_n H^1(C^\bullet_{X_n})$ is finitely-generated over $R$, and thereby verify (d).

Having verified all hypotheses of Proposition 4.11(iv), we can therefore now deduce from its conclusion that $M_X$ is a finitely-presented $\mathbb{A}$-module that verifies the hypotheses of Proposition 4.5 and, in addition, that the natural map

$$(M_X)_{(n)} \to H^1(C^\bullet_{X_n}) = H^{1,\Sigma'}((C^\Sigma_{\mathcal{F}_n})_{\mathrm{W}}, \mathbb{G}_m)/X_n \tag{34}$$

is bijective. From Proposition 4.5, we can therefore deduce the existence of a canonical short exact sequence of $\mathbb{A}$-modules

$$0 \to \mathrm{Fit}^0_{\mathbb{A}}(M_X) \xrightarrow{\subseteq} \varprojlim\nolimits_{\varrho^n_{n-1}} \mathrm{Fit}^0_{\mathbb{A}_n}\big(H^{1,\Sigma'}((C^\Sigma_{\mathcal{F}_n})_{\mathrm{W}}, \mathbb{G}_m)/X_n\big) \to \varprojlim\nolimits^1_{\nu_n} \mathcal{T}_n(M_X) \to 0.$$

Given this, the displayed exact sequence in Theorem 6.5 follows directly from the second equality in claim (i) of the following result.

**Proposition 6.7.** *For each $n \in \mathbb{N}$, the following claims are valid.*

(i) $\mathrm{Fit}^0_{\mathbb{A}_n}\big((M_X)_{(n)}\big) = \mathrm{Fit}^0_{\mathbb{A}_n}\big(H^{1,\Sigma'}((C^\Sigma_{\mathcal{F}_n})_{\mathrm{W}}, \mathbb{G}_m)/X_n\big) = \theta_n \cdot \mathrm{Fitt}^0_{\mathbb{A}_n}(X_n)^{-1}$.

(ii) $\mathrm{Ann}_{\mathbb{A}_n}\big(\theta_n \cdot \mathrm{Fitt}^0_{\mathbb{A}_n}(X_n)^{-1}\big) = \mathrm{Ann}_{\mathbb{A}_n}(\theta_n)$.

(iii) *The $\mathbb{A}_n$-lattice $\mathcal{T}_n(M_X)$ has the same genus as $\mathrm{Ann}_{\mathbb{A}_n}(\theta_n)$ and is annihilated by $\theta_{\mathcal{F}}$.*

*Proof.* The first equality in (i) follows directly from the isomorphism (34). To prove the second equality we note that, since $X_n$ is finite and of order prime to $|G_n|$, one has

$$\mathrm{Fit}^0_{\mathbb{A}_n}\big(H^{1,\Sigma'}((C^\Sigma_{\mathcal{F}_n})_{\mathrm{W}}, \mathbb{G}_m)\big) = \mathrm{Fit}^0_{\mathbb{A}_n}\big(H^{1,\Sigma'}((C^\Sigma_{\mathcal{F}_n})_{\mathrm{W}}, \mathbb{G}_m)/X_n\big) \cdot \mathrm{Fit}^0_{\mathbb{A}_n}\big(X_n\big).$$

We are thereby reduced to recalling the key equality

$$\mathrm{Fit}^0_{\mathbb{A}_n}\big(H^{1,\Sigma'}((C^{\Sigma}_{\mathcal{F}_n})_{\mathrm{W}},\mathbb{G}_m)\big)=\theta_n\cdot\mathbb{A}_n. \tag{35}$$

Indeed, this equality follows directly by combining the results of Kurihara et al in [8, Th. 1.5(i) and Prop. 3.4] with the data $(K/k,S,T,V,r)$ taken to be $(\mathcal{F}_n/k,\Sigma,\Sigma',\emptyset,0)$ together with the observations in [8, Rem. 3.3(ii) and Rem. 5.3(i)].

In order to prove (ii) and (iii), we recall that if $Y$ and $Y'$ are any $\mathbb{A}_n$-lattices that have the same genus, then $\mathrm{Ann}_{\mathbb{A}_n}(Y)=\mathrm{Ann}_{\mathbb{A}_n}(Y')$.

In particular, since $\theta_n\cdot\mathrm{Fitt}^0_{\mathbb{A}_n}(X_n)^{-1}$ has the same genus as $\theta_n\cdot\mathbb{A}_n$ (because $\mathrm{Fitt}^0_{\mathbb{A}_n}(X_n)^{-1}$ is an invertible ideal), this fact directly implies the equality in (ii). In addition, since the first assertion of (iii) follows upon combining the equalities in (i) and (ii) with the result of Proposition 4.5(iii)(a), it also implies that $\mathrm{Ann}_{\mathbb{A}_n}(\mathcal{T}_n(M_X))$ is equal to $\mathrm{Ann}_{\mathbb{A}_n}\big(\mathrm{Ann}_{\mathbb{A}_n}(\theta_n)\big)$. Thus, since $\theta_{\mathcal{F}}$ acts on any element of $\mathbb{A}_n$ via the element $\varrho_n(\theta_{\mathcal{F}})=\theta_n\in\mathrm{Ann}_{\mathbb{A}_n}\big(\mathrm{Ann}_{\mathbb{A}_n}(\theta_n)\big)$, this verifies the second assertion of (iii). □

Next we note Proposition 6.7(i) implies that the limit $\varprojlim_{\varrho^n_{n-1}}(\theta_n\cdot\mathrm{Fit}^0_{\mathbb{A}_n}(X_n)^{-1})$ is an ideal of $\mathbb{A}=\varprojlim_{\varrho^n_{n-1}}\mathbb{A}_n$ and that $\theta_{\mathcal{F}}\cdot\varprojlim_{\varrho^n_{n-1}}\mathrm{Fit}^0_{\mathbb{A}_n}(X_n)^{-1}$ identifies naturally with a submodule of it. We can therefore derive the containment (31) by noting that

$$(\theta_{\mathcal{F}})^2\in\theta_{\mathcal{F}}\cdot(\theta_{\mathcal{F}}\cdot\varprojlim_{\varrho^n_{n-1}}\mathrm{Fit}^0_{\mathbb{A}_n}(X_n)^{-1})\subseteq\theta_{\mathcal{F}}\cdot\varprojlim_{\varrho^n_{n-1}}(\theta_n\cdot\mathrm{Fit}^0_{\mathbb{A}_n}(X_n)^{-1})\subseteq\mathrm{Fit}^0_{\mathbb{A}}\big(M_X\big).$$

Here the containment is valid since $\mathbb{A}\subseteq\varprojlim_{\varrho^n_{n-1}}\mathrm{Fit}^0_{\mathbb{A}_n}(X_n)^{-1}$ and the second inclusion follows from the exact sequence of Theorem 6.5 and the fact that Proposition 6.7(iii) implies $\theta_{\mathcal{F}}$ anihilates the $\mathbb{A}$-module $\varprojlim^1_{\nu_n}\mathcal{T}_n(M_X)$.

Having verified (31), all of the remaining assertions of Theorem 6.5 are now direct consequences of the following result.

**Lemma 6.8.** *The following claims are valid.*

(i) *$\theta_{\mathcal{F}}$ is a non-zero divisor of $\mathbb{A}$.*

(ii) *Let $Y$ denote either $\varprojlim^1_{\varrho^n_{n-1}}\mathrm{Ann}_{\mathbb{A}_n}(\theta_n)$ or $\varprojlim^1_{\nu_n}\mathcal{T}_n(M_X)$. Then $Y$ is divisible, $p$-torsion-free and fixed by an open subgroup of $G_{\mathcal{F}/F}$. In addition, $Y=(0)$ if and only if $|\Sigma|=1$ and the unique place in $\Sigma$ has full decomposition subgroup in $G_{\mathcal{F}/k}$.*

*Proof.* Set $Y_n:=\mathrm{Ann}_{\mathbb{A}_n}(\theta_n)$ and $Y'_n:=\mathcal{T}_n(M_X)$. Then to prove both (i) and the first assertion of (ii), we are reduced, by Lemma 4.7, to proving $\varprojlim_{\varrho^n_{n-1}}\big(\mathbb{Z}_p\otimes_{\mathbb{Z}}Y_n\big)$ and $\varprojlim_{\nu_n}\big(\mathbb{Z}_p\otimes_{\mathbb{Z}}Y'_n\big)$ vanish. To show this, we note that, for each $n\in\mathbb{N}$ and $\psi$ in $\mathrm{Hom}(G_n,\mathbb{C}^\times)$, there are equivalences

$$\begin{aligned}\dim_{\mathbb{C}}\big(e_\psi(\mathbb{C}\otimes_{\mathbb{Z}}Y'_n)\big)>0 &\iff \dim_{\mathbb{C}}\big(e_\psi(\mathbb{C}\otimes_{\mathbb{Z}}Y_n)\big)>0\\ &\iff e_\psi(\theta_n)=0\\ &\iff \dim_{\mathbb{C}}\big(e_\psi(\mathbb{C}\otimes_{\mathbb{Z}}B^0_n))\big)>0.\end{aligned} \tag{36}$$

Here the first equivalence is true since $Y_n$ and $Y'_n$ lie in the same $\mathbb{A}_n$-genus (and so span isomorphic $\mathbb{C}[G_n]$-modules), the second is obvious and the third follows from a comparison of the order of vanishing formula for Artin $L$-series proved by Tate in [30, Chap. I, Prop. 3.4] with the definition of Stickelberger elements.

Given the definition of $B^0_n$, the above equivalences imply that the lattices $Y_n$ and $Y'_n$ are only supported on characters of $G_n$ that are trivial on the decomposition subgroup of some place in $\Sigma$. In addition, (the final assertion of) Hypothesis 6.1(ii) implies the existence of a natural number $n_0$ such that the decomposition subgroup in $G_{\mathcal{F}/k}$ of every place in $\Sigma$ contains $G(n_0):=G_{\mathcal{F}/\mathcal{F}_{n_0}}$.

Hence, for all $m > n \geq n_0$, the equalities in Proposition 6.7(i) and (ii) combine with the argument of Proposition 4.5(iii) to imply the existence of a commutative diagram

$$\begin{array}{ccccc} \mathbb{Z}_p \otimes_{\mathbb{Z}} Y'_m & \xrightarrow{\sim} & \mathbb{Z}_p \otimes_{\mathbb{Z}} Y_m & \xrightarrow{\subseteq} & H^0(G(n_0), \mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{\Lambda}_m) \\ {\scriptstyle \nu_n^m}\downarrow & & {\scriptstyle \varrho_n^m}\downarrow & & {\scriptstyle \varrho_n^m}\downarrow \\ \mathbb{Z}_p \otimes_{\mathbb{Z}} Y'_n & \xrightarrow{\sim} & \mathbb{Z}_p \otimes_{\mathbb{Z}} Y_n & \xrightarrow{\subseteq} & H^0(G(n_0), \mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{\Lambda}_n) \end{array}$$

with $\nu_n^m := \nu_m \circ \nu_{m-1} \circ \cdots \circ \nu_{n+1}$. In particular, since $\varrho_n^m\big(H^0(G(n_0), \mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{\Lambda}_m)\big) \subseteq p^{m-n}(\mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{\Lambda}_n)$, upon passing to the limit (over $m > n$) of these diagrams, one deduces that $\varprojlim_{\nu_n}\big(\mathbb{Z}_p \otimes_{\mathbb{Z}} Y'_n\big)$ and $\varprojlim_{\varrho_{n-1}^n}\big(\mathbb{Z}_p \otimes_{\mathbb{Z}} Y_n\big)$ both vanish, as required.

We note next that the vanishing of $\varprojlim_{\nu_n}\big(\mathbb{Z}_p \otimes_{\mathbb{Z}} Y'_n\big)$ and $\varprojlim_{\varrho_{n-1}^n}\big(\mathbb{Z}_p \otimes_{\mathbb{Z}} Y_n\big)$ combines with the respective case of the exact sequence (18) to induce isomorphisms

$$\varprojlim\nolimits^1_{\nu_n} Y'_n \cong \varprojlim_{\nu_n}\big((\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} Y'_n\big) \text{ and } \varprojlim\nolimits^1_{\varrho_{n-1}^n} Y_n \cong \varprojlim_{\varrho_{n-1}^n}\big((\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} Y_n\big).$$

In addition, since $\mathbb{Z}_p/\mathbb{Z}$ is uniquely $p$-divisible, the above argument implies, for each $m' > m \geq n_0$, that the maps $(\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} Y'_{m'} \to (\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} Y'_m$ and $(\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} Y_{m'} \to (\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} Y_m$ induced by $\nu_m^{m'}$ and $\varrho_m^{m'}$ are bijective, thereby identifying $\varprojlim^1_{\nu_n} Y'_n$ and $\varprojlim^1_{\varrho_{n-1}^n} Y_n$ with $(\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} Y'_m$ and $(\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} Y_m$. It follows that $G(n_0)$ acts trivially on both $\varprojlim^1_{\nu_n} Y'_n$ and $\varprojlim^1_{\varrho_{n-1}^n} Y_n$ and further that these modules vanish if and only if $Y'_m$, respectively $Y_m$, vanishes for every $m \geq n_0$. In addition, by (36), one knows that the latter condition is satisfied if and only if $B^0_m$ vanishes for every $m \geq n_0$ and it is easily checked this occurs if and only if both $|\Sigma| = 1$ and the unique place in $\Sigma$ has full decomposition subgroup in $G_{\mathcal{F}/k}$. This completes the proof of (ii). □

**Remark 6.9.** Set $I_n := \mathrm{Fit}^0_{\mathbb{\Lambda}_n}(X_n)$ for each $n \in \mathbb{N}$ and also $J := \varprojlim_{\varrho_{n-1}^n} I_n^{-1}$. Then the inclusion of $\mathbb{\Lambda}$-ideals $\iota_{\mathcal{F}} : \theta_{\mathcal{F}} \cdot J \subseteq \varprojlim_{\varrho_{n-1}^n}(\theta_n \cdot I_n^{-1})$ used in the derivation of (31) is usually strict. To compute its cokernel, one can proceed as follows. Since each $\mathbb{\Lambda}_n$-module $I_n^{-1}$ is locally-free of rank 1, the equalities $\varrho_{n-1}^n(I_n) = I_{n-1}$ noted just before Theorem 6.5 can be used to show that $J$, and hence also (following Lemma 6.8(i)) $\theta_{\mathcal{F}} \cdot J$, is a finitely generated pro-discrete $\mathbb{\Lambda}$-module. In particular, the quotient module $M := \mathbb{\Lambda}/(\theta_{\mathcal{F}} \cdot J)$ has a presentation (13) in which $P = \theta_{\mathcal{F}} \cdot J$, $d = 1$ and every $\mathbb{\Lambda}_n$-module $(\theta_{\mathcal{F}} \cdot J)_{(n)} \cong J_{(n)}$ is isomorphic (via the final assertion of Proposition 2.4(i)) to $I_n^{-1}$ and so is locally-free of rank 1. By the same argument as in Remark 4.6, these facts can be combined to show that $\mathrm{cok}(\iota_{\mathcal{F}})$ is isomorphic, as a $\mathbb{\Lambda}$-module, to the derived limit $\varprojlim^1_{\nu_n} \mathcal{T}_n(M)$.

6.2.3. *The proof of Corollary 6.6.* Upon combining the second exact sequence in Lemma 6.4 with the isomorphism (33), one obtains an exact sequence of $R$-modules

$$0 \to \varprojlim\nolimits^1_{\mathrm{N}_n} U_n \to M_X \to M'_X \to 0.$$

In particular, since Theorem 6.5 implies (under the stated hypotheses) that the central module in this sequence is finitely-presented, the general results of [16, Th. 2.1.2, (2) and (3)] imply that the $R$-module $M'_X$ is finitely-presented if and only if the $R$-module $\varprojlim^1_{\mathrm{N}_n} U_n$ is finitely-generated.

Next we note that, with $n_0$ fixed as in the proof of Lemma 6.8, for all $n \in \mathbb{N}$ the action of $R$ on the module $B^0_n$ factors through the projection $R \to R_{n_0}$. Since each group $U_n$ is torsion-free, the central isomorphism in (32) then implies that the same is true of the action of $R$ on each group $U_n$. This in turn implies that the action of $R$ on $\varprojlim^1_{\mathrm{N}_n} U_n$ factors through $R_{n_0}$ and so, since $R_{n_0}$ is a finitely-generated abelian group and $\varprojlim^1_{\mathrm{N}_n} U_n$ is divisible (by Lemma 4.7), we deduce that the derived limit $\varprojlim^1_{\mathrm{N}_n} U_n$ is finitely-generated over $R$ (and hence over $R_{n_0}$) if and only if it vanishes.

Now, since each group $U_n$ is countable, one has $\varprojlim^1_{\mathrm{N}_n} U_n = (0)$ if and only if the inverse system $(U_n, \mathrm{N}_n)_n$ is Mittag-Leffler (cf. [18, p. 242, Prop.]). In addition, the first exact sequence in Lemma 6.3 implies $\varprojlim^1_{\mathrm{N}_n} U_n$ vanishes if and only if the natural map $H^{1,\Sigma'}_{\mathrm{dIw}}((C^{\Sigma}_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m) \to H^{1,\Sigma'}_{\mathrm{Iw}}((C^{\Sigma}_{\mathcal{F}})_{\mathrm{W}}, \mathbb{G}_m)$ is bijective.

Finally, as the action of $R$ on each $U_n$ factors through $R_{n_0}$, one has $U_n = U_{n_0}$ for all $n \geq n_0$. This in turn implies that the limit $\varprojlim_{\mathrm{N}_n}(\mathbb{Z}_p \otimes_{\mathbb{Z}} U_n)$ vanishes and then, just as in the proof of Lemma 6.8, the argument of Lemma 4.7 implies $\varprojlim^1_{\mathrm{N}_n} U_n$ vanishes if and only if $U_a$ vanishes for all $a \geq n_0$. This is equivalent, by (32), to requiring that $B^0_a$ vanishes for all $a \geq n_0$ and this occurs if and only if $|\Sigma| = 1$ and the unique place in $\Sigma$ has full decomposition group in $G_{\mathcal{F}/k}$.

This completes the proof of Corollary 6.6.

6.2.4. *Finite-generation.* Since $R$ is not Noetherian, the $R$-module $M'_X$ can be finitely-generated even if its submodule $A(\mathcal{F})/X$ is not. However, since $B^0(\mathcal{F})$ is finitely-generated over $R$ (as every place in $\Sigma$ has an open decomposition group in $G_{\mathcal{F}/k}$), the second exact sequence in Lemma 6.3 implies that $M'_X$ is finitely-generated whenever $A(\mathcal{F})/X$ is.

To analyse the latter condition, we write $A^{\ell}(\mathcal{F})$ for each prime $\ell$ for the maximal pro-$\ell$ subgroup $\varprojlim_{\mathrm{N}_n} A_n[\ell^{\infty}]$ of $A(\mathcal{F})$. We then note that, since each $X_n$ is finite of order prime-to-$p$, the $R$-module $A^p(\mathcal{F})$ is a direct summand of $A(\mathcal{F})/X$. In particular, if $A(\mathcal{F})/X$ is finitely-generated over $R$, then so is $A^p(\mathcal{F})$ and Remark 5.2 therefore implies the following restriction on characteristic polynomials

$$\mathrm{char}_{R^p}(A^p(\mathcal{F})) \text{ is not divisible by } \Phi_i(\gamma) \text{ for any } i \in \mathbb{N}_0. \tag{37}$$

We shall therefore henceforth assume this condition to be satisfied.

Next we note that, for each place $w$ in $\Sigma'(n) := \Sigma'_{\mathcal{F}_n}$, the complex $\mathsf{R}\Gamma(\kappa(w)_{\mathrm{W}}, \mathbb{G}_m)$ is canonically isomorphic to $\kappa(w)^{\times}[0]$ and so the long exact sequence of cohomology of (28) gives an exact sequence

$$\bigoplus\nolimits_{w \in \Sigma'(n)} \kappa(w)^{\times} \to A^{\Sigma'}_{\Sigma}(\mathcal{F}_n) \to A_{\Sigma}(\mathcal{F}_n) \to 0.$$

As $n$ increases, these sequences are compatible with respect to maps induced by the norm $\mathrm{N}_n$. In addition, since the first term in the sequence is finite, exactness is preserved when passing to the limit over $n$ and so, setting $T_{\Sigma'}(\mathcal{F}) := \varprojlim_{\mathrm{N}_n}\big(\bigoplus_{w \in \Sigma'(n)} \kappa(w)^{\times}\big)$, one obtains a canonical exact sequence of $R$-modules

$$T_{\Sigma'}(\mathcal{F}) \to A(\mathcal{F}) \to A_{\Sigma}(\mathcal{F}) \to 0. \tag{38}$$

**Lemma 6.10.** $\mu_R\big(T_{\Sigma'}(\mathcal{F})\big) \leq |\Sigma'| \cdot |G_{F/k}|$.

*Proof.* We set $\Delta := G_{\mathcal{F}/F}$ and $\Delta_n := G_{\mathcal{F}_n/F}$ for $n \in \mathbb{N}$, and note that (under Hypothesis 6.1(i)) the restriction map $G_{\mathcal{F}/k} \to G_{\mathcal{K}/k}$ induces an identification of $\mathbb{Z}[\![\Delta]\!] = \varprojlim_{\epsilon^n_{n-1}} \mathbb{Z}[\Delta_n]$ with $R$, where we write $\epsilon^n_{n-1}$ for the natural projection $\mathbb{Z}[\Delta_n] \to \mathbb{Z}[\Delta_{n-1}]$.

Via the $R$-module decomposition $T_{\Sigma'}(\mathcal{F}) = \bigoplus_{v \in \Sigma'} T_{\{v\}}(\mathcal{F})$, we may assume $|\Sigma'| = 1$. In this case, we fix $v' \in \Sigma'_F$ and write $T(v')$ for $\varprojlim_{\mathrm{N}_n}\big(\bigoplus_{w \in \{v'\}_{\mathcal{F}_n}} \kappa(w)^{\times}\big)$, so that $T_{\Sigma'}(\mathcal{F})$ is isomorphic to $\mathbb{Z}[H] \otimes_{\mathbb{Z}[H(v')]} T(v')$, with $H(v')$ the decomposition subgroup of $v'$ in $H (= G_{F/k})$.

We then assume, firstly, that $v'$ splits completely in $\mathcal{F}$ so that there are isomorphisms

$$T(v') \cong \varprojlim_{\epsilon^n_{n-1} \otimes \mathrm{id}} (\mathbb{Z}[\Delta_n] \otimes_{\mathbb{Z}} \kappa(v')^{\times}) \cong R \otimes_{\mathbb{Z}} \kappa(v')^{\times}.$$

In particular, in this case, if $x$ is any generator of the (cyclic) group $\kappa(v')^{\times}$, then $T(v')$ is generated over $R$ by $1 \otimes x$ and so $\mu_R(T_{\Sigma'}(\mathcal{F})) = [H : H(v')] \leq |H|$.

We are thereby reduced to considering the case that the decomposition subgroup $\Delta(v')$ of $v'$ in $\Delta$ is non-trivial (and hence open) so that $\mathcal{F}^{\Delta(v')} = \mathcal{F}_{n_1}$ for some $n_1 \in \mathbb{N}$. We then fix $w \in \{v'\}_{\mathcal{F}_{n_1}}$

and for $n \geq n_1$ write $w_n$ for the unique place of $\mathcal{F}_n$ above $w$ so that one has an isomorphism of $R$-modules

$$T(v') \cong R \otimes_{\mathbb{Z}[\![\Delta(v')]\!]} \varprojlim_{n \geq n_1} \kappa(w_n)^\times,$$

in which the limit is taken with respect to the maps $\mathrm{N}_n : \kappa(w_n)^\times \to \kappa(w_{n-1})^\times$. To prove the claimed inequality $\mu_R(T_{\Sigma'}(\mathcal{F})) \leq |H|$ it is therefore enough to show $\varprojlim_{n \geq n_0} \kappa(w_n)^\times$ is a cyclic $\mathbb{Z}[\![\Delta(v')]\!]$-module. In addition, this is clear since for $n > n_1$, the map $\mathrm{N}_n$ acts by raising each element of $\kappa(w_n)^\times$ to its $(|\kappa(w_n)^\times|/|\kappa(w_{n-1})^\times|)$-th power and so any generating element of the (cyclic) group $\kappa(w_{n-1})^\times$ has a pre-image under $\mathrm{N}_n$ that is a generator of $\kappa(w_n)^\times$. □

For each $b \in \mathbb{N}$, we now define a (possibly infinite) set of rational primes

$$V_b(\mathcal{F}) := \{q \in \mathcal{P} \setminus \{p\} : \exists\, m \in \mathbb{N}\,,\, \mathrm{r}_q(A_\Sigma(\mathcal{F}_m)) > (b/t_m(q))p^m\}.$$

We then assume that, for some fixed $b$ and every $n \in \mathbb{N}_0$, one has

$$\bigoplus_{q \in V_b(\mathcal{F})} (\mathbb{Z}_q \otimes_{\mathbb{Z}} A_\Sigma^{\Sigma'}(\mathcal{F}_n)) \subseteq X_n.$$

We write $\overline{X_n}$ and $\overline{X}$ for the images of $X_n$ and $X$ under the canonical surjective homomorphisms $A_\Sigma^{\Sigma'}(\mathcal{F}_n) \to A_\Sigma(\mathcal{F}_n)$ and $A_\Sigma^{\Sigma'}(\mathcal{F}) \to A_\Sigma(\mathcal{F})$.

Then the explicit definition of $V_b(\mathcal{F})$ ensures that $(t_n(\ell)/p^n) \cdot \mathrm{r}_\ell(A_\Sigma(\mathcal{F}_n)/\overline{X_n}) \leq b$ for every $n \in \mathbb{N}_0$ and $\ell \in \mathcal{P} \setminus \{p\}$. Upon combining this fact with the assumption (37) and the argument of Theorem 5.1(v), we may therefore deduce that $A_\Sigma(\mathcal{F})/\overline{X}$ is finitely-generated over $R$. In view of Lemma 6.10 and the exact sequence (38), this then implies that the module $A(\mathcal{F})/X$ is finitely-generated over $R$, as desired.

In this regard, we also note that, since the number of automorphisms of a finite abelian $\ell$-group grows exponentially with its $\ell$-rank, Cohen-Lenstra-Martinet-type heuristics make it very likely that, for large $b$, the set $V_b(\mathcal{F})$ is comparatively small.

For example, if $\mathcal{F} = F^{\mathrm{con}}$, then the proof of Theorem 5.1(vi) shows that $V_b(F^{\mathrm{con}}) = \emptyset$ for every $b \geq 2 \cdot \mathrm{gen}(C_F) + 1$ and so the above argument can be applied with $X = (0)$. We shall discuss this case in detail in the next section.

## 7. Constant extensions

In this section, we fix data $(F/k, \Sigma, \Sigma')$ that satisfies Hypothesis 6.1 with respect to $\mathcal{K} := k^{\mathrm{con}}$. (In particular, since no place of $k$ either ramifies or splits completely in $k^{\mathrm{con}}$, Hypothesis 6.1(ii) now simplifies to requiring that $\Sigma$ is finite, non-empty and contains all places that ramify in $F$). We set

$$H := G_{F/k},\ \mathbb{A} := R[H] \cong \mathbb{Z}[\![\mathbb{Z}_p \times H]\!] \text{ and } \mathbb{A}_n := R_n[H] \cong \mathbb{Z}[(\mathbb{Z}/p^n) \times H] \text{ for } n \in \mathbb{N}_0.$$

For an intermediate field $E$ of $F/k$, we write $E_n$ for the constant extension $E_n^{\mathrm{con}}$ of $E$ of degree $p^n$.

Our aim is to establish links between the Stickelberger element $\theta_{F^{\mathrm{con}},\Sigma}^{\Sigma'} = \varprojlim_{\varrho_{n-1}^n} \theta_{F_n,\Sigma}^{\Sigma'}$ and the derived Iwasawa Weil-étale cohomology group

$$M_\Sigma^{\Sigma'}(F) := H_{\mathrm{dIw}}^{1,\Sigma'}((C_{F^{\mathrm{con}}}^\Sigma)_{\mathrm{W}}, \mathbb{G}_m)$$

(cf. Definition 6.2) that are stronger than those given by the general result of Theorem 6.5.

7.1. **Statement of the main results.** Since the $R$-module $A_\Sigma(F^{\mathrm{con}})$ is finitely-generated (by Theorem 5.1(vi)), Theorem 6.5 applies with $X = (0)$ to show both that $M_\Sigma^{\Sigma'}(F)$ is finitely-presented and torsion over $\mathbb{A}$, and that there exists a short exact sequence of $\mathbb{A}$-modules

$$0 \to \mathrm{Fit}_{\mathbb{A}}^0(M_\Sigma^{\Sigma'}(F)) \xrightarrow{\subseteq} \varprojlim_{\varrho_{n-1}^n} (\theta_{F_n,\Sigma}^{\Sigma'} \cdot \mathbb{A}_n) \xrightarrow{\xi} \varprojlim_{\nu_n}{}^1 \mathcal{T}_n(M_\Sigma^{\Sigma'}(F)) \to 0. \tag{39}$$

On the other hand, combining Lemma 6.8(i) with the argument of Remark 4.6 directly gives a short exact sequence of $\mathbb{\Lambda}$-modules

$$0 \to \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} \cdot \mathbb{\Lambda} \xrightarrow{\subseteq} \varprojlim_{\varrho^n_{n-1}} (\theta^{\Sigma'}_{F_n,\Sigma} \cdot \mathbb{\Lambda}_n) \xrightarrow{\xi'} \varprojlim^1_{\varrho^n_{n-1}} \mathrm{Ann}_{\mathbb{\Lambda}_n}(\theta^{\Sigma'}_{F_n,\Sigma}) \to 0. \tag{40}$$

The argument of Lemma 6.8(ii) shows the modules $\varprojlim^1_{\nu_n} \mathcal{T}_n(M^{\Sigma'}_{\Sigma}(F))$ and $\varprojlim^1_{\varrho^n_{n-1}} \mathrm{Ann}_{\mathbb{\Lambda}_n}(\theta^{\Sigma'}_{F_n,\Sigma})$ to be very closely related and, in addition, $\xi$ and $\xi'$ are both canonical connecting homomorphisms arising in the theory of pro-discrete modules. The strategy of this section will thus be to leverage the similarities between (39) and (40) in order to refine Theorem 6.5 (with $X = (0)$). In particular, this approach will be used in §7.3 to prove the following result.

**Theorem 7.1.** *Set $\mathcal{I}^{\Sigma'}_{\Sigma}(F) := \mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma}(F)) \cap (\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} \cdot \mathbb{\Lambda})$. Then the following claims are valid.*

(i) *The modules $\mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma}(F))/\mathcal{I}^{\Sigma'}_{\Sigma}(F)$ and $(\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} \cdot \mathbb{\Lambda})/\mathcal{I}^{\Sigma'}_{\Sigma}(F)$ are finite and have order that is prime-to-p and divisible only by primes that divide $|H|$.*

(ii) *One has $\mathrm{Fit}^0_{\mathbb{\Lambda}}\big(M^{\Sigma'}_{\Sigma}(F)\big) = \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} \cdot \mathbb{\Lambda}$ if either*
   (a) *$H$ is a $p$-group, or*
   (b) *$F/k$ is ramified at a single place, this place is totally ramified in $F$ and its degree (over $k$) is prime-to-p.*

**Remark 7.2.** Let $k_1$ be any function field of characteristic $p$, fix a place $v_\infty$ of $k_1$ and write $A$ for the subring of $k_1$ comprising elements that are regular at every place other than $v_\infty$. Fix a finite (possibly empty) set $\Sigma_1$ of prime ideals of $A$ and write $\mathfrak{m}$ for the product of ideals in $\Sigma_1$. Then the real (relative to $v_\infty$) ray class field $H(\mathfrak{m})$ of conductor $\mathfrak{m}$ is a finite abelian extension of $k_1$ in which $v_\infty$ splits completely and all places that ramify belong to $\Sigma_1$. In particular, if we set $k := H(A)$, then $H(\mathfrak{m}^\infty) := \bigcup_{t\in\mathbb{N}} H(\mathfrak{m}^t)$ is an abelian extension of $k$ that ramifies at every place in $\Sigma_{1,k}$ (and at no other place), contains no unramified extension of $k$ and is such that $G_{H(\mathfrak{m}^\infty)/k}$ has an open subgroup (and hence also a quotient) that is isomorphic to $\mathbb{Z}_p^{\mathbb{N}}$, thereby giving an abundant supply of finite extensions $F/k$ satisfying the hypotheses of Theorem 7.1(ii)(b). We further note that the extensions $H(\mathfrak{m}^\infty)/k_1$ incorporate the Drinfeld modular towers considered by Bley and Popescu in [4] and hence also the Carlitz-Hayes cyclotomic extensions considered by Anglès et al in [1].

In any situation in which the equality in Theorem 7.1(ii) is valid, Lemma 6.8(i) implies that the $\mathbb{\Lambda}$-module $\mathrm{Fit}^0_{\mathbb{\Lambda}}\big(M^{\Sigma'}_{\Sigma}(F)\big)$ is free of rank 1. If $p$ is not exceptional (in the sense recalled at the beginning of §4.1), then, without needing any hypotheses on either $|H|$ or the ramification of $F/k$, it is possible to interpret this condition on $\mathrm{Fit}^0_{\mathbb{\Lambda}}\big(M^{\Sigma'}_{\Sigma}(F)\big)$ in terms of a natural Euler characteristic of $\mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)$. To state this result, we write $\mathfrak{M}^{\mathrm{lc}}_H$ for the category of finitely-generated $\mathbb{\Lambda}$-modules that are locally-free (and hence projective) of constant rank. We denote the Grothendieck group of $\mathfrak{M}^{\mathrm{lc}}_H$ by $K^{\mathrm{lc}}_0(\mathbb{\Lambda})$ and use the canonical 'determinant' homomorphism $\delta^{\mathrm{lc}}_{\mathbb{\Lambda}} : K^{\mathrm{lc}}_0(\mathbb{\Lambda}) \to \mathrm{Pic}(\mathbb{\Lambda})$. We shall also say that an object $X$ of $\mathsf{D}(\mathbb{\Lambda})$ is 'strictly-perfect' if it is isomorphic (in $\mathsf{D}(\mathbb{\Lambda})$) to a bounded complex of modules in $\mathfrak{M}^{\mathrm{lc}}_H$ and we note that each such $X$ has a well-defined 'Euler characteristic' $\chi^{\mathrm{lc}}_{\mathbb{\Lambda}}(X)$ in $K^{\mathrm{lc}}_0(\mathbb{\Lambda})$ (that depends only on $X$ up to isomorphism in $\mathsf{D}(\mathbb{\Lambda})$). Finally, we write $\Delta_H$ for the natural (diagonal) projection map $\mathrm{Pic}(\mathbb{\Lambda}) \to \varprojlim_n \mathrm{Pic}(\mathbb{\Lambda}_n)$.

The following result will be proved in §7.4.

**Proposition 7.3.** *Set $C^{\Sigma'}_{\Sigma}(F) := \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)$ and $\mathcal{J}^{\Sigma'}_{\Sigma}(F) := \mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma}(F))$. Then, if $p$ is not exceptional, the following claims are valid.*

(i) *$C^{\Sigma'}_{\Sigma}(F)$ is strictly-perfect as a complex of $\mathbb{\Lambda}$-modules.*

(ii) *$\mathcal{J}^{\Sigma'}_{\Sigma}(F)$ is an invertible ideal of $\mathbb{\Lambda}$ and its class in $\mathrm{Pic}(\mathbb{\Lambda})$ is equal to $\delta^{\mathrm{lc}}_{\mathbb{\Lambda}}\big(\chi^{\mathrm{lc}}_{\mathbb{\Lambda}}(C^{\Sigma'}_{\Sigma}(F))\big)$ and belongs to $\ker(\Delta_H)$.*

(iii) $\bigwedge_R^{|H|}(\mathcal{J}_\Sigma^{\Sigma'}(F))$ *is a free $R$-module of rank* 1.

7.2. **Reduction results.** In order to prove Theorem 7.1, we first establish several useful reduction results. To do this, we use the ideals $\mathcal{I}_\Sigma^{\Sigma'}(F)$ that are defined in Theorem 7.1.

**Lemma 7.4.** *Let $v$ be a place of $k$ outside $\Sigma\cup\Sigma'$. Set $\Sigma_1 := \Sigma\cup\{v\}$ and, noting $v$ is unramified in $F^{\mathrm{con}}$, write $\sigma_v = \sigma_v(F)$ for its Frobenius automorphism in* $\mathrm{Gal}(F^{\mathrm{con}}/k)$. *Then the following claims are valid.*

(i) $1-\sigma_v$ *is a non-zero divisor in* $\mathbb{\Lambda}$.
(ii) $\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma_1} = (1-\sigma_v)\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}$.
(iii) $\mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma_1}(F)) = (1-\sigma_v)\mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma}(F))$.
(iv) $\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}\cdot\mathbb{\Lambda} = \mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma}(F)) \Longleftrightarrow \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma_1}\cdot\mathbb{\Lambda} = \mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma_1}(F))$.
(v) *Multiplication by $1-\sigma_v$ induces isomorphisms of $\mathbb{\Lambda}$-modules*

$$\mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma}(F))/\mathcal{I}^{\Sigma'}_{\Sigma}(F) \cong \mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma_1}(F))/\mathcal{I}^{\Sigma'}_{\Sigma_1}(F) \quad \textit{and} \quad (\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}\cdot\mathbb{\Lambda})/\mathcal{I}^{\Sigma'}_{\Sigma}(F) \cong (\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma_1}\cdot\mathbb{\Lambda})/\mathcal{I}^{\Sigma'}_{\Sigma_1}(F).$$

*Proof.* To prove (i), it is enough to note $\sigma_v^{|H|}$ generates an open subgroup of $G_{F^{\mathrm{con}}/F}$. Indeed, this implies that the product $(1-\sigma_v)\sum_{i=0}^{i=|H|-1}\sigma_v^i = 1-\sigma_v^{|H|}$ is a non-zero divisor in $\mathbb{Z}_p[\![G_{F^{\mathrm{con}}/F}]\!] \cong \mathbb{Z}_p[\![\mathbb{Z}_p]\!]$, and hence also in $\mathbb{\Lambda} = (\mathbb{Z}[\![\mathbb{Z}_p]\!])[H]$.

Claim (ii) is a well-known property of truncated Stickelberger elements (and follows directly from their definition in terms of truncated $L$-series). To prove (iii) we note that, for each $n\in\mathbb{N}_0$, there exists a canonical exact triangle in $\mathsf{D}^{\mathrm{perf}}(\mathbb{\Lambda}_n)$

$$\mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F_n})_{\mathrm{W}},\mathbb{G}_m) \to \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma_1}_{F_n})_{\mathrm{W}},\mathbb{G}_m) \to \mathsf{R}\mathrm{Hom}_{\mathbb{\Lambda}_n}(\mathsf{R}\Gamma(\kappa(v)_{\mathrm{W}},\mathbb{\Lambda}_n),\mathbb{\Lambda}_n[-1]) \to \cdot$$

that is compatible with the natural projection morphisms as $n$ varies (cf. [8, Prop. 2.4(ii) and Rem. 2.5(i)]). In addition, for each $n\in\mathbb{N}$, the third complex in this triangle is canonically isomorphic to the complex $\mathbb{\Lambda}_n \to \mathbb{\Lambda}_n$ in which the first term occurs in degree 0 and the differential sends each $x$ to $(1-\sigma_v)x$, and the natural projection isomorphism

$$\mathbb{\Lambda}_{n-1}\otimes^{\mathsf{L}}_{\mathbb{\Lambda}_n}\mathsf{R}\mathrm{Hom}_{\mathbb{\Lambda}_n}(\mathsf{R}\Gamma(\kappa(v)_{\mathrm{W}},\mathbb{\Lambda}_n),\mathbb{\Lambda}_n[-1]) \cong \mathrm{Hom}_{\mathbb{\Lambda}_{n-1}}(\mathsf{R}\Gamma(\kappa(v)_{\mathrm{W}},\mathbb{\Lambda}_{n-1}),\mathbb{\Lambda}_{n-1}[-1])$$

is induced by the morphism of complexes of $\mathbb{\Lambda}_n$-modules

$$\begin{array}{ccc} \mathbb{\Lambda}_n & \xrightarrow{x\mapsto(1-\sigma_v)x} & \mathbb{\Lambda}_n \\ \downarrow & & \downarrow \\ \mathbb{\Lambda}_{n-1} & \xrightarrow{x\mapsto(1-\sigma_v)x} & \mathbb{\Lambda}_{n-1} \end{array}$$

in which both vertical maps are the canonical projection. Hence, upon passing to the inverse limit over $n$ of the above triangles, one deduces from Proposition 4.11(ii) (and the argument given at the end of the proof of Proposition 4.11(iv)), the existence of an exact triangle in $\mathsf{D}(\mathbb{\Lambda})$ of the form

$$\mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}},\mathbb{G}_m) \to \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma_1}_{F^{\mathrm{con}}})_{\mathrm{W}},\mathbb{G}_m) \to \big[\mathbb{\Lambda}\xrightarrow{\phi_v}\mathbb{\Lambda}\big] \to \cdot$$

in which the first module in the third complex occurs in degree 0 and $\phi_v$ sends $x$ to $(1-\sigma_v)x$. In particular, since $\phi_v$ is injective (by (i)), the long exact cohomology sequence of this triangle combines with Lemma 6.4 with $X=(0)$ to give an exact sequence $0\to M^{\Sigma'}_{\Sigma}(F)\to M^{\Sigma'}_{\Sigma_1}(F)\to \mathrm{cok}(\phi_v)\to 0$. Hence, since the first two modules in this sequence are finitely-presented (by Theorem 6.5 with $X=(0)$) and the third lies in a short exact sequence

$$0\to\mathbb{\Lambda}\xrightarrow{\phi_v}\mathbb{\Lambda}\to\mathrm{cok}(\phi_v)\to 0, \tag{41}$$

a general algebraic result of Cornacchia and Greither [11, Lem. 3] implies that

$$\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma_1}(F)) = \mathrm{Fit}^0_{\mathbb{A}}(\mathrm{cok}(\phi_v))\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F)).$$

To deduce the equality in (iii) it is therefore now enough to note that the short exact sequence (41) combines with the explicit definition of Fitting ideals to imply $\mathrm{Fit}^0_{\mathbb{A}}(\mathrm{cok}(\phi_v))$ is equal to the $\mathbb{A}$-ideal generated by $\det(\phi_v) = 1 - \sigma_v$.

The forward implication in (iv) is an immediate consequence of the equalities in (ii) and (iii) and the reverse implication a consequence of the same equalities and the fact $1 - \sigma_v$ is a non-zero divisor of $\mathbb{A}$. In a similar way, the latter fact also implies the assignment $x \mapsto (1 - \sigma_v)x$ is injective on $\mathbb{A}$ and so combines with (ii) and (iii) to imply that

$$\begin{aligned}\mathcal{I}^{\Sigma'}_{\Sigma_1}(F) &= (1 - \sigma_v)\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F)) \cap (1 - \sigma_v)(\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}\cdot\mathbb{A})\\ &= (1 - \sigma_v)\big(\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F)) \cap (\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}\cdot\mathbb{A})\big)\\ &= (1 - \sigma_v)\mathcal{I}^{\Sigma'}_{\Sigma}(F).\end{aligned}$$

The isomorphisms in (v) then follow directly from these facts and the equalities in (ii) and (iii). □

We now set $c := |H|$, $\mathbb{Z}' := \mathbb{Z}[1/c]$, $\mathbb{A}' := \mathbb{A}[1/c]$ and $\mathbb{A}'_n := \mathbb{A}_n[1/c] = \mathbb{Z}'[G_{F_n/k}]$ for $n \in \mathbb{N}_0$.

**Lemma 7.5.** *Theorem 7.1(i) is valid if and only if* $\mathrm{Fit}^0_{\mathbb{A}}\big(M^{\Sigma'}_{\Sigma}(F))\big)[1/c] = \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}\cdot\mathbb{A}'$.

*Proof.* For brevity, we write $\theta$ for $\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}$. Then the forward implication is clear since Theorem 7.1(i) implies that $\mathrm{Fit}^0_{\mathbb{A}}\big(M^{\Sigma'}_{\Sigma}(F))\big)[1/c]$ and $\theta\cdot\mathbb{A}'$ are both equal to $\mathcal{I}^{\Sigma'}_{\Sigma}(F)\cdot\mathbb{A}'$.

To prove the reverse implication, we note that the $\mathbb{A}$-module $M^{\Sigma'}_{\Sigma}(F)$ is finitely-presented (by Theorem 6.5), and hence that the $\mathbb{A}$-module $\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F))$ is finitely-generated. Thus, if the stated equality is valid, then for any sufficiently large integer $t$ one has

$$c^t(\theta\cdot\mathbb{A}) \subseteq \mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F)) \subseteq c^{-t}(\theta\cdot\mathbb{A}). \tag{42}$$

In particular, if we set $Z := \varprojlim^1_{\nu_n}\mathcal{T}_n(M^{\Sigma'}_{\Sigma}(F))$, then the first inclusion here combines with the exact sequences (39) and (40) to imply $\xi$ sends the submodule $\theta\cdot\mathbb{A}$ of $\varprojlim_{\varrho^n_{n-1}}(\theta^{\Sigma'}_{F_n,\Sigma}\cdot\mathbb{A}_n)$ into the submodule $Z[c^t]$ of $Z$ comprising elements that are annihilated by $c^t$. However, from the proof of Lemma 6.8(ii), one knows $Z$ is isomorphic to $(\mathbb{Z}_p/\mathbb{Z})\otimes_{\mathbb{Z}} Y'_m$ for a (finitely-generated) lattice $Y'_m$ and so $Z[c^t]$ is both finite and of order prime-to-$p$. The quotient $\mathbb{A}$-module

$$(\theta\cdot\mathbb{A})/\mathcal{I}^{\Sigma'}_{\Sigma}(F) \cong \big(\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F)) + \theta\cdot\mathbb{A}\big)/\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F)) \overset{\xi}{\hookrightarrow} Z[c^t]$$

is therefore also finite and of order prime to $p$, as required.

Finally, we observe that the analogous result for the quotient $\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F))/\mathcal{I}^{\Sigma'}_{\Sigma}(F)$ is derived from the second inclusion of (42) and the argument of Lemma 6.8(ii) in an entirely similar way. □

For an intermediate field $E$ of $F/k$, we now set $H^E := G_{F/E}$, $H_E := G_{E/k}$, $\mathbb{A}_E := R[H_E]$ and $\mathbb{A}'_E := \mathbb{A}_E[1/c]$. We also write $e_E$ for the idempotent $(1/|H^E|)\sum_{h\in H^E} h$ of $\mathbb{A}'$ and note that the natural projection map $\pi_E : \mathbb{A}' \to \mathbb{A}'_E$ identifies $\mathbb{A}'_E$ with the direct factor $e_E\mathbb{A}'$ of $\mathbb{A}'$.

**Lemma 7.6.** *Let $E$ be an intermediate field of $F/k$. Then, in $e_E\mathbb{A}' = \mathbb{A}'_E$, one has both*

$$e_E\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} = \theta^{\Sigma'}_{E^{\mathrm{con}},\Sigma} \quad \textit{and} \quad e_E(\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F))[1/c]) = \mathrm{Fit}^0_{\mathbb{A}'_E}(M^{\Sigma'}_{\Sigma}(E)[1/c]).$$

*Proof.* The first claimed equality follows directly from the fact $\pi_E(\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}) = \theta^{\Sigma'}_{E^{\mathrm{con}},\Sigma}$ (which is itself a well-known consequence of the invariance of Dirichlet $L$-functions under inflation of characters).

To prove the second equality, we note the functor $N \mapsto \mathbb{A}'_E \otimes_{\mathbb{A}'} N$ is exact on the category of $\mathbb{A}'$-modules and that $\mathbb{A}'_E \otimes_{\mathbb{A}'} N$ identifies with the submodule $e_E(N)$ of $H^E$-invariant elements of $N$. In particular, since for each $n \in \mathbb{N}$ there exists a canonical isomorphism in $\mathsf{D}^{\mathrm{perf}}(\mathbb{Z}[G_{E_n/k}])$

$$\mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{E_n})_{\mathrm{W}}, \mathbb{G}_m) \cong \mathbb{Z}[G_{E_n/k}] \otimes^{\mathsf{L}}_{\mathbb{Z}[G_{F_n/k}]} \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F_n})_{\mathrm{W}}, \mathbb{G}_m) \cong \mathsf{R}\mathrm{Hom}_{H^E}(\mathbb{Z}, \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F_n})_{\mathrm{W}}, \mathbb{G}_m))$$

(cf. [6, Lem. 1(iv)]), one also has an induced composite isomorphism in $\mathsf{D}(\mathbb{A}'_E)$

$$\begin{aligned}
\mathbb{A}'_E \otimes^{\mathsf{L}}_{\mathbb{A}} \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m) &\cong \mathrm{Hom}_{\mathbb{Z}'[H^E]}\big(\mathbb{Z}', (\underleftarrow{\mathsf{R}\mathrm{lim}}_{\rho_n} \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F_n})_{\mathrm{W}}, \mathbb{G}_m))[1/c]\big) \\
&\cong \big(\underleftarrow{\mathsf{R}\mathrm{lim}}_{\rho_n} \mathsf{R}\mathrm{Hom}_{H^E}(\mathbb{Z}, \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F_n})_{\mathrm{W}}, \mathbb{G}_m))\big)[1/c] \\
&\cong \underleftarrow{\mathsf{R}\mathrm{lim}}_{\rho_n} \big(\mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{E_n})_{\mathrm{W}}, \mathbb{G}_m)\big)[1/c] \\
&= \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{E^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)[1/c].
\end{aligned}$$

Then, since $\mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)$ is acyclic in degrees greater than 1 (by Lemma 6.4 with $X = (0)$), this isomorphism induces an isomorphism of $\mathbb{A}'_E$-modules

$$\begin{aligned}
\mathbb{A}'_E \otimes_{\mathbb{A}'} M^{\Sigma'}_{\Sigma}(F)[1/c] &= \mathbb{A}'_E \otimes_{\mathbb{A}} H^1(\mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)) \\
&\cong H^1(\mathbb{A}'_E \otimes^{\mathsf{L}}_{\mathbb{A}} \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)) \qquad (43) \\
&\cong H^1(\mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{E^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)[1/c]) = M^{\Sigma'}_{\Sigma}(E)[1/c].
\end{aligned}$$

We can therefore now obtain the claimed equality via the calculation

$$\begin{aligned}
e_E(\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F))[1/c]) &= \pi_E(\mathrm{Fit}^0_{\mathbb{A}'}(M^{\Sigma'}_{\Sigma}(F)[1/c])) \\
&= \mathrm{Fit}^0_{\mathbb{A}'_E}(\mathbb{A}'_E \otimes_{\mathbb{A}'} M^{\Sigma'}_{\Sigma}(F)[1/c]) \\
&= \mathrm{Fit}^0_{\mathbb{A}'_E}(M^{\Sigma'}_{\Sigma}(E)[1/c]).
\end{aligned}$$

Here the first equality is clear, the second follows from a standard property of Fitting ideals under ring projection and the third is a consequence of the isomorphism (43). □

7.3. **The proof of Theorem 7.1.** At the outset, we observe (ii)(a) follows directly from the validity of (i). Indeed, if $|H|$ is a power of $p$, then (i) implies that both quotient modules $\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F))/\mathcal{I}^{\Sigma'}_{\Sigma}(F)$ and $(\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}\cdot\mathbb{A})/\mathcal{I}^{\Sigma'}_{\Sigma}(F)$ vanish, and so one has $\mathrm{Fit}^0_{\mathbb{A}}(M^{\Sigma'}_{\Sigma}(F)) = \mathcal{I}^{\Sigma'}_{\Sigma}(F) = \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}\cdot\mathbb{A}$, as required.

The following result next verifies (ii)(b) and, in addition, an important special case of (i).

**Proposition 7.7.** *The following claims are valid.*

(i) *If either $F = k$ or the conditions of Theorem 7.1(ii)(b) are satisfied, then one has*

$$\mathrm{Fit}^0_{\mathbb{A}}\big(M^{\Sigma'}_{\Sigma}(F)\big) = \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma}\cdot\mathbb{A}.$$

(ii) *If $F/k$ is unramified, then Theorem 7.1(i) is valid.*

*Proof.* To prove (i), we label the places in $\Sigma$ as $\{v_i : 1 \leq i \leq |\Sigma|\}$.

We first assume the conditions of Theorem 7.1(ii)(b). Then the (unique) place that ramifies in $F$ belongs to $\Sigma$ and we assume the above labelling is such that this place in $v_1$. Since every place in $\Sigma \setminus \{v_1\}$ is unramified in $F$, we may now apply Lemma 7.4(iv) $(|\Sigma| - 1)$-times, with the pair $(\Sigma, v)$ in the latter result taken successively to be $(\Sigma \setminus \{v_2\}, v_2), (\Sigma \setminus \{v_2, v_3\}, v_3)$ etc., in order to deduce it is enough to prove the claimed equality with $\Sigma = \{v_1\}$. Then, since $v_1$ is both totally ramified in $F$ (by assumption) and also inert in $k^{\mathrm{con}}$ (since $\mathrm{d}_k(v)$ is assumed to be prime-to-$p$), it has full decomposition group in $G_{F^{\mathrm{con}}/k}$. In this case, therefore, Lemma 6.8(ii) implies that the derived limits

$\varprojlim^1_{\nu_n} \mathcal{T}_n(M_\Sigma^{\Sigma'}(F))$ and $\varprojlim^1_{\varrho^n_{n-1}} \mathrm{Ann}_{\mathbb{A}_n}(\theta^{\Sigma'}_{F_n,\Sigma})$ in the exact sequences (39) and (40) both vanish. The latter sequences thus combine directly to imply the required equality via the computation

$$\mathrm{Fit}^0_{\mathbb{A}}(M_\Sigma^{\Sigma'}(F)) = \varprojlim_n(\theta^{\Sigma'}_{F_n,\Sigma} \cdot \mathbb{A}_n) = \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} \cdot \mathbb{A}.$$

To complete the proof of (i), we now assume $F = k$ and recall that the known validity of the Riemann Hypothesis for function fields implies, for any sufficiently large integer $n$, there exists a place $v_0$ of $k$ with $v_0 \notin \Sigma \cup \Sigma'$ and $\mathrm{d}_k(v_0) = n$ (cf. [25, Th. 5.12]). We fix such a place $v_0$ for which $d_k(v_0)$ is prime-to-$p$. Then, by applying Lemma 7.4(iv) with $v = v_0$, it is enough to verify the claimed equality after replacing $\Sigma$ by $\Sigma_0 := \Sigma \cup \{v_0\}$. Then, by $|\Sigma|$ further uses of Lemma 7.4(iv), with the pair $(\Sigma, v)$ in the latter result taken successively to be $(\Sigma_0 \setminus \{v_1\}, v_1), (\Sigma_0 \setminus \{v_1, v_2\}, v_2)$ etc., we are further reduced to verifying the claimed equality in the case that $\Sigma = \{v_0\}$. In addition, since $v_0$ is inert in $k^{\mathrm{con}}/k$ (as $\mathrm{d}_k(v_0)$ is prime-to-$p$), it has full decomposition group in $G_{k^{\mathrm{con}}/k}$. Hence, by the same argument as above, the claimed equality can be directly derived in this case from the exact sequences (39) and (40). This proves (i).

We assume henceforth that $F/k$ is unramified and we set $\mathcal{J}_\Sigma^F := \mathrm{Fit}^0_{\mathbb{A}}(M_\Sigma^{\Sigma'}(F))[1/c]$, $\mathcal{T}_\Sigma^F := \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} \cdot \mathbb{A}'$ and $\mathcal{I}_\Sigma^F := \mathcal{I}_\Sigma^{\Sigma'}(F)[1/c]$. Then, by Lemma 7.5, to prove (ii) it is enough for us to show that $\mathcal{J}_\Sigma^F = \mathcal{T}_\Sigma^F$. To do this, we note Lemma 7.6 implies that $e_k\mathcal{J}_\Sigma^F = \mathcal{J}_\Sigma^k$, $e_k\mathcal{T}_\Sigma^F = \mathcal{T}_\Sigma^k$ and $e_k\mathcal{I}_\Sigma^F = \mathcal{I}_\Sigma^k$, where $e_k$ is the idempotent of $\mathbb{Z}'[H]$ defined just before the latter result. In particular, if $H$ acts trivially on the quotient module $\mathcal{J}_\Sigma^F/\mathcal{I}_\Sigma^F$, then one would have

$$\mathcal{J}_\Sigma^F/\mathcal{I}_\Sigma^F = e_k\big(\mathcal{J}_\Sigma^F/\mathcal{I}_\Sigma^F\big) = (e_k\mathcal{J}_\Sigma^F)/(e_k\mathcal{I}_\Sigma^F) = \mathcal{J}_\Sigma^k/\mathcal{I}_\Sigma^k.$$

Since (i) implies $\mathcal{J}_\Sigma^k/\mathcal{I}_\Sigma^k$ vanishes, we could therefore deduce that $\mathcal{J}_\Sigma^F = \mathcal{I}_\Sigma^F$. In a similar way, if $H$ also acts trivially on $\mathcal{T}_\Sigma^F/\mathcal{I}_\Sigma^F$, we could deduce from (i) that $\mathcal{T}_\Sigma^F = \mathcal{I}_\Sigma^F$, and hence that $\mathcal{J}_\Sigma^F = \mathcal{T}_\Sigma^F$, as required. To complete the proof of (ii), we are therefore reduced to showing $H$ acts trivially on $\mathcal{X}_\Sigma^F/\mathcal{I}_\Sigma^F$, where $\mathcal{X}_\Sigma^F$ denotes either $\mathcal{J}_\Sigma^F$ or $\mathcal{T}_\Sigma^F$.

To do this, we note that there exists an open subgroup $V$ of $G_{F^{\mathrm{con}}/F}$ that acts trivially on $\mathcal{X}_\Sigma^F/\mathcal{I}_\Sigma^F$ (see the proof of Lemma 7.5). We now fix an element $h$ of $H$. Recalling that $F \cap k^{\mathrm{con}} = k$, we then use the Tchebotarev density theorem to choose a place $v$ of $k$ with $v \notin \Sigma'$ for which its Frobenius element $\sigma_v$ in $G_{F^{\mathrm{con}}/k} \cong G_{F^{\mathrm{con}}/F} \times H$ belongs to $V \times \{h\}$. By repeatedly applying the isomorphisms in Lemma 7.4(v) (in the same manner we applied the equivalences of Lemma 7.4(iv) in the above proof of (i)), we can then deduce the existence of an isomorphism of $\mathbb{A}$-modules

$$\mathcal{X}_\Sigma^F/\mathcal{I}_\Sigma^F \cong \mathcal{X}_{\{v\}}^F/\mathcal{I}_{\{v\}}^F.$$

We now note the argument of Lemma 6.8(ii) (with $\mathcal{K} = k^{\mathrm{con}}, \Sigma = \{v\}$ and $X = (0)$) implies $\sigma_v$ acts trivially on both $\mathbb{A}$-modules $\varprojlim^1_{\nu_n} \mathcal{T}_n(M_{\{v\}}^{\Sigma'}(F))$ and $\varprojlim^1_{\varrho^n_{n-1}} \mathrm{Ann}_{\mathbb{A}_n}(\theta^{\Sigma'}_{F_n,\{v\}})$. From the respective short exact sequence (39) or (40), we can thus deduce $\sigma_v$ acts trivially on $\mathcal{X}_{\{v\}}^F/\mathcal{I}_{\{v\}}^F$. Then, since $\sigma_v \in V \times \{h\}$ and $V$ acts trivially on $\mathcal{X}_\Sigma^F/\mathcal{I}_\Sigma^F$, the last displayed isomorphism implies $h$ acts trivially on $\mathcal{X}_\Sigma^F/\mathcal{I}_\Sigma^F$. As $h$ is an arbitrary element of $H$, this therefore completes the proof of (ii). □

At this point, to complete the proof of Theorem 7.1, we are reduced to proving (i). To do this, we shall prove the equality in Lemma 7.5 by means of an induction on the number $r_F$ of places of $k$ that ramify in $F$.

If, firstly, $r_F = 0$, then $F/k$ is unramified and so this case is proved in Proposition 7.7(ii). We shall therefore assume $r_F > 0$. By the argument of Lemma 7.4(iv), we are then further reduced to considering the case that $\Sigma$ is equal to the (non-empty) set of places that ramify in $F$.

To deal with this case, we write $\mathfrak{S} = \mathfrak{S}_{F/k}$ for the set of intermediate fields of $F/k$ and, for $E \in \mathfrak{S}$, denote the subset of $\Sigma$ comprising places that ramify in $E$ by $\Sigma(E)$ (so $\Sigma(F) = \Sigma$). We set

$H^* := \mathrm{Hom}(H, \mathbb{Q}^{c,\times})$ and, for each $\psi \in H^*$, write $F_\psi \in \mathfrak{S}$ for the fixed field of $F$ under $\ker(\psi)$. Then, for each $E \in \mathfrak{S}$, the set

$$\Xi(E) := \{\psi \in H^* : F_\psi = E\}$$

is a (possibly empty) conjugacy class for the action of $G_{\mathbb{Q}^c/\mathbb{Q}}$ on $H^*$ and so the idempotent

$$\varepsilon_E := \sum\nolimits_{\psi \in \Xi(E)} e_\psi$$

belongs to $\mathbb{Z}'[H]$. Since $\varepsilon_E = \varepsilon_E e_E$ (for the idempotent $e_E$ defined just before Lemma 7.6), each $\varepsilon_E$ can be regarded as an idempotent of $\mathbb{Z}'[H_E]$. In addition, in $\mathbb{Q}^c[H]$, one has

$$1 = \sum\nolimits_{\psi \in H^*} e_\psi = \sum\nolimits_{E \in \mathfrak{S}} \big(\sum\nolimits_{\psi \in \Xi(E)} e_\psi\big) = \sum\nolimits_{E \in \mathfrak{S}} \varepsilon_E = \sum\nolimits_{E \in \mathfrak{S}_1} \varepsilon_E + \sum\nolimits_{E \in \mathfrak{S}_2} \varepsilon_E, \tag{44}$$

with $\mathfrak{S}_1 := \{E \in \mathfrak{S} : \Sigma(E) = \Sigma\}$ and $\mathfrak{S}_2 := \{E \in \mathfrak{S} : \Sigma(E) \neq \Sigma\}$.

We first consider a field $E$ in $\mathfrak{S}_1$ and, for every $n \in \mathbb{N}_0$, regard $\varepsilon_E$ as an idempotent of $\mathbb{Z}'[H] \subseteq \mathbb{Z}'[G_{F_n/F} \times H] = \mathbb{Z}'[G_{F_n/k}]$ (or, equivalently, each $\psi \in \Xi(E)$ as a character of $G_{F_n/k}$). Then, for every $\psi \in \Xi(E)$, the field $F_n^{\ker(\psi)} = E$ is ramified at all places in $\Sigma(E) = \Sigma$ and so $\psi$ cannot be trivial on the decomposition subgroup in $G_{F_n/k}$ of any place in $\Sigma$. In particular, from the equivalence (36) (and the first sentence of the new paragraph that follows it) one finds that $\varepsilon_E$ annihilates both modules $\mathcal{T}_n(M_\Sigma^{\Sigma'}(F))[1/c]$ and $\mathrm{Ann}_{\mathbb{A}_n}(\theta_{F_n,\Sigma}^{\Sigma'})[1/c]$ (for every $n$) and hence also the derived limits $(\varprojlim^1_{\nu_n} \mathcal{T}_n(M_\Sigma^{\Sigma'}(F)))[1/c]$ and $(\varprojlim^1_{\varrho^n_{n-1}} \mathrm{Ann}_{\mathbb{A}_n}(\theta_{F_n,\Sigma}^{\Sigma'}))[1/c]$. Upon combining the (scalar extensions of the) exact sequences (39) and (40), we can therefore deduce that

$$\varepsilon_E\big(\mathrm{Fit}^0_{\mathbb{A}}(M_\Sigma^{\Sigma'}(F))[1/c]\big) = \varepsilon_E\big((\varprojlim_n(\theta_{F_n,\Sigma}^{\Sigma'} \cdot \mathbb{A}_n))[1/c]\big) = \varepsilon_E\big(\theta_{F^{\mathrm{con}},\Sigma}^{\Sigma'} \cdot \mathbb{A}'\big). \tag{45}$$

We next consider a field $E$ in $\mathfrak{S}_2$. In this case, we use the second equality in Lemma 7.6 to compute that, in $\varepsilon_E \mathbb{A}' = \varepsilon_E(e_E \mathbb{A}') = \varepsilon_E \mathbb{A}'_E$, one has

$$\begin{aligned}
\varepsilon_E(\mathrm{Fit}^0_{\mathbb{A}}(M_\Sigma^{\Sigma'}(F))[1/c]) &= \varepsilon_E \cdot \mathrm{Fit}^0_{\mathbb{A}_E}(M_\Sigma^{\Sigma'}(E))[1/c] \\
&= \varepsilon_E\big(\prod\nolimits_{v \in \Sigma \setminus \Sigma(E)}(1 - \sigma_v)\big) \cdot \mathrm{Fit}^0_{\mathbb{A}_E}\big(M_{\Sigma(E)}^{\Sigma'}(E))\big)[1/c] \\
&= \varepsilon_E\big(\prod\nolimits_{v \in \Sigma \setminus \Sigma(E)}(1 - \sigma_v)\big)\theta_{E^{\mathrm{con}},\Sigma(E)}^{\Sigma'} \cdot \mathbb{A}'_E \\
&= \varepsilon_E(\theta_{F^{\mathrm{con}},\Sigma}^{\Sigma'} \cdot \mathbb{A}').
\end{aligned}$$

Here the second equality follows from Lemma 7.4(iii) (with $F$ replaced by $E$), the third from the inductive hypothesis (since $E \in \mathfrak{S}_2$ implies $r_E < r_F$) and the last from Lemma 7.4(ii) (with $F$ replaced by $E$) and the first equality of Lemma 7.6.

The explicit equality in Lemma 7.5 now follows upon combining the last displayed equality (for each $E \in \mathfrak{S}_2$) with (45) (for each $E \in \mathfrak{S}_1$) and the decomposition (44) of 1 as a sum of the mutually orthogonal idempotents $\varepsilon_E$.

This completes the proof of Theorem 7.1.

7.4. **The proof of Proposition 7.3.** In this argument we set $C_F^\bullet := \mathsf{R}\Gamma^{\Sigma'}((C_{F^{\mathrm{con}}}^\Sigma)_{\mathrm{W}}, \mathbb{G}_m)$ and assume that $p$ is not exceptional.

In this case, the observations made at the beginning of §6.2.2 (with $X = (0)$ and $\mathcal{F} = F^{\mathrm{con}}$) combine with Proposition 4.11(iv) to imply the existence of an isomorphism in $\mathsf{D}(\mathbb{A})$

$$C_F^\bullet \cong \big[P \xrightarrow{\theta} \mathbb{A}^d\big], \tag{46}$$

in which $P$ is a finitely-generated projective $\mathbb{A}$-module placed in degree 0 and $d$ is a suitable natural number. From Lemma 6.4, it then follows that $\ker(\theta)$ and $\mathrm{cok}(\theta)$ are respectively isomorphic to $\varprojlim_{\mathrm{N}_n} \mathcal{O}_{F_n,\Sigma}^{\times,\Sigma'}$ and $M_\Sigma^{\Sigma'}(F)$. In particular, since the isomorphisms (32) combine with the observations

made in the paragraph that follows (36) to imply that $\mathcal{O}^{\times,\Sigma'}_{F_n,\Sigma} = \mathcal{O}^{\times,\Sigma'}_{F_{n+1},\Sigma}$ for all sufficiently large $n$, the module $\ker(\theta)$ vanishes. Hence, since Theorem 6.5 implies the $\mathbb{\Lambda}$-module $\mathrm{cok}(\theta) \cong M^{\Sigma'}_{\Sigma}(F)$ is torsion, the (projective) $\mathbb{\Lambda}$-module $P$ is locally-free of constant rank $d$ and so the isomorphism (46) implies $C^{\bullet}_F$ is a strictly-perfect complex of $\mathbb{\Lambda}$-modules, thereby proving (i).

We now fix a natural number $d' \geq d$ and a surjective morphism of $\mathbb{\Lambda}$-modules $\pi : \mathbb{\Lambda}^{d'} \to P$, and consider the commutative diagram

$$\begin{tikzcd} \bigwedge^d_{\mathbb{\Lambda}}(\mathbb{\Lambda}^{d'}) \arrow[d, "\bigwedge^d_{\mathbb{\Lambda}}\pi"'] \arrow[dr, "\bigwedge^d_{\mathbb{\Lambda}}(\theta\circ\pi)"] & \\ \bigwedge^d_{\mathbb{\Lambda}} P \arrow[r, "\bigwedge^d_{\mathbb{\Lambda}}\theta"] & \bigwedge^d_{\mathbb{\Lambda}}(\mathbb{\Lambda}^d) \cong \mathbb{\Lambda}, \end{tikzcd}$$

in which the isomorphism is with reference to the standard basis of $\mathbb{\Lambda}^d$. Then, since $\theta \circ \pi$ defines a free presentation of $M^{\Sigma'}_{\Sigma}(F)$, if one computes $\mathrm{im}(\bigwedge^d_{\mathbb{\Lambda}}(\theta\circ\pi))$ in terms of the matrix representing $\theta\circ\pi$ with respect to the standard bases of $\mathbb{\Lambda}^{d'}$ and $\mathbb{\Lambda}^d$, the explicit definition of Fitting ideal implies that

$$\mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma}(F)) = \mathrm{im}(\bigwedge\nolimits^d_{\mathbb{\Lambda}}(\theta\circ\pi)) = \mathrm{im}(\bigwedge\nolimits^d_{\mathbb{\Lambda}}\theta) \cong \bigwedge\nolimits^d_{\mathbb{\Lambda}} P.$$

Here the second equality follows from the surjectivity of $\bigwedge^d_{\mathbb{\Lambda}}\pi$ in the above commutative diagram and the isomorphism (of $\mathbb{\Lambda}$-modules) is induced by the fact $\bigwedge^d_{\mathbb{\Lambda}}\theta$ is injective since $\bigwedge^d_{\mathbb{\Lambda}} P$ is an invertible $\mathbb{\Lambda}$-module. From the isomorphism (46), we can therefore deduce that $\delta^{\mathrm{lc}}_{\mathbb{\Lambda}}\big(\chi^{\mathrm{lc}}_{\mathbb{\Lambda}}(C^{\bullet}_F)\big)$ is the isomorphism class of the invertible $\mathbb{\Lambda}$-module

$$(\bigwedge\nolimits^d_{\mathbb{\Lambda}} P) \otimes_{\mathbb{\Lambda}} \mathrm{Hom}_{\mathbb{\Lambda}}(\bigwedge\nolimits^d_{\mathbb{\Lambda}} \mathbb{\Lambda}^d, \mathbb{\Lambda}) \cong \bigwedge\nolimits^d_{\mathbb{\Lambda}} P \cong \mathrm{Fit}^0_{\mathbb{\Lambda}}(M^{\Sigma'}_{\Sigma}(F)),$$

as claimed in (ii). To complete the proof of (ii), it is now enough to show, for each $n$, that $\chi^{\mathrm{lc}}_{\mathbb{\Lambda}}(C^{\bullet}_F)$ belongs to the kernel of the projection map $\pi'_n : K^{\mathrm{lc}}_0(\mathbb{\Lambda}) \to K_0(\mathbb{\Lambda}_n)$. In addition, since Proposition 4.11(iv) implies $\pi'_n(\chi^{\mathrm{lc}}_{\mathbb{\Lambda}}(C^{\bullet}_F))$ is equal to the Euler characteristic in $K_0(\mathbb{\Lambda}_n)$ of the complex $\mathbb{\Lambda}_n \otimes^{\mathsf{L}}_{\mathbb{\Lambda}} C^{\bullet}_F \cong \mathsf{R}\Gamma^{\Sigma'}((C^{\Sigma}_{F_n})_{\mathrm{W}}, \mathbb{G}_m)$ in $\mathsf{D}^{\mathrm{perf}}(\mathbb{\Lambda}_n)$, this required vanishing follows from [8, Rem. 3.3(ii)].

Turning to (iii), we recall results of Kervaire and Murthy [20, Th. 1.3] and Ullom [32, Cor. 2.6] combine to imply that, unless $p$ is exceptional, the canonical map $\mathrm{Pic}(R) \to \varprojlim_n \mathrm{Pic}(R_n)$ is injective (cf. Stolin [27, Th. 4.1] and [7, Rem. 3.9]). To prove (iii), it is thus enough to show, for each $n$, that the class of $\bigwedge^{|H|}_R(\mathcal{J}^{\Sigma'}_{\Sigma}(F))$ belongs to the kernel of the projection map $\mathrm{Pic}(R) \to \mathrm{Pic}(R_n)$. Moreover, since (ii) implies that the class of $\bigwedge^{|H|}_R(\mathcal{J}^{\Sigma'}_{\Sigma}(F))$ is the image of $\chi^{\mathrm{lc}}_{\mathbb{\Lambda}}(C^{\bullet}_F)$ under the composite morphism $K^{\mathrm{lc}}_0(\mathbb{\Lambda}) \to K^{\mathrm{lc}}_0(R) \to \mathrm{Pic}(R)$, in which the first map is induced by restriction of scalars and the second is $\delta_H$ for $H$ the trivial group, the required vanishing follows from the vanishing of $\pi'_n(\chi^{\mathrm{lc}}_{\mathbb{\Lambda}}(C^{\bullet}_F))$ observed above.

This completes the proof of Proposition 7.3.

7.5. **An integral main conjecture.** Taken together, the observations and results of this section prompt us to formulate the following 'main conjecture of equivariant integral Iwasawa theory'.

**Conjecture 7.8.** *If $(F/k, \Sigma, \Sigma')$ is any data satisfying Hypothesis 6.1 with respect to $\mathcal{K} = k^{\mathrm{con}}$, then the (finitely-presented, torsion) $\mathbb{Z}[\![\mathbb{Z}_p \times H]\!]$-module $H^{1,\Sigma'}_{\mathrm{dIw}}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)$ is such that*

$$\mathrm{Fit}^0_{\mathbb{Z}[\![\mathbb{Z}_p\times H]\!]}\big(H^{1,\Sigma'}_{\mathrm{dIw}}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)\big) = \theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} \cdot \mathbb{Z}[\![\mathbb{Z}_p \times H]\!].$$

*In particular, for all such data the ideal $\mathrm{Fit}^0_{\mathbb{Z}[\![\mathbb{Z}_p\times H]\!]}\big(H^{1,\Sigma'}_{\mathrm{dIw}}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)\big)$ is principal.*

**Remark 7.9.** The link between Stickelberger elements (and hence Dirichlet $L$-series) and the Galois structure of the (derived Iwasawa) Weil-étale cohomology of $\mathbb{G}_m$ predicted by the above conjecture is formulated relative to a ring $\mathbb{Z}[\![\mathbb{Z}_p \times H]\!] = \mathbb{Z}[\![\mathbb{Z}_p]\!][H]$ that is neither coherent, compact nor regular and has infinite Krull dimension. Nevertheless, Theorem 7.1(ii) proves the link in many cases for which the exact sequence (40) combines with the final assertion of Theorem 6.5 to imply

$$\theta^{\Sigma'}_{F^{\mathrm{con}},\Sigma} \cdot \mathbb{Z}[\![\mathbb{Z}_p \times H]\!] \subsetneq \varprojlim_n (\theta^{\Sigma'}_{F_n,\Sigma} \cdot \mathbb{Z}[(\mathbb{Z}/p^n) \times H]) \ \text{ in } \ \mathbb{Z}[\![\mathbb{Z}_p \times H]\!] = \varprojlim_n \mathbb{Z}[(\mathbb{Z}/p^n) \times H],$$

and hence that the equality of Conjecture 7.8 is strictly finer than the simultaneous validity of the equivariant Tamagawa number conjecture for $\mathbb{G}_m$ over all $F_n/k$. In addition, Proposition 7.3 gives further support for the prediction that $\mathrm{Fit}^0_{\mathbb{Z}[\![\mathbb{Z}_p \times H]\!]}\big(H^{1,\Sigma'}_{\mathrm{dIw}}((C^{\Sigma}_{F^{\mathrm{con}}})_{\mathrm{W}}, \mathbb{G}_m)\big)$ should be principal.

**Remark 7.10.** We set $\mathbb{A} := \mathbb{Z}[\![\mathbb{Z}_p \times H]\!]$. Then, if Conjecture 7.8 is valid, the exact sequences (39) and (40) combine to imply the existence of an isomorphism of $\mathbb{A}$-modules $\kappa : \varprojlim^1_{\nu_n} \mathcal{T}_n(M^{\Sigma'}_{\Sigma}(F)) \cong \varprojlim^1_{\varrho^n_{n-1}} \mathrm{Ann}_{\mathbb{A}_n}(\theta^{\Sigma'}_{F_n,\Sigma})$. Conversely, if such an isomorphism exists, then the equality in Conjecture 7.8 would follow directly from (39) and (40) if $\kappa$ could be chosen such that $\kappa \circ \xi = \xi'$. (A closer analysis shows it would suffice that $\kappa \circ \xi = \lambda \cdot \xi'$ for some $\lambda \in \mathbb{A}$ with the property that, for any fixed integer $n \geq n_0$, with $n_0$ as in the proof of Lemma 6.8, the quotient module $\mathbb{A}_n/(\theta^{\Sigma'}_{F_n,\Sigma}, \varrho_n(\lambda))$ is finite and of $p$-power order). However, even in this case, we cannot prove that a suitable $\kappa$ exists and, in general, the explicit analysis of connecting homomorphisms in the theory of pro-discrete modules seems difficult. To help clarify this point, we recall the infinite chain $J(1) \supsetneq J(2) \supsetneq \cdots$ of principal ideals of $\mathbb{Z}[\![\mathbb{Z}_p]\!]$ constructed in the proof of Proposition 3.2(iii). We then fix natural numbers $m$ and $m'$ with $m' > m$ and a finite abelian group $H$ of order equal to the (finite, prime-to-$p$) index of $J(m')$ in $J(m)$. Then Proposition 3.2(iii) implies that the $\mathbb{A}$-ideals $\mathcal{J}(m) := J(m) \cdot \mathbb{A} = J(m) \otimes_{\mathbb{Z}} \mathbb{Z}[H]$ and $\mathcal{J}(m') := J(m') \cdot \mathbb{A} = J(m') \otimes_{\mathbb{Z}} \mathbb{Z}[H]$ have all of the following properties.

(i) In analogy to Theorem 7.1(i), one has $\mathcal{J}(m') \subseteq \mathcal{J}(m)$ and $\mathcal{J}(m)/\mathcal{J}(m')$ is finite and has order that is prime-to-$p$ and divisible only by primes that divide $|H|$.

(ii) In analogy to (35), for every $n \in \mathbb{N}$ one has $\varrho_n(\mathcal{J}(m')) = \varrho_n(\mathcal{J}(m))$.

(iii) In analogy to (39) and (40), for both $i = m$ and $i = m'$ there exists an exact sequence

$$0 \to \mathcal{J}(i) \to I(0) \otimes_{\mathbb{Z}} \mathbb{Z}[H] \to (\mathbb{Z}_p/\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}[H] \to 0$$

of $\mathbb{A}$-modules in which the third arrow denotes the respective connecting homomorphism that is induced by the canonical short exact sequence in the lower row of (6).

(iv) In analogy to Conjecture 7.8, the $\mathbb{A}$-ideals $\mathcal{J}(m)$ and $\mathcal{J}(m')$ are both principal.

(v) For every $\ell \in \mathcal{P}$ one has $\mathbb{Z}_\ell[\![\mathbb{Z}_p]\!] \cdot \mathcal{J}(m') = \mathbb{Z}_\ell[\![\mathbb{Z}_p]\!] \cdot \mathcal{J}(m)$ in $\mathbb{Z}_\ell[\![\mathbb{Z}_p \times H]\!]$.

However, despite all of these close links, the ideals $\mathcal{J}(m)$ and $\mathcal{J}(m')$ are distinct.

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